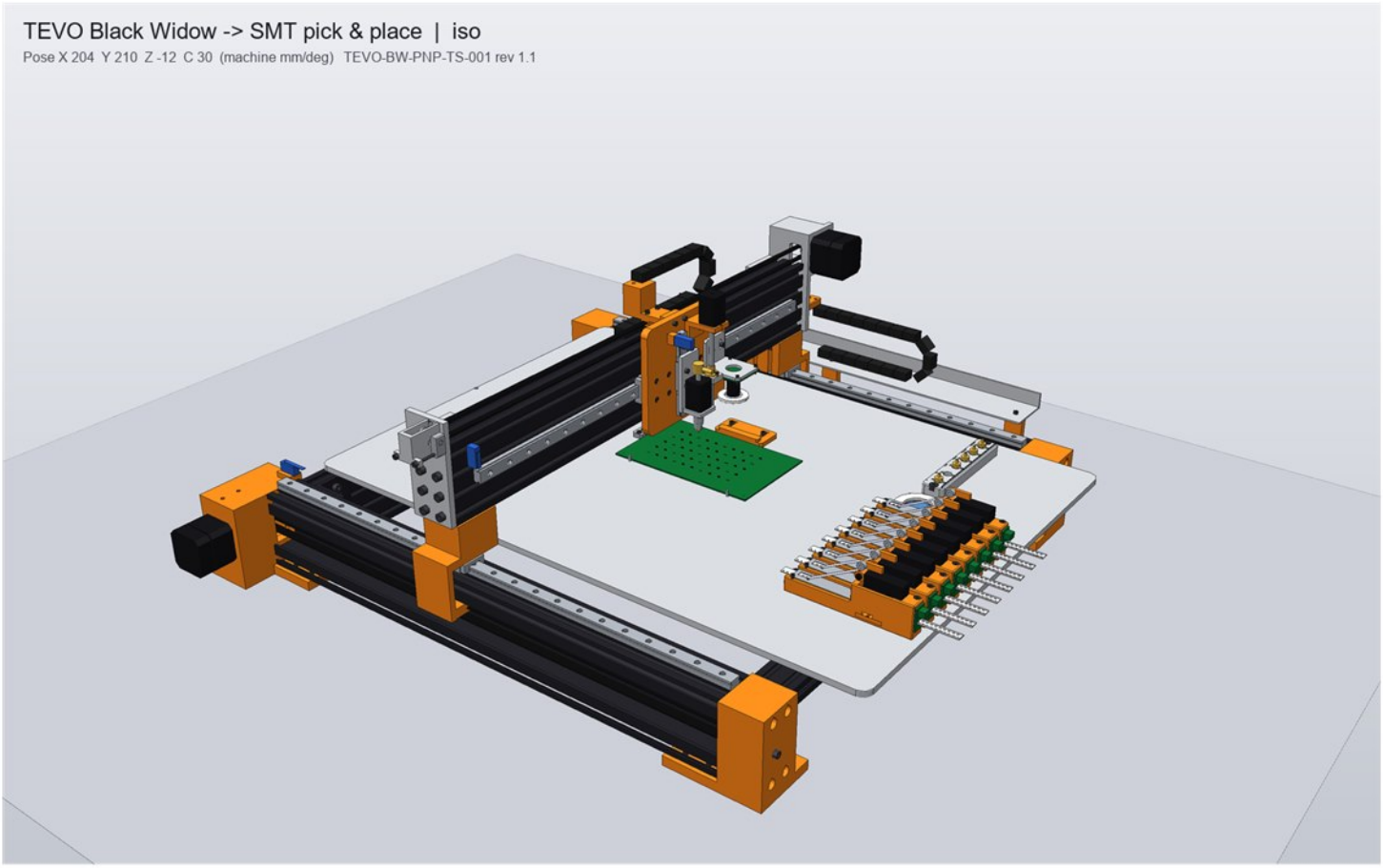
TEVO BLACK WIDOW

Инженерное досье

Компоновка · сборка · расчётные проверки · риски



CAD-рендер текущей компоновки. Плата и feeder bank неподвижны; портал движется по X, каретка — по Y.

ХОД X/Y	ПОДАТЧИКИ	МАССА ГОЛОВЫ
453 × 382,6 мм	8 дорожек, шаг 22 мм	499 г по модели / 500 г бюджет

Предварительная оценка по исходникам проекта и CAD-отчёту. Геометрическая проверка столкновений не доказывает жёсткость, точность или ресурс физической машины.

Содержание · полный комплект документации

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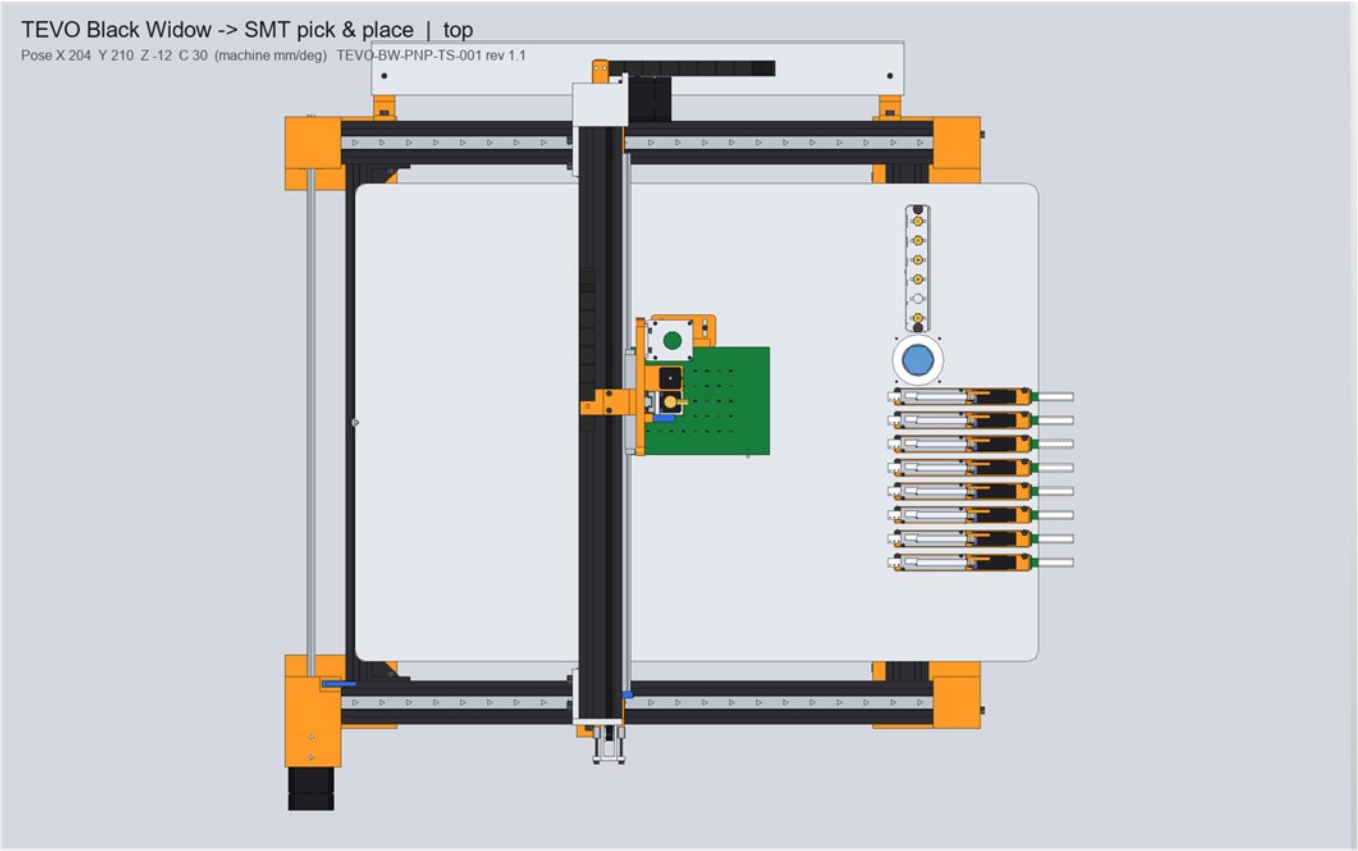
Примечание: техническая спецификация исходной ревизии сохранена как историческая база требований. При расхождениях ориентироваться на актуальные модель, отчёт и руководство сборки. Архивные feeder v2.1 и экспериментальные детали в этот том не включены.

1 · Архитектура и компоновка

Две C-Beam балки уложены горизонтально и работают как X-рельсы. Профиль 40×80×550 мм образует подвижную Y-балку. Рабочая плита, PCB, камеры, магазин и ленты остаются неподвижными.



Вид сверху: инструменты вынесены за правую границу хода портала.

Узел	Реализация	Что проверить
X	Две MGN12, два GT2 ремня и общий вал 8 мм	Синхронность ремней и yaw портала
Y	MGN12 и штатный ременной привод на балке 40×80	Боковое качание головы и тяга цепи
Z / C	NEMA8 linear T5×2; NEMA8 hollow-shaft на оси сопла	Перекос Z, люфт, биение наконечника
Tooling	Плита 635×444×6 мм на трёх опорах	Плоскостность PCB и прогиб выноса

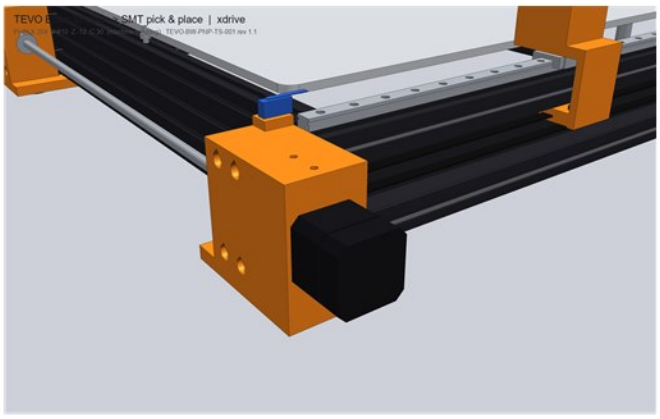
2 · Сборочная схема машины

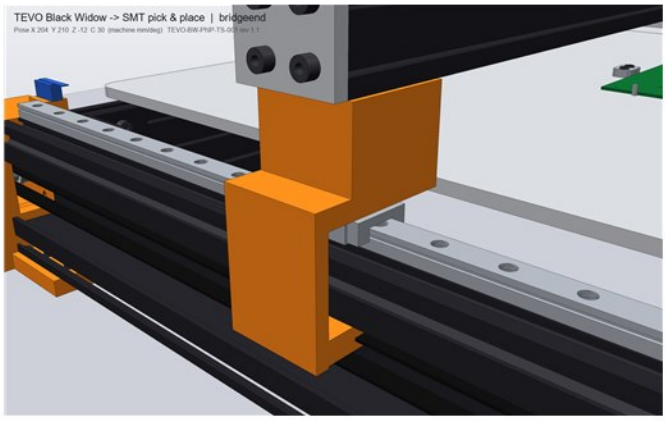
1 Основание
C-Beam + угловые блоки
поперечины → выверка

2 Портал X
MGN12 + торцы
балка + два ремня

3 Каретка Y
рельс + ремень
голова + drag chain

Сначала выставить базу и рельсы; ремни натягивать после проверки свободного хода.





Слева: общий вал и X-привод. Справа: мостовой торец на MGN12H и крепление к балке.

Этап	Контроль перед продолжением
Рама	Измерить диагонали, зазор C-Beam, плоскость крепления рельсов.
Портал	Свести фазу ремней; измерить перекос на обоих концах хода.
Каретка	Проверить ход без заедания и повторить измерения с установленной головой.

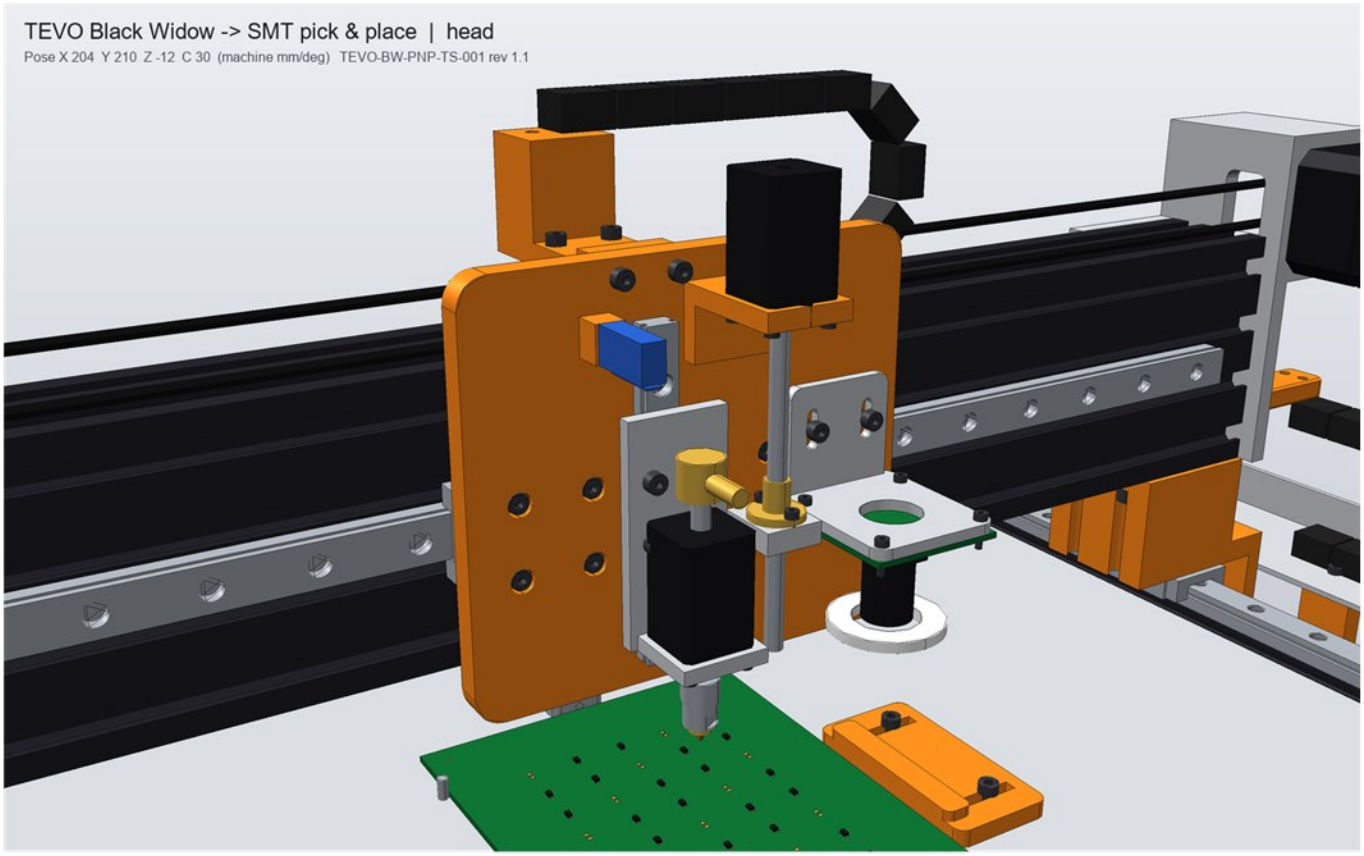


Схема сборки головы (последовательность, не взрывной CAD-вид)

Y-блоки MGN12	**Каретка PETG-CF**	**Z rail + slide MGN9 / Al**	**C motor hollow shaft**	**Compliance + nozzle**

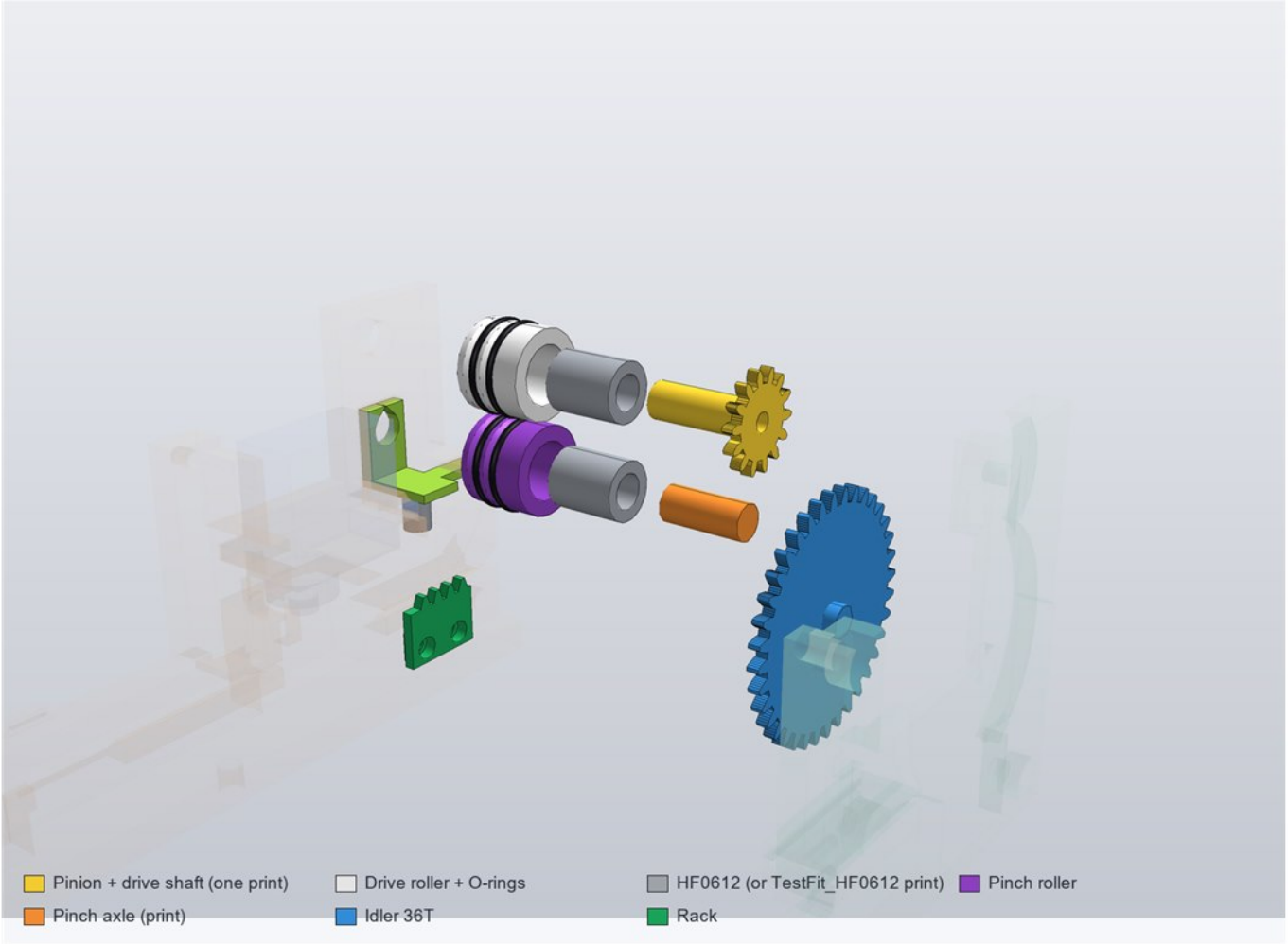
Подсистема	Модель	Проверка
Z	NEMA8 linear actuator, винт T5×2, фланцевая гайка	Измерить Z perpendicularity, backlash и повторяемость home.
C	NEMA8 hollow-shaft, вакуум через ось	Измерить TIR наконечника по обороту и при боковой нагрузке.
Vision	Down camera смещена на (2,57) мм от сопла	Калибровать offset по нескольким реперным точкам.

4 · Податчик TBW-Feeder v3.0

Ниже — настоящий CAD-взрывной вид peel head. Ориентация двух обгонных подшипников HF0612 определяет, какой ролик тянет ленту, а какой удерживает натяжение на обратном ходе.

TBW-Feeder v3.0 | peel head, exploded along the assembly axis

Drive: HF0612 into the drive roller, roller onto the pinion's printed shaft, shaft into the +Y cheek.
Pinch: HF0612 into the pinch roller, printed axle through it into the yoke. Then the idler, then the side plate (its pin enters the shaft end).



Кинематические цепи

Серво MG90S	**Сектор 13T + слайдер**	**Рейка + 36T idler**	**14T pinion + HF0612**	**Peel roller + nip → bin**

Отдельный пружинный pawl перемещает carrier tape на F2/F4; detent фиксирует следующий шаг. Одна дорожка проходит физические тесты до изготовления оставшихся семи.

Шаги на единицу — при заданных двигателях, шкивах и микрошаге:

Проверка	Расчёт	Результат модели
XY	200 × 16 / (2 мм × 20T)	80 steps/mm
Z	200 × 4 / 2 мм lead	400 steps/mm
C	200 × 16 / 360°	8,8889 steps/°
X torque margin	0,252 / 0,088 N·m; базовый расчёт 24 В	2,9
Y torque margin	0,252 / 0,042 N·m; базовый расчёт 24 В	6,1
Цикл PR-10	4,97 с/шт; 3600 / 4,97	724 шт/ч; до 814 при перекрытии подачи

Ходы и рабочие пределы

X travel
453 мм

Y travel
382,6 мм

Z usable
35 мм

Модель считает X: 3,08 кг при 3000 мм/с² и 400 мм/с; Y: около 0,50 кг при 4000 мм/с². Для torque curve используется PSU 24 В и типовой NEMA17. При 12 В отчёт указывает запас X около 1,9 — ниже проектного коэффициента 2.

PR-10: базовый расчёт 724 шт/ч не достигает цели 800 шт/ч; вариант с перекрытием подачи даёт 814 шт/ч только по оценке модели. Цикл на реальной машине не измерен.

Границы расчёта

- Нет FEA и modal analysis, нет проверки bolt preload или деформации печатных деталей.
- Каталожные массы/моменты, фактический драйвер и усилие кабельной цепи ещё не измерены.

6 · Слабые места и меры усиления

Риск	Наблюдение по CAD/расчёту	Что сделать до полной сборки
P0 · Стол	Плита 635×444×6 мм, около 4,5 кг, три опоры; прогиб не посчитан.	Снять карту плоскости PCB-индикатором; добавить нижние рёбра/опорные профили, сохранив кинематические три точки.
P0 · Углы рамы	Печатные PETG/ASA блоки совмещают опору, привод/подшипник, tie и ногу.	Металлические пластины/кронштейны в силовой цепи X-мотора и опор вала; контроль соосности после натяжения.
P0 · Масса головы	499 г при лимите 500 г; расчётный запас 1 г.	Взвесить голову вместе с проводами и вакуумным шлангом; обновить модель.
P1 · Портал	Один MGN12H на каждом X-рельсе, стойки портала 32 мм; моменты не рассчитаны.	Измерить прогиб/качание на крайних позициях; проверить моментные рейтинги блока и креплений.
P1 · Питание	Расчёт движения основан на 24 В; при 12 В запас X ниже цели.	Подтвердить PSU, ток моторов и драйвер; разгонять только после тестов на потерю шагов.
P1 · Камера lane 1	Расчётная камера имеет примерно 5,5 мм запаса до края Y-хода.	Физически проверить предел, кронштейн и допуски; при необходимости сдвинуть узел.
P1 · C-axis	Hollow-shaft мотор задаёт биение сопла; выбранный экземпляр не подтверждён.	Измерить runout наконечника; при превышении допуска добавить отдельную прецизионную опору.

Общая оценка компоновки

Размещение логичное: плата и ленты неподвижны; feeder bank находится вне траектории балки; общий вал синхронизирует X. Главные неопределённости относятся к жёсткости, деформации, вибрациям и метрологии, а не к базовой логике расположения узлов.

7 · План сборки и приёмки

#	Сборочный этап	Выходной контроль
1	Обмер до разборки	PSU/плата/драйверы, геометрия рамы, плоскость стола и фотографии разъёмов.
2	Рама и база	Диагонали, параллельность рельсов, плоскость tooling plate и повторяемость измерений.
3	Портал/каретка	Натяжение двух X-ремней, yaw, полный ход, прогиб под массой головы.
4	Голова	Фактическая масса; Z perpendicularity/home; C runout; nozzle-to-camera offset.
5	Feeder first article	Одна дорожка: канал, подача F2/F4, peel, удержание, endurance из feeder/TEST.md.
6	Электрика и OpenPnP	Отдельные контроллеры motion и feeder; проверки E-stop, vacuum, камеры, Safe Z.
7	Приёмка	PR-8 размещение 0603, PR-10 цикл, PR-11 lost picks, PR-12 homing — записать измерения.

Подтверждено / не подтверждено

CAD-ПРОВЕРЕНО
Reach по рассчитанной геометрии; отсутствие пересечений на выбранных позах; расчётная кинематика и BOM.

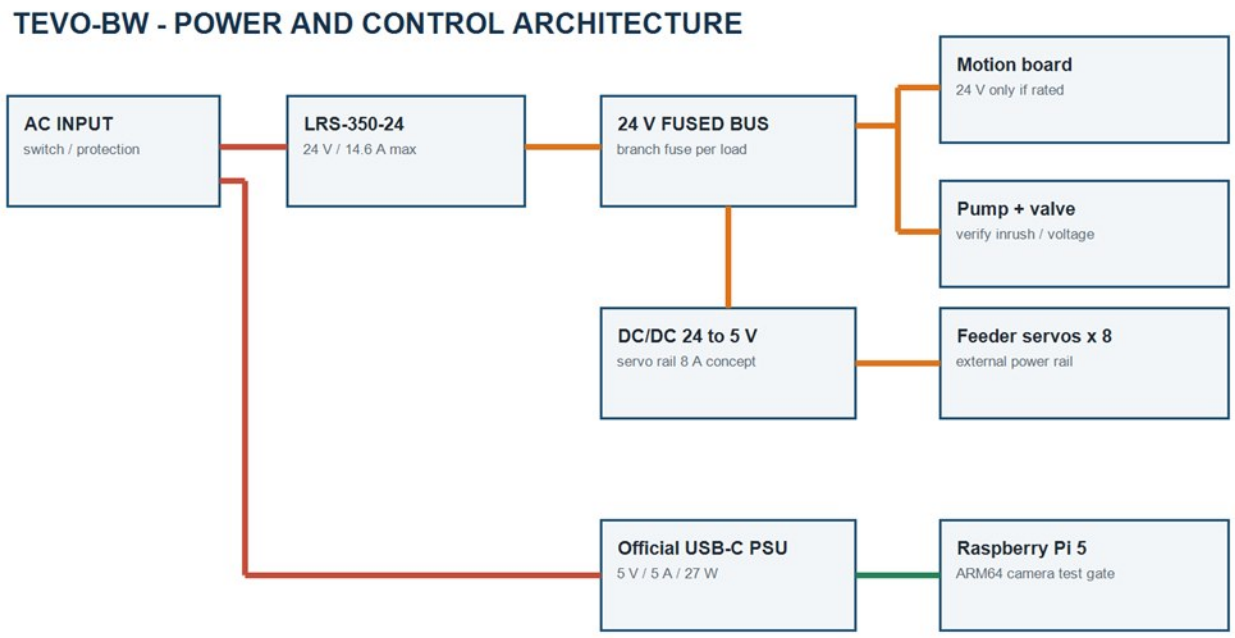
НУЖНО ИЗМЕРИТЬ
Плоскость, прогиб, вибрации, реальный torque margin, runout сопла, repeatability, feeder endurance и placement scatter.

Сборку полного станка начинать после проверки actual donor hardware и ревизии спецификации. Опорные документы: pnp/BUILD_AND_ASSEMBLY.md, pnp/out/report.md, pnp/feeder/ASSEMBLY.md, ELECTRONICS.md, OPENPNP.md, TEST.md.

Документ подготовлен по файлам проекта D:\TEVO_BW; CAD-рендеры взяты из pnp/out/render и pnp/feeder/out.

8 · Питание и компьютер управления

Проектное решение: общая 24 V шина для motion board и периферии после проверки их паспортов; отдельный DC/DC 24→5 V для сервоприводов; Raspberry Pi 5 питается своим официальным USB-C БП. Feeder Nano остаётся независимым USB serial контроллером.



PE bonds enclosure, door and PSU. E-stop drops drive power/enable; Pi remains powered for logging.

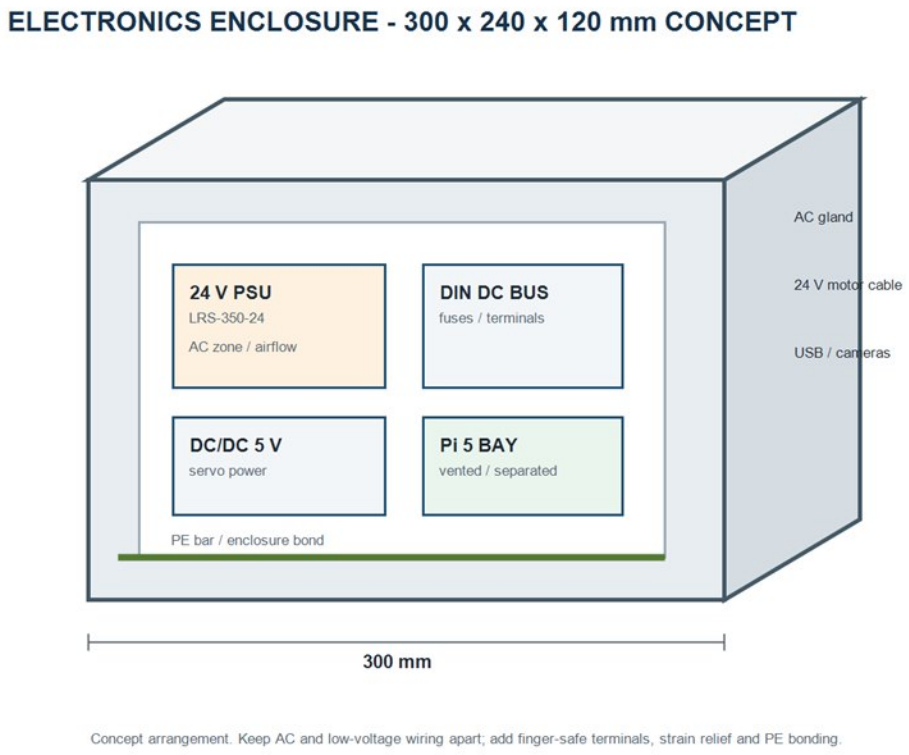
Concept: rate branch fuses, cables and AC protection from measured loads and local code.

Узел	Концепция	Условие до сборки
24 V PSU	LRS-350-24, 14.6 A / 350.4 W max	Только предварительный выбор; подтвердить суммарную нагрузку и совместимость платы.
Motion / vacuum	Раздельные ветви с защитой	Номиналы защит и кабелей выбрать по фактическим платам, току и пуску насоса.
Feeder servos	24→5 V, целевой DC/DC ≥8 A	Не питать сервоприводы через Pi, Nano или USB; проверить stall/просадку.
Host	x86-64 mini-PC is production recommendation; Pi 5 optional auxiliary	Official OpenPnP Linux packages list x86/x64. Pi/Orange ARM64 requires a tested custom build and two-camera qualification.

E-stop снимает аппаратный DRIVER_EN и питание приводов; SBC остаётся включённым для журнала. PE соединяет металлический корпус, дверь и PE источника. Все сетевые защиты, сечения и клеммы уточнить по шильдикам и местным нормам.

9 · Корпус блока и ввод в эксплуатацию

Компоновочный конверт 300×240×120 мм. На картинке показаны зоны размещения: PSU и сетевой ввод, защищённая DC-разводка, отдельный 5 V buck, SBC в охлаждаемом низковольтном отсеке, вводы кабелей и шина PE.

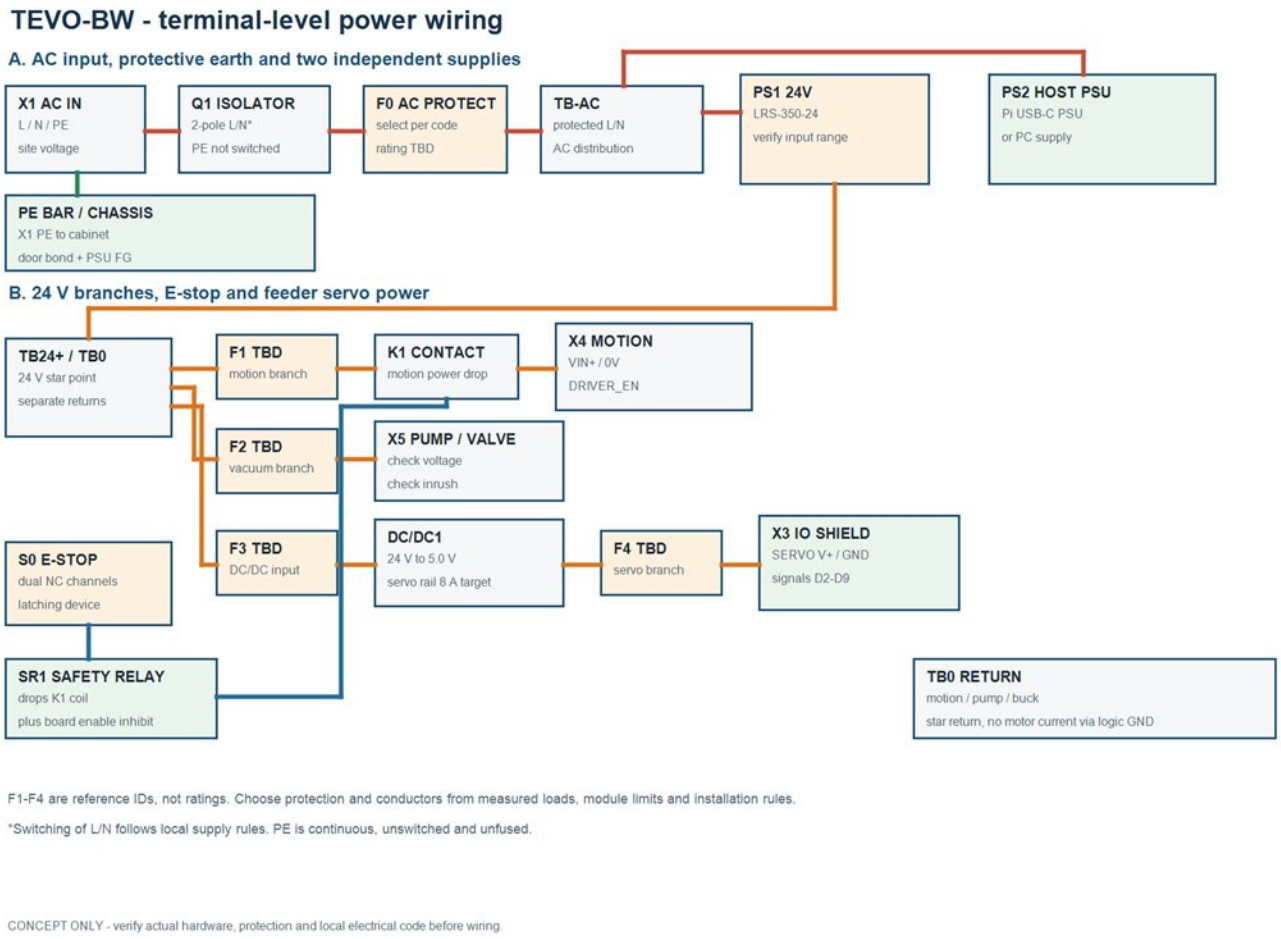


Concept arrangement. Keep AC and low-voltage wiring apart; add finger-safe terminals, strain relief and PE bonding.

#	Пусковой контроль
1	Сверить donor board, drivers и маркировку всех нагрузок; подтвердить 12/24 V.
2	Проверить полярность и выставить 24.0 V / 5.0 V с ограничением тока.
3	Проверить E-stop без движения: аппаратный запрет приводов, Pi остаётся online.
4	Проверить Nano и сервошину отдельно, затем измерить просадку на дальнем servo.
5	На x86-64 host проверить OpenPnP, обе камеры и serial motion/feeder; Pi ARM64 оставить экспериментальным.
6	Тепловой прогон 30 минут; записать температуры, напряжения, brownout и сбросы.

10 · Подробная клеммная схема питания

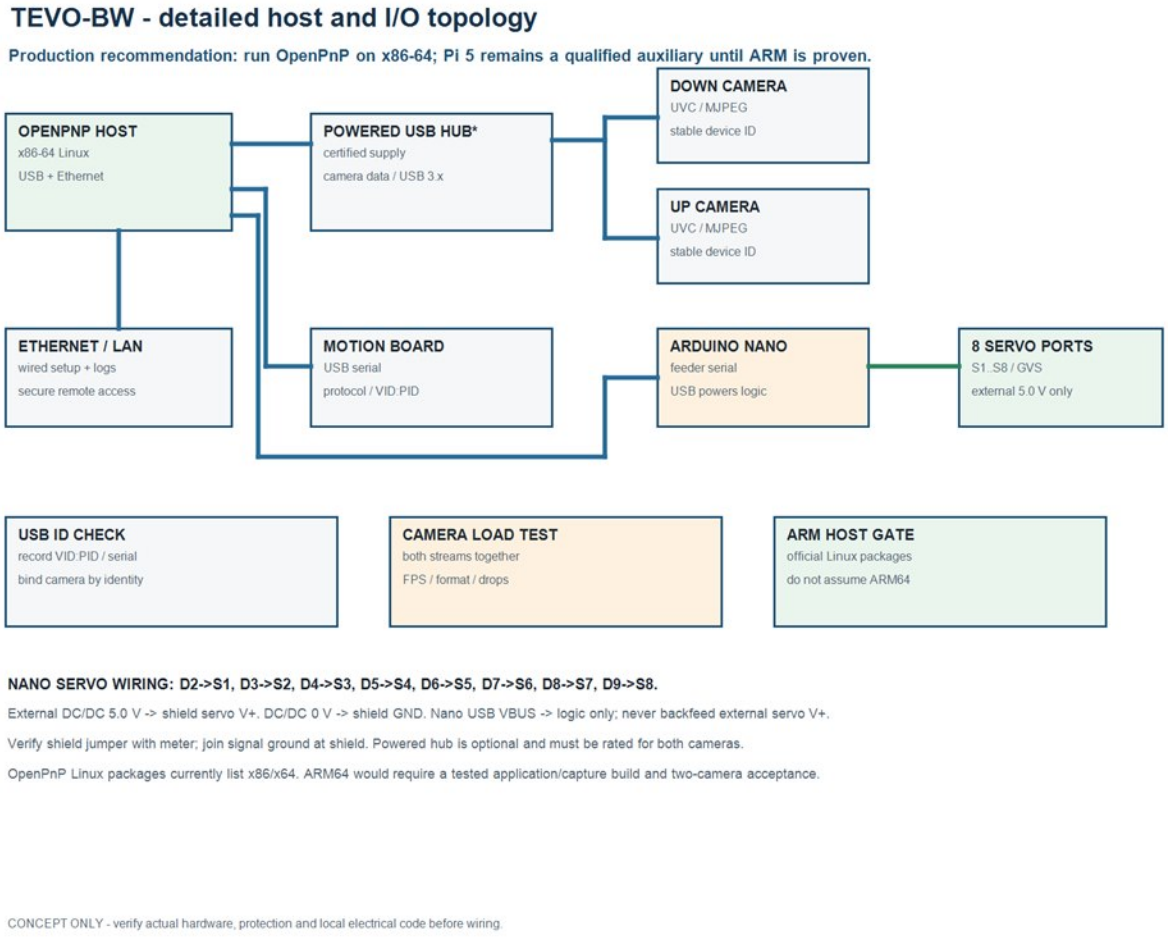
Схема показывает назначение коннекторов X1-X5, распределение AC/PE/24 V, идентификаторы ответвлений F1-F4, E-stop и отдельную 5 V сервошину. Номиналы F0-F4 намеренно оставлены TBD до идентификации реальных нагрузок и проводов.



Возврат моторов и насоса вести отдельными проводами на TB0; не пропускать рабочий ток через логику Nano/Pi. PE непрерывен, без предохранителя и коммутации. Перед подключением плат прозвонить X3/X4/X5, проверить полярность и отсутствие перемычки между внешним servo V+ и USB VBUS.

11 · Host, камеры и USB

Производственный host — x86-64 mini-PC с Linux 64-bit и официальной OpenPnP Linux сборкой. Raspberry Pi 5/Orange Pi с ARM64 можно использовать как вспомогательную плату; запуск OpenPnP на ARM требует отдельной сборки и квалификации.



Nano: D2-D9 управляют дорожками 1-8. USB питает только логику; DC/DC1 питает сервошину через отдельный вход shield. Камеры назначать по устойчивому device ID и проверять одновременно под выбранной частотой кадров.

Порядок подготовки Pi OS, SHA-256 официального образа, ссылки на OpenPnP и результаты сетевой попытки скачивания записаны в pnp/electronics/PI5_SOFTWARE.md.

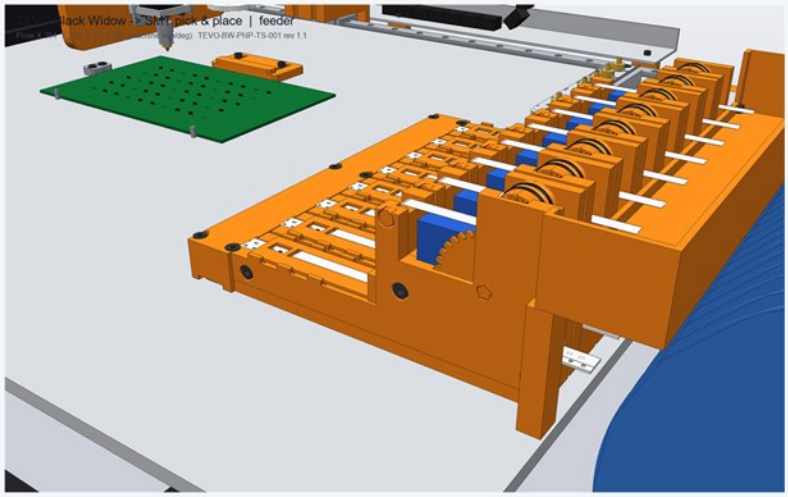
12 · TBW-Feeder v3.0: свежие CAD-рендеры

Лист собран из обновлённых FreeCAD-рендеров: feeder bank в контексте машины, механизм одной дорожки и отдельный взрывной peel head.

TBW-FEEDER v3.0 - CURRENT CAD RENDER SHEET

Generated from the current FreeCAD source model - 2026-10-31

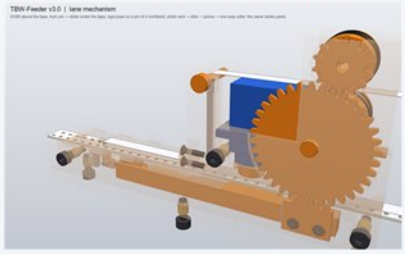
1 Eight-lane feeder bank in the machine layout



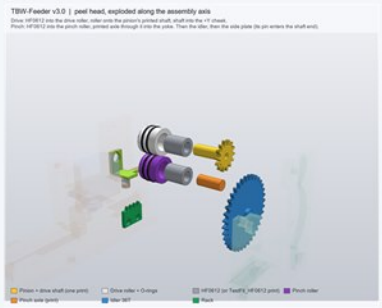
8 lanes | 22 mm pitch | each lane has its own servo

One Nano controls servo signals D2-D9; external 5 V rail supplies servo current.

2 Lane mechanism: servo, rack and one-way feed



3 Peel head exploded view



Feed path: SG90 crank -> slider rack -> idler -> pinion -> HF0612 drive roller.
Peel path uses the second one-way bearing, pinch roller, idler and side plate.

Render sources: pnp.py render --views feeder,feeder_mech; feeder.ps1 explode.

The electronics/schematic diagrams are separate. This sheet describes current v3.0 geometry, not tolerances or a printed-part acceptance report.

Исходные виды: render/feeder.png и render/feeder_mech.png; взрывной вид: feeder/out/peel_exploded.png. Все четыре файла перегенерированы из текущей геометрии v3.0 31.10.2026.

Project map and getting started

Исходный файл: README.md

TEVO Black Widow pick-and-place conversion

This repository contains a CAD-led conversion of a TEVO Black Widow 3D printer into a desktop SMT pick-and-place machine, plus the independent TBW-Feeder v3.0 design. The main machine is generated from Python models with FreeCAD; exported files are manufacturing outputs, not the editable source.

Start here

- Build and assembly guide (pnp/BUILDANDASSEMBLY.md): software setup, machine build sequence, assembly schemes, commissioning gates.
- Technical specification (TEVOBWPickandPlaceTechnicalSpecification.md): requirements and original work packages. Some sections describe an earlier design; see the guide's revision notes before building.
- PnP model README (pnp/README.md): model modules, CLI options, generated outputs.
- Feeder documentation (pnp/feeder/README.md): feeder design, assembly, electronics, OpenPnP and tests.
- Power/control design (pnp/electronics/POWERCONTROLDESIGN.md): 24 V distribution, x86-64 OpenPnP host, optional Raspberry Pi, wiring and enclosure concept.
- Detailed wiring sheets (pnp/electronics/WIRING_DETAILS.md): AC/PE, fused DC branches, E-stop, feeder servo pin map and host/camera USB.
- Board software setup (pnp/electronics/PI5_SOFTWARE.md): Raspberry Pi OS image/checksum, OpenPnP architecture compatibility and installation procedure.
- Combined technical dossier (pnp/out/TEVOBWPnP_Dossier.pdf): machine assembly, calculations, current feeder v3 exploded view and power-panel concept.
- Generated model report (pnp/out/report.md): current model dimensions, reach, mass/dynamics estimates, BOM and print orientations.

Build the CAD outputs

Install FreeCAD 1.1 for Windows. In PowerShell:

```
cd D:\TEVO_BW\pnp
.\pnp.ps1 info
.\pnp.ps1 build
.\pnp.ps1 check
```

The wrappers detect either %LOCALAPPDATA%\Programs\FreeCAD11\bin or %LOCALAPPDATA%\Programs\FreeCAD 1.1 \bin; set $env:FREECAD_BIN if your install uses another path. build writes the assembly document, STEP/STL exports, BOM, report and OpenPnP coordinate JSON under pnp/out/. check checks modeled moving/fixed interference at its defined poses; it does not certify a physical machine. Use .\pnp.ps1 all to build, check and render.

The feeder has its own CLI in pnp/feeder:

```
cd D:\TEVO_BW\pnp\feeder
.\feeder.ps1 build
.\feeder.ps1 audit
```

Build one feeder lane as a first article and complete feeder TEST.md (pnp/feeder/TEST.md) before producing the remaining seven. The model and audit are design checks; hardware fit, tape compatibility, clutch orientation and repeatability need physical verification.

Repository map

Path | Role

- Path: `pnp/params.py`, `pnp/model/` | Role: Parametric machine dimensions and FreeCAD geometry
- Path: `pnp/pnp.py` | Role: Machine build/check/render/info CLI
- Path: `pnp/feeder/` | Role: Current v3.0 lane, bank, firmware and build documents
- Path: `pnp/legacy/` | Role: Archived feeder v2.1 and obsolete head experiments
- Path: `Tevo Black Widow 3d printer - 1933864/` | Role: Stock printer reference CAD
- Path: `TEVO BW/` | Role: Additional carriage and hotend reference designs
- Path: `pnp/out/` | Role: Generated deliverables; regenerate after source changes

The source specification and current model have drifted. Treat pnp/params.py, current model code and the generated report as the implemented geometry, while resolving any safety- or fit-critical disagreement against the real printer before cutting material.

Original technical specification (historical baseline)

Исходный файл: TEVO_BW_Pick_and_Place_Technical_Specification.md

Technical Specification

NOTE: Implementation status: This is the original rev 1.1 work specification, not a fully synchronized as-built
NOTE: manual. The current CAD has since changed several decisions, especially the Z/C head and feeder architecture.
NOTE: Before fabrication, read pnp/BUILDANDASSEMBLY.md (pnp/BUILDANDASSEMBLY.md), the current generated
NOTE: pnp/out/report.md (pnp/out/report.md), and the feeder's ASSEMBLY.md (pnp/feeder/ASSEMBLY.md). Where this
NOTE: document conflicts with the current model, treat it as historical intent and verify dimensions against the
NOTE: actual printer. A controlled rev 2 specification is still needed.

Conversion of the Tevo Black Widow into a Single-Head SMT Pick-and-Place Machine

Field | Value

- Field: Document | Value: TEVO-BW-PNP-TS-001
- Field: Revision | Value: 1.1
- Field: Date | Value: 2026-10-16
- Field: Language | Value: English
- Field: Machine baseline | Value: Tevo Black Widow, OpenBuilds V-slot Cartesian, stock envelope about 370 × 250 × 300 mm
- Field: Project files | Value: Frame and plate models in `Tevo Black Widow 3d printer - 1933864/`, modular X mother plate in `TEVO BW/Tevo Black Widow - Modular X Carriage - 1829292/`

This document is the build specification and the implementation plan. Requirements use shall. Work packages in section 12 are the build instructions.

Revision 1.1 reorients the frame with its own members. The two vertical C-Beam Z columns become the horizontal X rails. The existing belt-driven X beam becomes the Y bridge. The Z leadscrews are removed and do not drive X.

1. Purpose

Convert the printer mechanism into a desktop placer for surface-mount components. The placement head is a fast XY carriage with a vacuum tweezer, a nozzle changer, a positioning camera, and a bank of tape holders. The existing controller board, stepper drivers, power supply, endstops, and wiring stay in service. New electronics are limited to the vacuum circuit, camera lighting, and any I/O the installed board does not already provide.

The machine places passive and small active parts from 8 mm carrier tape onto a bare PCB: 0603, 0805, 1206, SOT-23, SOD-123, and SOIC up to 8–16 pins. 0402 is a stretch goal after vision calibration. Fine-pitch QFP, BGA, and parts heavier than about 2 g are outside this revision.

2. Scope

2.1 In scope

- Kinematic conversion to a light, belt-driven XY carriage.
- Placement head: short Z stroke, nozzle rotation, spring compliance, vacuum tweezer.
- Passive nozzle magazine (the tool changer).
- Down-looking positioning camera on the head, plus one fixed up-looking camera so a picked part can be oriented before placement.
- Mechanical holders for 8 mm EIA-481 tapes, advanced by the head.
- PCB fixture on a fixed table.
- I/O map onto the existing board, firmware changes, and OpenPnP setup.
- Acceptance tests.

2.2 Out of scope

- Replacing the motion controller, power supply, or stepper drivers.
- Solder paste dispensing and reflow. Paste is applied outside this machine, or by a later dispenser tool.
- Motorized commercial feeders. The tape holders are head-actuated. A motorized feeder bus is a later option and must not require a new motion board.
- Automatic optical inspection after reflow.
- Enclosure HVAC and production-line safety certification. Section 10 covers workshop safety for this build.

2.3 Definitions

Term | Meaning

- Term: Carriage | Meaning: The moving XY head. It carries Z, rotation, the vacuum nozzle, and the down-looking camera.
- Term: Vacuum tweezer | Meaning: Spring-loaded nozzle on a vacuum line. It picks a part from a tape pocket and releases it on the pad.
- Term: Tool | Meaning: One nozzle tip. Tips differ by orifice and outer diameter.

- Term: Tool changer | Meaning: A fixed magazine. The head drops the current tip and picks up another. No extra motor.
- Term: Positioning camera | Meaning: The down-looking camera on the carriage. It finds PCB fiducials and tape pockets.
- Term: Up-looking camera | Meaning: A fixed camera under the travel path. The part flies over it so OpenPnP can correct X, Y, and angle.
- Term: Tape holder | Meaning: A lane that locates an 8 mm tape, peels or holds the cover tape, and advances one pocket when the head presses a lever.
- Term: Existing electronics | Meaning: The board, drivers, PSU, and harness already on this machine. Stock Black Widow class is an MKS Gen-class board running Marlin. The installed board is confirmed in WP0 and may differ.

3. Assets that stay

Asset | Use in the placer

- Asset: Two C-Beam Z columns, 40×80, assembly length 574 mm in the project model | Use in the placer: Laid horizontal. These are the X rails.
- Asset: 40×80×550 mm X beam, X motor plate, belt tension plate, X carriage | Use in the placer: Y bridge. The beam spans the two C-beams. Carriage travel along the beam is Y.
- Asset: 20×80×500 mm member | Use in the placer: Fixed tie that holds the C-beams parallel. A second tie comes from the retired bed extrusion if the frame has only one.
- Asset: Two 20×80×210 mm members | Use in the placer: Legs. Their length does not span the gap between the C-beams.
- Asset: X carriage body and the modular mother plate (`LP_MotherPlate` pattern: plate bolted to the OpenBuilds X carriage, daughter plates swapped with M3 hardware) | Use in the placer: Head mounting plate on the Y beam. Hotend daughter plates are removed.
- Asset: NEMA 17 now on the X beam | Use in the placer: Y motor. It stays on that beam.
- Asset: NEMA 17 now on the bed (stock Y) | Use in the placer: X motor, driving both C-beam belts through a jackshaft.
- Asset: NEMA 17 now on Z | Use in the placer: Head Z motor.
- Asset: Endstop switches | Use in the placer: Home X, Y, and head Z.
- Asset: Controller, drivers, 12 V or 24 V PSU, display | Use in the placer: Motion and valves. Heater outputs are reassigned.
- Asset: Z leadscrews, Delrin nut blocks, Z couplers | Use in the placer: Removed in WP1. Not an axis drive.
- Asset: Bed axis extrusion (Y-axis assembly, about 571 mm in the model) and heated bed plate | Use in the placer: Bed motion is removed. The plate becomes the tooling plate only if it is flat within 0.05 mm over the PCB area. Otherwise it is replaced. The extrusion is a spare tie.
- Asset: Bowden or direct-drive extruder, hotend, part cooling, bed heater, thermistors | Use in the placer: Removed from the motion path and disconnected.
- Asset: Electronics box, drag-chain brackets | Use in the placer: Kept. The chain is re-terminated on the moving bridge.

Sizes above are bounding sizes from Tevo Black Widow assembly.step. WP0 measures the machine and replaces them where they differ. Stock travels are about X 370 mm, Y 250 mm, Z 300 mm. After the reorientation the useful envelope is about 400 mm along the C-beams (X) and about 320–350 mm along the 550 mm beam (Y), before the camera, magazine, and tape bank take their edge space.

4. Selected architecture

The carriage moves in X and in Y. The PCB and the tapes stay still. The frame is rebuilt from its own extrusions: stock Z structure becomes X, stock X structure becomes Y.

1. Both C-Beam columns are unbolted from the vertical Z position and laid horizontal, parallel, as the X rails. The Z leadscrews and Delrin nuts come off. A leadscrew cannot meet PR-1.
2. The 550 mm X beam spans the two C-beams and is the moving bridge. Travel of that bridge along the C-beams is machine X.
3. The belt, motor plate, tension plate, and carriage that already live on the 550 mm beam stay there. Travel of that carriage along the beam is machine Y.
4. X is driven by two GT2 belts, one on each C-beam, turned by one motor through a jackshaft so the bridge cannot yaw. The motor is the former bed motor.
5. Y is the existing single GT2 belt on the 550 mm beam.
6. Both X and Y run on MGN12H linear rails bolted to those extrusions. Stock V-wheels may hold the bridge only while the rails are aligned, then they come off.
7. The fixed table sits in the rectangle between the C-beams, on the 20×80 ties. The 210 mm profiles are the legs. Bed motion is removed.
8. Pick and place strokes are a 35 mm Z axis on the head, driven by the former Z motor. The bridge has no vertical axis.
9. The nozzle rotates on its own axis (machine axis C, wired as Marlin axis A).
10. Vacuum is a diaphragm pump, a pick valve, a short blow-off pulse, and a vacuum switch.
11. Nozzle tips are Juki-compatible (CP40 / 500 series pattern) and live in a 6-pocket passive magazine.
12. The down-looking camera is bolted to the carriage at a fixed offset from the nozzle.
13. The up-looking camera is bolted to the table on the path between the tape bank and the PCB.
14. Tape holders sit along the front of the table, outside the PCB clamp zone. Each lane advances when the nozzle presses its lever.
15. A host PC runs OpenPnP, talks G-code to the existing board over USB, and captures both cameras over USB.

Coordinate system, right-handed, origin at the front-left home of the carriage:

- +X along the C-beams, from the home end toward the far end. Home is the left end when the operator faces the tape bank.
- +Y along the 550 mm beam, from the front tape bank toward the rear of the table.
- +Z up, nozzle descending toward the board as Z decreases.
- +C counterclockwise when looking down on the nozzle. Zero is the flat of a two-pin part parallel to X.

Every later offset (camera, magazine, feeders, fiducials) is stored in this frame.

5. Functional requirements

5.1 Carriage

- FR-1. The carriage shall move in X and Y with the PCB and the tape holders stationary.
- FR-2. X and Y shall be GT2 belt driven, 2 mm pitch, closed loop in the mechanical sense (tensioned belt, no spring compliance in the drive path).
- FR-3. X and Y shall home against limit switches. Homing direction is toward the switches. The firmware shall back off and re-approach at no more than 10 mm/s.
- FR-4. Moving mass of the carriage, including Z motor, C motor, nozzle, camera, and bracket, shall be at most 500 g.
- FR-5. The drag chain shall carry vacuum tube, camera USB, valve wires, and motor cables without tugging the carriage inside the work envelope.

5.2 Vacuum tweezer

- FR-6. The nozzle shall pick a part by vacuum and release it with a timed positive pressure pulse of 5–20 ms.
- FR-7. The nozzle shall have 1.0–1.5 mm of spring compliance so a slightly low Z command does not crush a part.
- FR-8. A vacuum sensor shall report "part present" after a pick and "part released" after a place. OpenPnP shall abort the place if the pick check fails.
- FR-9. The nozzle stem shall be interchangeable without tools beyond the magazine cycle. Stem runout at the tip shall be within 0.03 mm.
- FR-10. The vacuum pump shall remain on during a job. The pick valve shall open only for the pick-and-carry interval.

5.3 Head Z and rotation

- FR-11. Head Z travel shall be at least 35 mm, enough to clear the tallest nozzle in the magazine and the tallest tape cover.
- FR-12. Z shall be driven by a leadscrew or ball screw of 2 mm lead or finer, on a linear guide, with the nozzle axis parallel to Z within 0.05 mm over the stroke.
- FR-13. Axis C shall rotate the nozzle continuously through more than 360° and hold any commanded angle.
- FR-14. C and Z shall be coaxial: rotating C shall not shift the tip by more than 0.03 mm in X or Y.

5.4 Tool changer

- FR-15. The magazine shall hold 6 nozzle tips in known machine coordinates.
- FR-16. A change shall be a sequence of moves only: descend, twist to release, rise, move, descend, twist to engage, rise. No magazine actuator.
- FR-17. Each pocket shall reject a wrong engagement. A failed seat shall be detectable by a Z stall, a pocket switch, or a vacuum leak check before the next pick.
- FR-18. OpenPnP shall be the only authority for which tip is loaded. The operator shall be able to read the loaded tip name on the OpenPnP UI.

5.5 Positioning camera

- FR-19. The down-looking camera shall see PCB fiducials and tape pockets in the same machine coordinates as the nozzle, through a stored nozzle-to-camera offset.
- FR-20. The camera shall be global shutter, USB, with a lockable M12 lens focused at the PCB surface.
- FR-21. A light on the carriage shall illuminate the field of view. The light shall be switchable from the controller so it is off while the machine is idle.
- FR-22. The up-looking camera shall image the bottom of a part hovering at a fixed height. OpenPnP shall apply the measured X, Y, and angle error before the place move.
- FR-23. Both cameras shall be rigid. Focus, aperture, and mounting screws shall be thread-locked after calibration.

5.6 Tape holders

- FR-24. The bank shall provide at least 8 lanes of 8 mm tape. Lane pitch shall allow the lever of lane N to be pressed without striking lane N+1.
- FR-25. Each lane shall locate the tape by its sprocket holes, 4 mm pitch, per EIA-481.
- FR-26. Each lane shall advance exactly one pocket per head press, for part pitches of 2 mm and 4 mm. 2 mm pitch is two presses or a lever set to half index; the build record shall state which.
- FR-27. Cover tape shall be peeled back and held so the pocket is open at the pick point and the cover does not rise into the nozzle path.
- FR-28. The pick point of every lane shall be a single OpenPnP feeder coordinate, re-found by the positioning camera when a tape is changed.

5.7 Worktable and PCB

- FR-29. The PCB shall be clamped flat on a fixed plate. Datum is two pins or two edges against hard stops, plus a clamp.
- FR-30. Board thickness from 0.8 mm to 1.6 mm shall be supported. A change of thickness is a Z offset in OpenPnP, not a mechanical rebuild.

• FR-31. The plate shall be electrically idle. Heater power shall be disconnected and the heater MOSFET output reused or capped.

5.8 Existing electronics

• FR-32. The installed motion board shall generate steps for X, Y, Z, and C.

• FR-33. Two switched outputs shall drive the pick valve and the blow valve. A third output shall drive the camera light. The bed-heater and hotend-heater channels are the first candidates.

• FR-34. One digital input shall read the vacuum sensor.

• FR-35. Heaters, thermistors, and the extruder motor shall be removed from the configuration so a fault cannot turn them on.

• FR-36. Any shortfall in drivers or inputs shall be filled by a small add-on module on the existing supply. The motion board itself is not replaced under this specification.

6. Performance requirements

Targets apply after WP10, with the rails installed, at 20–25 °C, on the reoriented frame.

ID | Metric | Requirement

• ID: PR-1 | Metric: XY travel speed | Requirement: At least 400 mm/s in the clear area between feeders and board

• ID: PR-2 | Metric: XY acceleration | Requirement: At least 3000 mm/s² on X (the bridge) and 4000 mm/s² on Y (the light carriage)

• ID: PR-3 | Metric: Head Z speed | Requirement: At least 60 mm/s

• ID: PR-4 | Metric: C speed | Requirement: 360° in under 150 ms for a 90° index

• ID: PR-5 | Metric: Pick check | Requirement: Vacuum decides pass/fail within 30 ms of the pick dwell

• ID: PR-6 | Metric: Nozzle change | Requirement: Complete 6-tip cycle under 8 s each way, including moves from the board center

• ID: PR-7 | Metric: Fiducial repeatability | Requirement: Down-looking camera returns the same fiducial within 0.03 mm over 20 measurements

• ID: PR-8 | Metric: Placement, 0603 and larger | Requirement: At least 95% of 50 parts inside 0.10 mm of pad center and inside 2° of pad angle, after up-looking correction

• ID: PR-9 | Metric: Placement, 0402 | Requirement: Goal: 90% inside 0.08 mm. Failure does not fail the machine; it is recorded

• ID: PR-10 | Metric: Cycle rate | Requirement: At least 800 parts per hour on a demo board with one nozzle and parts from at least 4 lanes, vision on

• ID: PR-11 | Metric: Lost-part rate | Requirement: Under 2% picks aborted by the vacuum check on fresh tape

• ID: PR-12 | Metric: Homing repeatability | Requirement: Nozzle returns to a dial indicator within 0.03 mm after 10 home cycles

Stock marketing figures (XY 0.012 mm, Z 0.004 mm) are not acceptance criteria. PR-7, PR-8, and PR-12 are.

7. Mechanical design

7.1 Frame and X axis

The two C-beams are the X rails. The 550 mm beam is the bridge that travels on them. Square the rectangle before any rail is drilled.

• Remove both leadscrews, Delrin nut blocks, and couplers before the C-beams are laid down. Bag them with the lead written on the bag (measure it: turns of the nut over 20 mm of screw). A TR8×8 screw at a realistic NEMA 17 speed stays near 30–60 mm/s, which fails PR-1, so the screw is not refitted on X.

• Lay the C-beams horizontal and parallel. Set the gap so the 550 mm beam and its end plates bolt on as they do between the Z columns today. The model gap is inside a machine width of about 580 mm.

• Tie the beams with the 20×80×500 member. Add a second tie from the retired bed extrusion so the rectangle has two crossmembers. Stand the frame on the two 20×80×210 members as legs.

• Diagonals of the rectangle equal within 0.5 mm. Top faces of the two C-beams coplanar within 0.05 mm.

• Mount one MGN12H rail on each C-beam, reference edges facing the same way. Tighten from the center outward.

• Two blocks per rail carry a bridge plate. Bolt the 550 mm beam to those plates. Tram the beam: a pointer at the future nozzle changes height by no more than 0.05 mm from X minimum to X maximum.

• X drive is one NEMA 17, the former bed motor, and a jackshaft across the two C-beam ends. The shaft carries two 20-tooth GT2 pulleys, one aligned to each beam. Each beam has its own 9 mm GT2 loop, an idler at the far end, and the belt ends clamped to that side's bridge plate. Both belts are the same length and the same tooth phase.

• If a shaft cannot span the gap, use two matched NEMA 17 motors, one at each C-beam, wired in series on the single X driver. Pulleys still match tooth for tooth. Series wiring keeps the driver current equal to one motor. Parallel wiring is not used.

• X home switch is on the bridge at the left end.

• Both belt runs get a cover. Leave at least 10 mm of unused rail at each end so the blocks never hit the rail screws.

• Yaw check: move X through 200 mm and read a dial indicator on the bridge plate at the far C-beam. Drift above 0.05 mm means the two belts are out of phase or unequal in length. Correct that before any placement test.

7.2 Y axis

Y is the belt axis already built on the 550 mm beam. The beam now lies across the C-beams, so this travel runs from the tape bank toward the rear.

• Remove V-wheels, hotend, extruder, and any plate that exists only to carry them.

• Bolt one MGN12H rail to the face of the beam. The modular mother plate bolts to two MGN12H blocks. The nozzle axis is the hole on the plate centerline, drilled if the stock plate has none.

• Keep the X motor plate, the tension plate, and the GT2 belt. Use a 20-tooth pulley. Tension so the belt is one span and the carriage has no dead band when pushed 0.1 mm and released.

• The motor on this beam is the Y motor. Plug it into the board's Y driver (section 8.1).

• Y home switch is at the front end of the beam, on the tape-bank side.

• Leave at least 10 mm of unused rail at each end.

7.3 Head

Build the head as a stack on the mother plate. Keep it inside the 500 g budget. A practical stack, top to bottom:

1. Z motor, NEMA 17, on a printed or aluminum cage.

2. Z screw, TR8×2 (2 mm lead) with a spring-loaded anti-backlash nut, or a 2 mm lead ball screw if one is on hand. Travel 40 mm, soft stops at 2 mm and 38 mm.

3. Z carriage on an MGN9 or MGN12 short rail. This carriage holds the rotation assembly.

4. C motor, NEMA 11 or NEMA 14, driving the nozzle shaft by a 20-tooth / 20-tooth GT2 loop or by spur gears of equal count. Ratio 1:1 keeps the steps-per-degree integer. Use a closed belt, not an open one.

5. Nozzle shaft, 6 mm ground rod or a hollow stainless tube, in two bearings. The shaft is the vacuum path if hollow. If the shaft is solid, a side port and a flexible PU tube enter above the lower bearing through a swivel that does not twist the tube more than ±180° during a normal job. A hard stop on C at ±180° from zero is acceptable if jobs are planned inside that range; continuous rotation through a pneumatic slip ring is the upgrade when a job needs it.

6. Compliance spring between the shaft and the nozzle holder, preload about 1 N, stroke 1.0–1.5 mm. The holder is a Juki-style collet: the tip snaps in on a 1/4 turn or on the clutch geometry of the chosen tip family. Buy the tips and the holder as a set. Do not invent a collet that does not match a commercial tip.

7. Down-looking camera bracket on the mother plate, not on the Z carriage, so the camera stays in focus when Z moves. Nozzle-to-camera offset is constant in X and Y.

Z home is a switch at the top of the stroke. The nozzle then travels down by increasing negative Z, or the firmware uses a tool offset so the operator commands Z = 0 at the PCB surface. Pick one convention in WP9 and keep it:

• Machine Z zero is the home switch.

• OpenPnP "part height zero" is the top of the PCB, stored as a camera and nozzle Z offset.

Cable the Z and C motors in the drag chain. Vacuum tube is 4 mm OD PU, one continuous run from the valve block on the frame to the swivel. A tube that piggybacks on the Z carriage must have a service loop so Z can move 40 mm without lifting the nozzle.

7.4 Vacuum circuit

Mount the pump and valves on the frame, not on the carriage.

• Diaphragm pump, same supply voltage as the board (confirm 12 V or 24 V in WP0), free-air flow enough to reach −60 kPa in the nozzle line in under 100 ms with the valve open.

• Pick valve: 2-way normally closed, between pump and nozzle. Energized = vacuum at the nozzle.

• Blow valve: 3-way or a separate normally closed valve from a regulated 10–30 kPa puff into the nozzle line, for the release pulse. A simple arrangement that works: pick valve NC to vacuum, blow valve NC to a tee, and a small orifice leak so the line decays when both are off. Document the plumbing in the build record with a one-line diagram.

• Sensor: a vacuum switch set to about −40 kPa, or an analog sensor read by a spare thermistor input scaled in firmware. A digital switch is enough for FR-8.

• Inline filter, 10 µm, between nozzle and valves, so a dropped chip does not enter the pump.

• Muffler on the pump exhaust.

Nozzle set for the first articles:

Tip | Typical parts

• Tip: 0.4–0.5 mm | Typical parts: 0603, small SOT

• Tip: 0.7–0.8 mm | Typical parts: 0805, 1206, SOT-23

• Tip: 1.0–1.2 mm | Typical parts: SOD, tantalum A/B, SOIC

• Tip: 1.5–2.0 mm | Typical parts: Larger SOIC, inductors under 2 g

• Tip: Spare | Typical parts: Second 0.7 mm, the size that wears first

• Tip: Plug | Typical parts: Solid tip used only to prove the magazine

7.5 Tool changer magazine

• Aluminum or thick printed bar, bolted to the table at the left or right end, outside the PCB clamps and outside the tape bank.

• Six pockets on a straight line parallel to X, so a change is mostly one-axis motion.

• Pocket pitch at least 18 mm so the camera bracket clears the neighboring tip.

• Each pocket holds the tip the way the commercial holder releases it: for a twist-lock family, the pocket grips the tip collar and the head rotates C by the unlock angle (often 90°) while Z is at the engage height.

• Z engage height is the same for every pocket. A stepped pocket is a defect.

• Add a small leaf spring or magnet so a parked tip cannot bounce out when the machine homes.

• Engrave or mark pocket index 1 through 6 on the bar. OpenPnP nozzle tip names match those marks.

• Teach procedure is in WP6. The magazine origin is the center of pocket 1 at engage height.

7.6 Positioning camera

Down-looking camera:

• Global-shutter USB board camera, 720p or better. Rolling shutter is rejected because the carriage images on the move during setup and the blur is direction-dependent.

• M12 lens, focal length chosen so the field of view on the PCB is about 15–25 mm across. That is wide enough for a fiducial search and narrow enough that one pixel is under 0.03 mm. Compute it: FOV mm / pixel count = mm per pixel. Record the number after calibration; the calculation only sizes the lens.

• Working distance fixed by a slotted bracket. Set focus on a fiducial, then glue the focus ring.

• Light: a ring or two LED bars around the lens, diffuse, white or red. Red reduces glare on ENIG and HASL. The light is a MOSFET load, not a USB-powered always-on LED.

• Mount so the nozzle, at Z safe, is outside the field of view. The stored offset is the only link between camera center and nozzle center.

Up-looking camera:

• Same camera family, lens aimed up through a hole in the table or from a bracket at the front edge.

• Diffuse backlight or a low-angle ring so the part outline is a dark silhouette. A silhouette is more repeatable than a top light for angle measurement.

• Fixed hover height: nozzle tip 3–5 mm above the lens cover, recorded as the up-camera Z.

• Place the camera so the Y move from the tape bank to the board passes over it. The detour shall be under 80 mm.

7.7 Tape holders

One lane is a module. Eight modules bolt to a front rail.

Each module:

• Body locates a 8 mm tape width within 0.2 mm.

• Sprocket wheel, tooth pitch 4.0 mm, engages the tape holes. A 12-tooth or 16-tooth wheel is easy to print and to check with a pin: 4.0 mm chord between adjacent teeth.

• Ratchet so the wheel moves only in the feed direction.

• Lever on the top of the module. The nozzle comes down on the lever pad, travels 3–4 mm, and the sprocket advances one pitch. A hard stop limits the press so a low Z command cannot break the lever.

• Cover-tape peel: a fixed blade and a spring finger that fold the cover back over a take-up pin. The operator pulls the cover onto the pin when loading. The pick pocket is the first open pocket past the blade.

• Pick clearance: at the pick Z, the nozzle body clears the module walls by at least 1 mm.

Lane pitch 20–24 mm. Number the lanes 1–8 from left to right. Leave a 30 mm gap between the last lane and the magazine if they share a side.

Loading a lane is an operator procedure, written on a card at the machine:

1. Lift the cover finger.
2. Seat the tape, sprocket holes on the teeth, parts facing up.
3. Close the finger and peel 3 pockets onto the take-up pin.
4. In OpenPnP, select the feeder and run "vision home" so the positioning camera centers that pocket.

Head motion for one feed, before the pick: move to the lever XY, Z down to the press height, Z up, move to the pick pocket, Z down to pick, vacuum on, dwell, Z up, vacuum check.

7.8 PCB fixture

• Tooling plate, aluminum, at least 6 mm thick, bolted to the frame on three points so it can be trammed. Target: 0.05 mm flatness over a 200 × 150 mm board area.

• Two hardened pins, 3 mm, in a line parallel to X, as the board's rear or front datum. A slide clamp opposite the pins pushes the board against them.

• A third point, a spring plunger or a screw stop, sets the X datum.

• Record pin coordinates in the machine frame. OpenPnP board location uses those datums plus the fiducials.

• Do not rely on the old bed clips.

8. Electronics

WP0 writes the actual connector list. The map below is the assignment for an MKS Gen-class board, which is the stock Black Widow controller. If the installed board is a later 32-bit board, keep the same signal names and move them to the matching pins.

8.1 Axis map

Signal | Driver socket | Motor and mechanism

• Signal: X | Driver socket: board X | Motor and mechanism: Former bed motor, jackshaft, both C-beam belts. Series pair if WP2 uses two motors

• Signal: Y | Driver socket: board Y | Motor and mechanism: Motor that is already on the 550 mm beam

- Signal: Z | Driver socket: board Z | Motor and mechanism: Former Z motor, head screw
- Signal: A (extruder E0 driver) | Driver socket: existing E0 | Motor and mechanism: Nozzle rotation C
- Signal: E1 | Driver socket: left unplugged | Motor and mechanism: Do not assign a feeder motor in this revision

Rewire by the machine axis, not by the old printer label. The motor on the 550 mm beam moves to the Y socket. The bed motor moves to the X socket.

Steps per mm, starting values to correct by measurement in WP9:

- X and Y, 20-tooth pulley, 2 mm pitch: 80 steps/mm at 16 microsteps and 200 full steps. Formula: (200 × microsteps) / (teeth × 2).
- Z, TR8×2, 16 microsteps: (200 × 16) / 2 = 1600 steps/mm.
- C, 1:1, 16 microsteps: (200 × 16) / 360 = 8.888… steps/degree. Prefer 8 microsteps if the driver allows a cleaner number, and record the value used.

Set motor currents for the load, not for the old heated-bed era. X and Y lose the bed and the extruder mass, so current comes down until the motor is warm, not hot, after 20 minutes of PR-2 moves. C is a small motor; set its current at the driver pot or digipot before the first powered move.

8.2 Outputs and inputs

Function | First-choice pin | Notes

- Function: Pick valve | First-choice pin: former hotend MOSFET | Notes: 12/24 V, flyback diode at the valve
- Function: Blow valve | First-choice pin: former fan MOSFET | Notes: same
- Function: Camera light | First-choice pin: former part-fan MOSFET | Notes: same
- Function: Bed heater | First-choice pin: capped, connector removed | Notes: FR-31
- Function: Vacuum switch | First-choice pin: a spare endstop or probe input | Notes: active low, internal pull-up
- Function: X, Y, Z home | First-choice pin: existing endstops | Notes: NC switches preferred so a broken wire homes into the stop instead of crashing

Pump power is a manual switch on the supply, fused. The pump is not on a firmware MOSFET, so a locked-up board cannot leave a valve clicking with the pump off and cannot cycle the pump every pick. Valves do the per-pick work.

8.3 Firmware behavior

- Marlin (or the firmware already on the board) with four axes, DISTINCTEAXES unused, A axis enabled as a rotational axis if the firmware supports degrees. If it only supports linear E, command C in degrees by setting steps per mm equal to steps per degree and sending millimeters that mean degrees. Document the convention in the OpenPnP driver.
- Disable PREHEAT, bed PID, and thermal protection for the removed heaters so the board boots with no thermistor attached.
- Accelerations: start at 1500 mm/s², raise toward PR-2 only after the belts stay tracked.
- Jerk or junction deviation low enough that a full tape does not walk in its holder. Start at 8 mm/s junction speed.
- Add two M-codes or fan numbers for the valves, and use them from OpenPnP actuator G-code. Example pattern, final pins from WP0: pick on M42 P<pin> S255, pick off M42 P<pin> S0, blow as a short script M42 … S255 / G4 P10 / M42 … S0.
- Vacuum check: either a firmware macro that sets a pin OpenPnP can read, or OpenPnP reads the endstop state over M119. Choose M119 unless WP0 shows the board cannot report that pin.

8.4 Host

- PC next to the machine, USB to the board, USB to both cameras.
- OpenPnP current stable release.
- 5 V camera current comes from the PC, not from the motion board.

9. Software configuration

OpenPnP objects to create, in this order, after the axes move correctly by hand-jog:

1. Driver: G-code driver to the board's serial port. Motion command G1 X Y Z A F. Confirm the letter used for C.
2. Head: one head, nozzle on it, vacuum actuator, blow actuator.
3. Nozzle tip library: the six tips, with diameter and the pick/place Z offsets measured in WP6.
4. Cameras: down-looking units-per-pixel from a calibration square; up-looking the same way with a known pin.
5. Nozzle-to-camera offset: commanded move that puts a fiducial under the nozzle, then under the camera. The difference is the offset.
6. Axis C on the nozzle, with the unlock angle used by the magazine.
7. ReferenceNozzleTipChanger or the equivalent actuator, one drop location and one pick location per pocket.
8. Feeders: one ReferencePushPullFeeder or script feeder per lane. Vision calibration enabled.
9. Bottom vision pipeline on the up-looking camera, one pipeline per package outline (rectangle for chips, rectangle with leads for SOT and SOIC).
10. A test job: four fiducials optional, a 10 × 5 grid of 0603 or 0805, four feeders.

Homing script order: Z up first, then X and Y. A job start homes Z, X, Y, then moves to the up-looking camera and checks that the light works.

10. Safety

• Disconnect the bed heater at the PSU and remove the connector before any other rewire. A placer job must have no path that can heat the table.

• E-stop cuts motor power and both valve coils. The pump switch is next to it. After an E-stop the nozzle stays where it is; the operator lifts Z by hand if a part is under it. Do not home until the carriage is clear.

• Belt pinch points on both X loops and on the Y belt get a printed cover.

• Vacuum pump exhaust points away from the operator.

• Nozzle tips are small and sharp enough to pierce skin if a hand is under the head during a jog. Jog Z downward only from the up-looking camera view or from a measured height.

• Laser or super-bright LED ring: choose a diffuse LED. Do not use a focusable laser module as the camera light.

• The frame ground stays bonded to the PSU earth.

11. Acceptance

The machine is accepted when every row below is signed in the build record.

Test | Setup | Pass

• Test: T1 Homes | Setup: Cold start | Pass: X, Y, Z home without contact other than the switches. Repeat 10 times, PR-12

• Test: T2 Square | Setup: Dial indicator or camera on a straight edge | Pass: X motion changes Y by under 0.05 mm over 200 mm. Y motion changes X by under 0.05 mm over 150 mm

• Test: T3 Speed | Setup: OpenPnP jog script | Pass: PR-1 and PR-2 held for 5 minutes, belts stay on pulleys, carriage stays under 500 g on a kitchen scale check done once in WP4

• Test: T4 Vacuum | Setup: 20 picks of 0805 from lane 1 into a cup | Pass: PR-5 and PR-11

• Test: T5 Changer | Setup: 10 cycles through all 6 tips | Pass: PR-6, every tip seated, vacuum leak check passes on the open tip

• Test: T6 Fiducials | Setup: One fiducial, 20 finds | Pass: PR-7

• Test: T7 Up-camera | Setup: 20 parts of one 0805 reel, reported angle | Pass: Standard deviation under 0.5° on a square part after pipeline tuning

• Test: T8 Place | Setup: 50 parts, at least 0603 and SOT-23, on a scrap PCB with paste or on tacky flux | Pass: PR-8, checked under a microscope or by the down-looking camera against pad centers

• Test: T9 Rate | Setup: The T8 job timed | Pass: PR-10 on a longer run of 100 parts, or the T8 time scaled only if no faults occurred

• Test: T10 Power-off heaters | Setup: Thermistor channels and bed MOSFET inspected | Pass: No heater wired, board boots without thermal error

A failed row returns to the work package named in the fault note. Acceptance is the set of rows, not a single demo video.

12. Implementation plan

Do the work packages in order. WP4 and WP5 can be built on the bench in parallel with WP2 and WP3, and mounted only after the bridge homes. Do not skip WP0.

Each package lists the actions in the order to perform them.

WP0 — Survey the machine you have

Objective: a one-page baseline so later steps use measured numbers.

1. Photograph the frame from the front, the X carriage, the electronics box with the cover off, and the PSU label.
2. Record supply voltage, board name and revision, driver chips, and which connectors are occupied. This replaces the MKS assumption everywhere it differs.
3. Measure both C-beams, the 550 mm beam, the 500 mm tie, and the 210 mm legs. Write down the gap between the two C-beams while they are still the Z columns. That gap is the bridge span to reproduce.
4. Measure the leadscrew lead (nut travel for one motor turn) and bag the screws. Record how many Z motors and bed motors actually turn.
5. Weigh the current X carriage with a scale. The difference from 500 g is the mass budget for the new head.
6. List working motors (wire count, NEMA size) and working endstops.
7. Decide and write down: pump voltage, valve voltage, and whether the drag chain already reaches a moving gantry or only a moving bed. A bed chain must be re-terminated on the bridge.
8. Quarantine the hotend, extruder, bed power wires, and thermistors in a labeled bag. They are out of the build.

Done when: a single note contains voltage, board identity, travels, carriage mass, and the connector map, and the heater wires are off the machine.

WP1 — Reorient the frame and fix the table

Objective: both C-beams horizontal, the 550 mm beam spanning them, a trammed plate, no bed motion.

1. Remove the bed belt and the bed heater leads (already isolated in WP0). Keep the bed motor for the X jackshaft.
2. Remove both leadscrews, Delrin nuts, and couplers. Confirm they match the bag from WP0.
3. Unbolt the two C-beams and lay them horizontal at the gap recorded in WP0. Bolt the 550 mm beam across them with its existing end plates, still without the new rails.
4. Fit the 20×80×500 tie and a second tie from the bed extrusion. Fit the 210 mm profiles as legs. Equalize the diagonals within 0.5 mm.
5. Strip or replace the bed plate per section 7.8. Bolt it to the ties on three adjustable screws, inside the rectangle.

6. Put a dial indicator on a temporary arm on the 550 mm beam. Sweep along both beams. Shim the plate screws and the ties until the central 200 × 150 mm is within 0.05 mm.
7. Mark the PCB datum corner. Drill and ream two pin holes and one clamp slot. Fit the pins and the clamp from section 7.8.
8. Confirm a bare 100 × 80 mm PCB sits without rock. If it rocks, spot the plate. Do not shim the PCB with tape.

Done when: the C-beams do not shift when the bridge is pushed by hand, a PCB seats against the pins, and the indicator sweep meets 0.05 mm.

WP2 — X belts on the two C-beams

Objective: the bridge travels along the C-beams on rails, both belts in phase.

1. Mount both MGN12H rails as in section 7.1. Move the 550 mm beam from its temporary end plates onto the rail blocks.
2. Tram again. Twist shows up as a height change along the beam at two X positions. Shim the bridge plates.
3. Fit the jackshaft, two 20-tooth pulleys, two idlers, and two belts. Anchor each belt on its own bridge plate. Mark one tooth on each pulley that meets the belt at the same shaft angle.
4. Fit the X home switch at the left end. Confirm continuity unpowered, then M119.
5. Plug the bed motor into the X driver. Set steps/mm from section 8.1. Command a 100 mm move against a steel rule. Correct steps/mm by current × (commanded / measured).
6. Read yaw at the far bridge plate over 200 mm. Above 0.05 mm, re-phase the belts before raising speed.
7. Raise acceleration only after 100 mm moves are within 0.1 mm. Stop at 3000 mm/s² or at the first skipped step, whichever is lower, and record it.
8. Fit both belt covers.

Done when: X homes, a 200 mm move repeats within 0.05 mm, yaw stays within 0.05 mm, and the beam height stays inside 0.05 mm.

WP3 — Y carriage on the 550 mm beam

Objective: the light carriage on a rail along the bridge, belt driven, ready for the head.

1. Remove V-wheels and the extruder stack from the carriage on the 550 mm beam.
2. Mount the MGN12H rail on the beam. Mount the mother plate on two blocks. The nozzle hole is the plate centerline.
3. Refit the belt and the motor that belongs on this beam. Set tension as in section 7.2. Plug that motor into the Y driver.
4. Fit the Y home switch at the front end. Confirm with M119.
5. Calibrate Y steps/mm the same way as X.
6. Run a 150 mm move at 100 mm/s, then 200, then 400. At each speed, listen for pulley skip and watch the belt walk off the pulley flange. Flange contact means the pulley is skewed. Shim the motor plate before raising speed.
7. Install the drag chain from the frame to the mother plate. Leave a loop at full X and full Y. Nothing in the chain may go tight at the corners of travel.

Done when: X and Y home and interpolate a rectangle (command a 150 × 100 mm box) whose diagonal closes within 0.1 mm at the indicator, at 400 mm/s.

WP4 — Head, Z, and rotation

Objective: coaxial Z and C on the carriage, within the mass budget.

Build this on the bench before it goes on the rail.

1. Assemble the Z rail, screw, anti-backlash nut, and NEMA 17. One motor revolution shall move the carriage 2.0 mm ±0.05 mm on a rule.
2. Couple the nozzle shaft through two bearings on the Z carriage. Spin the shaft by hand. A dial indicator on the future tip shall read under 0.03 mm total runout. If it reads more, the bearings are not coaxial; fix the printed bores or switch that part to aluminum.
3. Add the compliance spring and the commercial nozzle holder. Fit one tip. Press the tip on a scale: it shall start moving at about 1 N and bottom at 1.0–1.5 mm.
4. Mount the C motor and the 1:1 belt. Command 90°, 180°, 360° with the driver on the bench. The tip angle shall match a protractor within 1° before OpenPnP backlash compensation, and the indicator runout shall stay under 0.03 mm while it turns (FR-14).
5. Fit the Z home switch at the top.
6. Weigh the finished head. If it is over 500 g, move the C motor up, shorten the bracket, or change NEMA 17 Z for NEMA 14 before installing it. Do not accept the overrun.
7. Bolt the head to the mother plate. The nozzle axis is vertical when the carriage sits on the rail: check with a small square against the table.
8. Connect Z as the board's Z driver and C as E0. Home Z upward only, by hand-jogging the first time with a finger on the power switch.
9. Set Z = 0 at the PCB surface using the convention in section 7.3. A feeler of 0.1 mm between tip and PCB, with the compliance just starting to move, is the place height for a 0 mm part. Record it.
10. Set acceleration on Z low (500 mm/s²) until the screw stays quiet, then raise it toward PR-3.

Done when: the nozzle touches a PCB, lifts 20 mm, rotates 90°, and returns to the same touch point within 0.05 mm, and the head mass is recorded under 500 g.

WP5 — Vacuum tweezer

Objective: pick and release with a sensor check.

1. Mount the pump, filter, pick valve, and blow valve on the rear frame. Wire the pump through its own fused switch.
2. Plumb 4 mm PU tube: pump, filter, pick valve, tee, blow valve, carriage loop, swivel, nozzle. Leak-check with soapy water or by pinching the nozzle: the sensor shall reach its trip point.
3. Wire pick and blow to the MOSFET outputs from WP0, with a diode across each coil.
4. Wire the vacuum switch to the spare input. M119 shall change when the nozzle is sealed with a finger.
5. Tune the blow pulse. Start at 10 ms. An 0805 shall leave the tip and fall, and shall not be blown sideways off a pad. If it sticks, add 5 ms. If it scatters, subtract 5 ms or lower the puff pressure.
6. In OpenPnP, write the pick actuator (valve on), the place actuator (blow script), and the pick-check that reads M119.
7. Pick 20 parts from a tape held in a vise, drop them in a cup. A missed sensor trip fails the test even if the part is on the nozzle.

Done when: T4 passes on the bench, and the blow time is written in the build record.

WP6 — Tool changer

Objective: six tips, exchanged by motion only.

1. Buy the tip family before cutting the magazine. The pocket is a copy of the holder's release geometry.
2. Machine or print the six-pocket bar. Bolt it to the table. Indicate the pocket centers: they shall lie on a line parallel to X within 0.1 mm, and at one height within 0.05 mm.
3. Jog the nozzle, with tip 1 loaded, into pocket 1. Lower Z until the clutch is at the release height. Record X, Y, Z.
4. Command the C unlock angle. Lift Z. The tip shall stay in the pocket.
5. Move to pocket 2, lower, command the lock angle, lift. Pull upward gently on the head: the tip shall come along. A tip that stays in the pocket is under-rotated or high.
6. Repeat for all pockets. Store the six coordinate triples in OpenPnP.
7. Add the seat check: after a pick-up, open the pick valve for 50 ms with the tip uncovered. A seated tip seals enough to trip the sensor or, on an open orifice, shows a stable vacuum different from a missing tip. Record both sensor states and use the missing-tip state as the failure.
8. Run 10 full exchanges.

Done when: T5 passes.

WP7 — Positioning camera and up-looking camera

Objective: fiducials and part angle in machine units.

1. Mount the down-looking camera on the mother plate. Set the lens to a 15–25 mm field at the PCB. Lock the lens.
2. Wire the light to its MOSFET. M42 shall turn it on and off.
3. In OpenPnP, set the camera resolution and run the units-per-pixel calibration on a scale or a known grid.
4. Find a fiducial with the camera. Jog the nozzle onto that fiducial. The jog delta is the nozzle-to-camera offset. Enter it and repeat from the other side of the board. The two results shall agree within 0.05 mm; average them.
5. Mount the up-looking camera on the fly path. Build the silhouette light.
6. Hover a known 0805 at the hover Z. Tune the bottom-vision pipeline until the reported center is stable to one pixel and the angle is stable to 0.5°.
7. Run a loop: pick, fly over the up camera, place on a marked cross, 20 times. Measure the scatter. This is the first look at PR-8, before the formal acceptance.

Done when: T6 and T7 pass.

WP8 — Tape holders

Objective: eight lanes the head can index.

1. Print or machine one lane. Check sprocket pitch with a 4.00 mm pin gauge or a stack of feeler checks: eight pitches shall be 32.0 mm ±0.2 mm.
2. Load a scrap 8 mm tape. Press the lever with the nozzle at the design stroke. Exactly one pocket shall advance. If it advances two, shorten the lever arm. If it advances none, lengthen the stroke or the arm.
3. Cycle 50 presses. The pick pocket shall not wander more than half a pocket. Ratchet backlash is the usual cause; add a drag spring on the wheel.
4. Duplicate the module to eight lanes. Bolt them to the front rail, numbers 1–8 left to right.
5. Set each lever XY and press Z, and each pick XY and pick Z, in OpenPnP.
6. Run the camera feeder calibration on each lane.
7. Pick 10 parts from each lane into a cup.

Done when: every lane feeds one pocket per press and 10 picks succeed, and the operator card from section 7.7 is at the machine.

WP9 — Firmware and OpenPnP integration

Objective: one homing script and one job file that runs the whole cycle.

1. Apply the firmware changes from section 8.3 on a copy of the current configuration. Keep the previous binary.
2. Remove every heater from the configuration. Boot and confirm T10: no thermal error, bed connector absent.
3. Enter the calibrated steps/mm, speeds, and accelerations from WP2, WP3, and WP4.
4. Home order in the start G-code: raise Z, home X, home Y, home Z to the switch, then apply the PCB Z offset.
5. Build the OpenPnP objects from section 9.
6. Dry-run a job with vacuum off and Z raised by 5 mm (a "Z probe" or a global placement offset). The nozzle shall trace picks, the up-camera hover, and places without touching tape or PCB.
7. Remove the 5 mm offset. Run 10 real parts.
8. Back up firmware configuration and the OpenPnP profile to the project folder.

Done when: a cold start, one button, places 10 parts, and both backups exist.

WP10 — Acceptance

Objective: sign section 11.

1. Run T1 through T10 in order.
2. For T8, use a scrap PCB with flux or paste, 50 parts, at least two package sizes.
3. Record speed, acceleration, blow time, steps/mm, nozzle-to-camera offset, and head mass in the build record next to the test results.
4. Any failed row goes back to the WP named in the note. Re-run that row and the rows after it.

Done when: the acceptance table is filled and PR-8 and PR-10 are met. PR-9 is recorded and does not gate acceptance.

13. Bill of materials, added parts

Quantities are for one machine. Stock motors, board, PSU, endstops, and frame hardware are not listed.

Item | Qty | Notes

- Item: MGN12H rail and 2 blocks | Qty: 2 sets | Notes: X, one on each C-beam, length to the measured 574 mm beam
- Item: MGN12H rail and 2 blocks | Qty: 1 set | Notes: Y, on the 550 mm beam
- Item: MGN9 or MGN12 short rail and block | Qty: 1 | Notes: Head Z
- Item: GT2 belt, 9 mm, 2 mm pitch | Qty: 2 loops plus spare | Notes: One loop per C-beam. The existing X belt stays on the Y beam
- Item: GT2 pulley, 20 tooth, 5 mm bore | Qty: 4 | Notes: Two on the X jackshaft, two idlers. The Y pulley is the one already on the 550 mm beam
- Item: Jackshaft, bearings, couplers | Qty: 1 | Notes: Spans the two C-beam ends. Dropped only if WP2 uses two series motors
- Item: NEMA 17 | Qty: 0 extra | Notes: Bed motor drives X. Beam motor drives Y. Z motor drives the head
- Item: NEMA 11 or 14 | Qty: 1 | Notes: Axis C
- Item: TR8×2 screw, nut, and anti-backlash nut | Qty: 1 | Notes: Head Z, or a 2 mm ball screw
- Item: Juki-compatible holder and 6 tips | Qty: 1 set | Notes: Section 7.4
- Item: Diaphragm vacuum pump | Qty: 1 | Notes: Matched to PSU voltage
- Item: Solenoid valve, NC | Qty: 2 | Notes: Pick and blow
- Item: Vacuum switch | Qty: 1 | Notes: About −40 kPa
- Item: PU tube 4 mm and filter | Qty: 1 set | Notes:
- Item: USB global-shutter camera and M12 lens | Qty: 2 | Notes: Down-looking and up-looking
- Item: LED ring or bar, 12/24 V | Qty: 2 | Notes: With series resistor or the driver it ships with
- Item: Aluminum tooling plate | Qty: 1 | Notes: If the old bed fails the flatness check
- Item: 3 mm datum pins, clamps | Qty: 1 set | Notes:
- Item: Tape-holder hardware | Qty: 8 lanes | Notes: Sprocket, ratchet, lever, peel finger
- Item: Fasteners, shims, thread locker | Qty: as needed | Notes: M3 and M5, the frame's existing sizes

14. Risks and the response built into this plan

Risk | What the plan does

- Risk: V-wheel backlash is too large for 0603 | What the plan does: Rails are part of the specified machine, installed in WP2 and WP3, before vision tuning
- Risk: The Z leadscrew is reused as the fast X drive | What the plan does: WP1 removes both screws. X is a dual GT2 belt. A screw on X fails PR-1
- Risk: The bridge yaws because the two C-beam belts differ | What the plan does: One jackshaft and matched tooth phase. WP2 stops on a 0.05 mm yaw reading
- Risk: The board has only five drivers | What the plan does: Two X motors, if used, share the X driver in series. C uses E0. Feeders are head-actuated
- Risk: Belt stretch under acceleration | What the plan does: PR-2 is a ceiling reached by test, not a starting value. WP2 and WP3 stop at the first skipped step
- Risk: Tip family does not match the magazine | What the plan does: WP6 step 1 buys the tips before the pockets are made

- Risk: Camera mm/pixel too coarse | What the plan does: Lens field of view is sized in section 7.6 and checked in WP7 before placement tests
- Risk: Heater output left live | What the plan does: WP0 disconnects it, WP9 removes it from firmware, T10 confirms it

15. Build record

Keep one page with the machine and fill it during the work packages.

- Board name, voltage, firmware binary name.
- Measured C-beam length, beam length, rail gap, X travel, and Y travel.
- Head mass.
- Steps/mm for X, Y, Z, and steps/degree for C.
- Accelerations actually used.
- Blow pulse width.
- Nozzle-to-camera offset.
- Pocket coordinates 1–6.
- Lever and pick coordinates for lanes 1–8.
- Acceptance table from section 11, with dates.

End of specification TEVO-BW-PNP-TS-001.

Current machine build and assembly guide

Исходный файл: pnp/BUILD_AND_ASSEMBLY.md

TEVO-BW PnP: build and assembly guide

This guide connects the parametric CAD to a physical first machine. It describes the implementation represented by params.py and the generated out/report.md, including the dual-belt X drive, 8-lane motorized feeder, light Z slide and one-piece tooling plate. It is a shop build plan, not a claim that the machine has passed physical acceptance.

1. What is buildable now

The repository has source models and exports for the frame conversion, bridge/X-Y motion, head and tooling layout, feeder bank, printed and machined custom parts, and a generated BOM/report. The feeder has unusually complete separate assembly, electronics and first-article test documentation. pnp/out/report.md reports a successful modeled interference check at selected poses and current model estimates.

The build is not yet production-ready from documentation alone. The original technical specification still contains earlier choices (head-actuated feeders, stock NEMA17 Z, different carriage/table concepts); current model notes D1-D14 supersede several of them. Physical measurements, part selections and first-article test evidence are still needed. Use the model report as an estimate and the actual printer as the authority for dimensions.

2. CAD generation and deliverables

Use FreeCAD 1.1's bundled Python. On Windows, set the executable directory once per PowerShell session if needed:

```
$env:FREECAD_BIN = "$env:LOCALAPPDATA\Programs\FreeCAD 1.1\bin"
cd D:\TEVO_BW\pnp
.\pnp.ps1 info
.\pnp.ps1 build
.\pnp.ps1 check
.\pnp.ps1 render
```

The wrapper invokes python.exe -I pnp.py; FREECADBIN must be the bin folder, not the executable. If the installed FreeCAD uses another folder name, set the variable to that location. Outputs are under pnp/out/: pnpmachine.FCStd, pnpmachine.step, partsstep/, parts_stl/, bom.csv, report.md, and OpenPnP coordinate JSON. render writes review images. check covers modeled geometry and selected poses only. Review the exported STEP and BOM before release to a shop; do not hand-edit generated outputs as source.

Feeder source/build is separate:

```
cd D:\TEVO_BW\pnp\feeder
.\feeder.ps1 build
.\feeder.ps1 audit
```

It produces the lane and bank outputs under pnp/feeder/out/. For a slicer project, run python orca_project.py after build; this may need an OrcaSlicer project template as described in feeder/README.md.

3. Mechanical layout

```
flowchart TB
  subgraph FRAME[Fixed frame and work surface]
    X1[Left 40x80 C-beam: X rail + belt]
    X2[Right 40x80 C-beam: X rail + belt]
    T[Cross ties and corner blocks]
    PL[One-piece tooling plate]
    PCB[PCB fixture]
    BANK[8 feeder lanes + reel rack + waste handling]
    CAM[Fixed up-looking camera]
    X1 --- T --- X2
    T --- PL
    PL --- PCB
    PL --- BANK
    PL --- CAM
  end
  subgraph MOVING[Moving mechanism]
    DRIVE[Bed motor -> jackshaft -> matched left/right X belts]
    BR[550 mm bridge moves in X]
    Y[Bridge MGN12 + stock belt moves head in Y]
    HEAD[Head carriage + Z + C nozzle + down camera]
    DRIVE --> BR --> Y --> HEAD
```

```
  end
  X1 --- BR
  X2 --- BR
```

Coordinates and clearances are in params.py; the machine is modeled from the frame origin, while generated OpenPnP coordinates use the nozzle-at-home origin. Do not mix those coordinate frames when locating parts or entering camera offsets.

Axis and control paths

```
flowchart LR
  subgraph MOTION[Motion system: existing printer controller, verify board first]
    GX[X motor driver] --> JS[Motor + jackshaft]
    JS --> BL[Left X belt]
    JS --> BR[Right X belt]
    GY[Y motor driver] --> YB[Bridge belt]
    GZ[Z driver, 1/4 microstep target] --> ZM[NEMA8 linear motor + T5x2]
    GC[C / A driver] --> CM[NEMA8 hollow-shaft C motor]
  end
  PC[OpenPnP host] -->|G-code over USB| MC[Motion controller]
  MC --> GX
  MC --> GY
  MC --> GZ
  MC --> GC
  PC -->|USB cameras| V[Down camera + fixed up camera]
  PC -->|separate USB serial| FN[Feeder Nano]
  FN --> S1[Servo lane 1]
  FN --> S8[Servo lane 8]
  PSU[Separate regulated 5 V servo supply] --> FN
  PSU --> S1
  PSU --> S8
```

The existing printer controller and the feeder Nano are separate devices. The machine controller handles motion and the feeder controller advances tape; do not connect servo power to the Nano's USB 5 V rail. The installed board, driver outputs, voltage and pin mapping must be documented during the initial survey before wiring.

Feeder force and tape path

```
flowchart LR
  SERVO[MG90S servo] --> GEAR[13T gear sector]
  GEAR --> SLIDER[4.2 mm slider stroke]
  SLIDER --> PAWL[Spring-loaded feed pawl]
  PAWL --> TAPE[Carrier tape advances 2 or 4 mm]
  TAPE --> DETENT[Spring-loaded detent holds position]
  SLIDER --> RACK[Rack + 36T idler + 14T pinion]
  RACK --> DRIVE[HF0612 drive roller peels cover tape]
  DRIVE --> NIP[Spring-loaded nip limits tension]
  NIP --> BIN[Cover tape waste bin]
  TAPE --> EXIT[Empty carrier tape chute]
```

This is a functional path, not an exploded drawing. The lane's printable part orientation, hardware, assembly order, bearing direction and adjustment procedure are specified in feeder/ASSEMBLY.md (feeder/ASSEMBLY.md).

4. Physical assembly sequence

Follow the specification's WP0-WP10 as a useful staged checklist, but apply the revision mapping below before following individual steps. The sequence below reflects the current modeled architecture.

1. Survey and baseline. Photograph and label the installed board, drivers, PSU voltage, connectors, motors and endstops. Measure the actual C-beam gap, extrusion lengths, tie positions, plate flatness and usable travel. Record replacements for the WP0 assumptions in params.py; preserve all settings and wiring before disassembly.
2. Rebuild the base. Isolate mains and PSU power. Remove the bed/heater and Z screw drive. Lay the two C-beams parallel as X rails, install ties and four corner/drive blocks, level the frame, then mount the tooling plate and standoffs. Verify diagonal equality and plate flatness before attaching motion hardware.
3. Install X motion. Align the two MGN12 rails in a common plane, fit bridge blocks/ends and the 550 mm bridge. Mount both GT2 belts, jackshaft, motor and tensioners. Phase the belts together and measure bridge yaw at both ends of travel before tightening.
4. Install Y motion. Fit the MGN12 rail and belt to the bridge, carriage and motor/idler hardware. Route the chain and confirm the full carriage path clears the frame, plate, feeders and camera.
5. Build the head on the bench. Assemble the printed carriage, MGN9 Z guide, NEMA8 linear motor/T5x2 slide, C rotation motor, nozzle shaft/compliance, camera and endstop. Check free Z travel and nozzle runout before mounting. Keep the moving head below the modeled 500 g target; weigh the real assembly.

6. Mount tooling and optics. Install the PCB fixture/clamp, up camera, nozzle magazine, feeder bank and tape/waste routing. Align all mounting faces to the plate datum. Validate camera/nozzle offsets and each target's reach from actual home positions.

7. Wire and configure motion. Verify the actual controller pinout and voltage before connecting. Configure X/Y/Z/C steps, directions, limits, microstepping and safe homing for the installed board. Current modeled Z uses 1/4 microstepping and 400 steps/mm; do not copy legacy 1/16 settings. Ensure heater outputs are disabled and repurposed safely before connecting valves or other loads.

8. Feeder first article. Build one v3.0 lane exactly per pnp/feeder/ASSEMBLY.md, wire it per ELECTRONICS.md, and run TEST.md. Confirm 2/4 mm feed, cover-tape take-up, clutch direction, reliable pocket registration and servo temperature before replicating seven lanes.

9. Build the power/control panel. Use pnp/electronics/POWERCONTROLDESIGN.md, the wiring diagram and enclosure render. First identify the donor board/drivers and every load; do not wire from assumed 12/24 V. Keep the motion board, vacuum, feeder servo rail and SBC on separate protected branches. Test E-stop hardware before enabling motion.

10. Commission slowly. Start with motor power limited and conservative acceleration. Home each axis independently, check travel limits, belt retention, cable chains and emergency stop. Validate vacuum, blow-off, camera calibration, nozzle changes, feeder positions and OpenPnP safe Z before any automatic cycle.

11. Acceptance and release. Record physical measurements and results against specification section 11 / WP10. Do not call cycle-rate, placement accuracy or unattended operation validated until those tests have been run on hardware.

5. Head and feeder subassembly schemes

```
flowchart TB
  YBLOCK[Two Y MGN12 blocks] --> CARR[Printed head carriage]
  CARR --> ZRAIL[MGN9 rail + Z block]
  CARR --> ZM[NEMA8 linear motor / T5x2]
  ZRAIL --> ZSLIDE[Machined Z slide]
  ZM --> ZSLIDE
  ZSLIDE --> CM[Nozzle C motor]
  CM --> ROTOR[Hollow shaft + compliance + nozzle]
  CARR --> DOWN[Down camera + LED]
  PCB[Fixed PCB fixture] --> PLACE[Pick / place work zone]
  LANE[Servo lane] --> POCKET[Indexed tape pocket]
  POCKET --> ROTOR
  ROTOR --> UPCAM[Fixed up camera]
  UPCAM --> PLACE
  MAG[Passive nozzle magazine] --> ROTOR
```

The feeder's detailed part order, fastener sizes, wiring, firmware and calibration already live in pnp/feeder/ASSEMBLY.md, ELECTRONICS.md, OPENPNP.md and TEST.md. Use those as the controlling lane-level procedures; this overview does not replace them.

6. Power/control panel

The preliminary electrical design, block-level schematic, enclosure concept render, bill of materials and bring-up gates are in electronics/POWERCONTROLDESIGN.md (electronics/POWERCONTROLDESIGN.md). The proposed SBC is a Raspberry Pi 5 with a separate official supply; verify OpenPnP and both cameras on ARM64 before committing to it. The feeder Nano stays a separate USB serial device, with servo current supplied by the dedicated 5 V branch. The enclosure itself is not yet part of the FreeCAD machine assembly; fit, ventilation, electrical protection and actual donor hardware remain open checks.

7. Revision map and known design changes

Topic | Earlier technical specification | Current modeled direction

- Topic: Tape advance | Earlier technical specification: Nozzle presses passive lever lanes | Current modeled direction: Eight independent servo-driven TBW-Feeder v3.0 lanes, separate Nano/controller
- Topic: Z | Earlier technical specification: Former NEMA17 and TR8x2 | Current modeled direction: NEMA8 linear motor and T5x2; lightweight carriage
- Topic: X bridge | Earlier technical specification: Dual belts/jackshaft architecture was proposed | Current modeled direction: Implemented in current model with two MGN12 X rails and dual GT2 belts
- Topic: Tooling plate | Earlier technical specification: Reuse bed only if flat, otherwise replace | Current modeled direction: One-piece 6 mm plate carries PCB and feeder/tooling strip
- Topic: Head | Earlier technical specification: Modular plate concept | Current modeled direction: Integrated printed head carriage plus machined Z slide
- Topic: Z microsteps | Earlier technical specification: Legacy assumptions vary | Current modeled direction: Current parameters/report use 1/4 microstep, 400 steps/mm

Before fabrication, reconcile the specification's sections 2, 3, 7, 8, 11 and WP1-WP10 against params.py, current model, report and measured machine. The model report's own design notes D1-D14 are the current decision record but are not a substitute for a revised, controlled specification.

7. Changes that would make the project easier to build

Prioritize these updates before a complete machine build:

1. Publish a rev 2 specification. Move D1-D14 into a dated change log, replace obsolete head-actuated feeder instructions and update BOM/electrical assumptions. Identify the controller/motor/PSU variant explicitly.

2. Add fit-critical drawings. For every machined item, publish dimensioned PDF/DXF drawings with datum scheme, tolerances, stock size, finish and inspection dimensions. STEP alone is not a machining drawing. Add rail alignment and tooling-plate drilling templates.

3. Make the BOM procurement-ready. Add manufacturer/part number, quantity, source, voltage/rating, substitutions and "verify before purchase" status. Separate reused printer parts, bought parts, printed parts and machined parts; resolve any generated BOM omissions before ordering.

4. Create a wiring pack. Provide one board-specific pin map, connector view, power distribution diagram, fuse/wire sizing and valve/servo isolation details. Current documentation spans two control systems: the original motion board and the feeder Nano.

5. Add a whole-machine commissioning checklist. Record homing direction, endstop logic, travel limits, steps/mm, belt tension, squareness, Z safe height, vacuum response and camera calibration with pass/fail evidence.

6. Capture physical validation. Weigh the head, measure runout/flatness/reach, test belt yaw and repeated placement, and update model assumptions from those measurements. Distinguish modeled checks from hardware-verified checks in reports.

7. Improve release packaging. Add a versioned release folder or archive containing source revision, FreeCAD version, generated STEP/STL/BOM/report, slicer profile and firmware revision. Keep generated outputs traceable to the source revision.

Current manufacturing split

Use the generated BOM as the detailed quantity list. The model report currently identifies 150 printed part instances and 9 machined custom items, plus 411 fasteners. These are CAD-derived counts and need a purchase review against the actual donor printer before ordering.

Work type | Examples | Release input

• Work type: Reuse and measure | Examples: C-beams, bridge extrusion, motors, controller, PSU, belts and selected hardware | Release input: WP0 survey sheet and actual dimensions

• Work type: Print | Examples: Corner blocks, bridge ends, head carriage, feeder lanes, reel rack and camera housing | Release input: `parts_stl/`, slicer profile, material and orientation table in `out/report.md`

• Work type: Machine | Examples: Tooling plate, Z slide, Y motor/idler brackets, nozzle magazine, camera angle and tension blocks | Release input: `parts_step/` plus dimensioned drawings to be added

• Work type: Buy | Examples: Linear rails/blocks, bearings, motors, screws, springs, optics, vacuum parts and feeder electronics | Release input: `bom.csv` with vendor and electrical ratings verified

• Work type: Wire/configure | Examples: Motion board, feeder Nano, cameras, vacuum valves, endstops and emergency stop | Release input: Board -specific wiring pack to be added

Build gates

Gate | Evidence required before continuing

• Gate: Frame accepted | Evidence required before continuing: Measured rail gap, diagonals, level, plate support and PCB-area flatness recorded

• Gate: Motion accepted | Evidence required before continuing: X/Y homes and limits correct; bridge yaw measured; no belt derailment or missed steps at the chosen profile

• Gate: Head accepted | Evidence required before continuing: Z stroke and homing correct; head mass weighed; C runout and nozzle repeatability measured

• Gate: Feeder accepted | Evidence required before continuing: One lane passes `pnp/feeder/TEST.md`, including feed, peel, tape registration and endurance checks selected for first article

• Gate: Integration accepted | Evidence required before continuing: OpenPnP coordinates generated from the as-built model; camera offsets taught; Safe Z physically verified

• Gate: Machine accepted | Evidence required before continuing: Specification WP10/section 11 test records complete with pass/fail and measured data

8. Practical limits

• The reported no-interference result applies to the modeled parts and sampled poses, not cable snagging, assembly tolerance stack-up or deflection under acceleration.

• PETG/PETG-CF printed load-bearing parts may creep; verify fastener retention, temperature and belt loads on hardware. The large corner blocks and bridge ends deserve inspection after initial motion tests.

• The feeder audit reports geometry/mechanism checks, while the feeder README still calls for a physical first article. Run that article before printing the full bank.

• Cycle time, torque margins, reach, mass and printability are model estimates dependent on actual motors, power supply, rail alignment, slicer and firmware settings.

• The stock CAD archive includes several printer revisions. Confirm every reused part against the actual machine before ordering or machining around it.

Parametric model and CAD workflow

Исходный файл: pnp/README.md

TEVO-BW pick-and-place: FreeCAD model (scripts + CLI)

A parametric FreeCAD model of the TEVO Black Widow pick-and-place conversion. The current implementation has changed from parts of the original specification; use BUILDANDASSEMBLY.md (BUILDANDASSEMBLY.md) and the generated out/report.md to understand the current architecture. Everything runs headless with the Python bundled with FreeCAD 1.1. You don't need the FreeCAD GUI.

```
.\pnp.ps1 build    # FCStd, STEP/STL parts, BOM, report and OpenPnP coordinate JSON in out/
.\pnp.ps1 check    # modeled interference checks at selected poses
.\pnp.ps1 render   # review PNGs in out/render/
.\pnp.ps1 all      # build + check + render
.\pnp.ps1 info     # derived dimensions and OpenPnP coordinates as JSON
.\pnp.ps1 gui      # open the model in the FreeCAD GUI
options: --pose X Y Z C   machine pose (mm from home, Z <= 0, deg)   e.g. --pose 200 150 -30 90
         --views iso,head --size 1600x1000 --ss 2 --zoom 1.0 --prefix name_
```

Use pnp.cmd in cmd or .\pnp.ps1 in PowerShell. The wrappers detect the common FreeCAD11 and FreeCAD 1.1 install folders. If FreeCAD is installed somewhere else, set FREECAD_BIN.

File | Contents

- File: `params.py` | Contents: All dimensions and choices (TS sections cited). Edit this file first.
- File: `model/frame.py` | Contents: Reoriented C-beam frame, ties and corner/drive blocks
- File: `model/xaxis.py` | Contents: Dual GT2 X belts, jackshaft, bridge ends and bridge beam
- File: `model/yaxis.py` | Contents: MGN12 Y rail, carriage and stock belt drive
- File: `model/head.py` | Contents: Lightweight carriage, MGN9 Z, NEMA8 linear motor/T5x2 slide, C nozzle rotation and down camera
- File: `model/tooling.py` | Contents: Tooling plate, PCB fixture, up camera, feeder bank and 6-pocket magazine
- File: `model/services.py` | Contents: Vacuum panel, E-stop and pump switch, X and Y drag chains (built for each pose)
- File: `model/assembly.py` | Contents: Motion groups (bridge / carriage / zslide / rotor), pose placements, FCStd writer
- File: `render.py` | Contents: Coin3D offscreen renderer to PNG
- File: `model/feeder.py` | Contents: TBW-Feeder v3.0 lane + bank geometry (exit chute, waste bin, reel rack)
- File: `feeder/` | Contents: Feeder build, audit, firmware and production documents, see `feeder/README.md`
- File: `electronics/` | Contents: Preliminary 24 V power distribution, SBC and enclosure concept, diagrams and BOM

Colours: pnp gui (or freecad.exe open_gui.py) applies them and saves them into the FCStd. Every object also has MoveGroup, Kind, MassGram and BOM properties.

See BUILDANDASSEMBLY.md (BUILDANDASSEMBLY.md) for the physical build sequence, assembly diagrams, changes from the original specification and prioritized project improvements. The CAD interference check does not replace physical fit, wiring or acceptance tests.

The combined PDF dossier is out/TEVOBWPnPDossier.pdf. See out/render/RENDERSTATUS.md for the current-vs-legacy feeder render manifest. Electrical architecture, detailed terminal wiring and software setup are documented in electronics/POWERCONTROLDESIGN.md, WIRINGDETAILS.md and PI5SOFTWARE.md.

Printed parts (rev E)

model/printing.py handles every kind="print" part: it is coloured orange, given a print orientation (UP / LANE_UP tables), horizontal holes over 5.2 mm get a 45 deg teardrop roof, and the STL in out/parts_stl/ is written already lying on the bed. report.md -> "Printing" lists orientation, size, bed contact, overhang and bridge area per part. No part needs supports.

Generated machine model report and design decisions

Исходный файл: pnp/out/report.md

TEVO-BW PnP - model report

Generated by pnp build from params.py. Spec: TEVO-BW-PNP-TS-001 rev 1.1. Date: 2026-10-24 17:36.

Envelope (TS 3, 7.1, 7.2)

Quantity | Value

- Quantity: Frame footprint (C-beams, without drive blocks) | Value: 574 x 560 mm
- Quantity: C-beam inner gap (measured in stock STEP) | Value: 480 mm
- Quantity: X travel (bridge on MGN12, 10 mm rail margin) | Value: 453.0 mm
- Quantity: Y travel (carriage on MGN12, 10 mm rail margin) | Value: 382.6 mm
- Quantity: Z stroke / usable (FR-11 >= 35) | Value: 40 / 35 mm
- Quantity: Z soft limit (tip 1.0 mm into compliance at PCB top) | Value: -37.4 mm
- Quantity: Bridge beam underside | Value: Z 153 mm (frame)
- Quantity: Table top / PCB top | Value: Z 114.0 / 115.6 mm (frame)
- Quantity: Nozzle tip at Z home | Value: Z 152.0 mm (frame)
- Quantity: Machine origin (nozzle home) in frame coords | Value: (114.0, 88.7, 152.0)

Steps per unit (TS 8.1)

Axis | Value

- Axis: X, Y (GT2, 20T, 1/16) | Value: 80.000 steps/mm
- Axis: Z (T5x2 lead 2, 1/4) | Value: 400.0 steps/mm
- Axis: C (1:1, 1/16) | Value: 8.8889 steps/deg

Carriage mass budget (FR-4, <= 500 g)

Part | Group | Mass g

- Part: Head_Carriage | Group: carriage | Mass g: 106.6
- Part: ZMotor_NEMA8_LIN30 | Group: carriage | Mass g: 60.0
- Part: CMotor_NEMA8_HOLLOW30 | Group: zslide | Mass g: 60.0
- Part: YBlock_A | Group: carriage | Mass g: 37.0
- Part: YBlock_B | Group: carriage | Mass g: 37.0
- Part: ZRail_MGN9 | Group: carriage | Mass g: 34.2
- Part: ZBlock_MGN9H | Group: zslide | Mass g: 26.0
- Part: Z_Slide | Group: zslide | Mass g: 25.1
- Part: Head_BeltClamp | Group: carriage | Mass g: 21.7
- Part: Camera_Angle | Group: carriage | Mass g: 18.8
- Part: ZScrew_T5x2 | Group: carriage | Mass g: 13.9
- Part: DownCam_Board | Group: carriage | Mass g: 10.0
- Part: Vacuum_Swivel | Group: zslide | Mass g: 10.0
- Part: DownCam_Lens_M12 | Group: carriage | Mass g: 8.0
- Part: DownCam_LEDRing | Group: carriage | Mass g: 6.0
- Part: ZNut_T5x2_Flange | Group: zslide | Mass g: 5.0
- Part: Hollow_Shaft | Group: rotor | Mass g: 4.7
- Part: DownCam_LensHolder | Group: carriage | Mass g: 4.0
- Part: Juki_Holder | Group: rotor | Mass g: 3.3
- Part: ZHome_Switch | Group: carriage | Mass g: 3.0
- Part: Compliance_Sleeve | Group: rotor | Mass g: 2.7
- Part: Nozzle_Tip_loaded | Group: rotor | Mass g: 1.5
- Part: **Total moving in Y** | Group: | Mass g: **499**
- Part: Total without the two MGN12H blocks | Group: | Mass g: 425

Status: PASS (Z motor NEMA8_LIN30). With the stock NEMA17 40 mm on Z the total is 719 g (+220 g). Purchased-part masses are catalogue typicals; printed parts use PETG 1.27 g/cm3 x 70% effective fill. WP4 step 6 weighs the real head.

Reach (machine coordinates)

Nozzle reach: X 114.0..567.0, Y 88.7..471.3 (frame).

Target | X, Y | Reachable

- Target: lane 1 pick | X, Y: 426.0, 62.5 | Reachable: yes
- Target: lane 2 pick | X, Y: 426.0, 84.5 | Reachable: yes
- Target: lane 3 pick | X, Y: 426.0, 106.5 | Reachable: yes

- Target: lane 4 pick | X, Y: 426.0, 128.6 | Reachable: yes
- Target: lane 5 pick | X, Y: 426.0, 150.6 | Reachable: yes
- Target: lane 6 pick | X, Y: 426.0, 172.6 | Reachable: yes
- Target: lane 7 pick | X, Y: 426.0, 194.6 | Reachable: yes
- Target: lane 8 pick | X, Y: 426.0, 216.6 | Reachable: yes
- Target: magazine pocket 1 | X, Y: 434.0, 288.3 | Reachable: yes
- Target: magazine pocket 2 | X, Y: 434.0, 306.3 | Reachable: yes
- Target: magazine pocket 3 | X, Y: 434.0, 324.3 | Reachable: yes
- Target: magazine pocket 4 | X, Y: 434.0, 342.3 | Reachable: yes
- Target: magazine pocket 5 | X, Y: 434.0, 360.3 | Reachable: yes
- Target: magazine pocket 6 | X, Y: 434.0, 378.3 | Reachable: yes
- Target: up camera | X, Y: 434.0, 249.3 | Reachable: yes
- Target: PCB front-left | X, Y: 136.0, 161.3 | Reachable: yes
- Target: PCB rear-right | X, Y: 296.0, 261.3 | Reachable: yes
- Target: PCB area 200x150 far corner | X, Y: 336.0, 311.3 | Reachable: yes
- Target: down camera over lane 1 pick | X, Y: 424.0, 5.5 | Reachable: yes
- Target: down camera over lane 8 pick | X, Y: 424.0, 159.6 | Reachable: yes

Z heights (machine Z, 0 = home switch)

Lowest tip (soft limit): Z -37.4. Lowest fixed head part at home: frame Z 146.0; tallest tooling the head passes over (X < 592, feeder towers excluded): frame Z 140.8; clearance 5.2 mm.

Operation | Z needed | Reachable with >= 1 mm margin

- Operation: place on PCB (top) | Z needed: -36.4 | Reachable with >= 1 mm margin: yes
- Operation: pick from tape pocket | Z needed: -22.6 | Reachable with >= 1 mm margin: yes
- Operation: magazine engage height | Z needed: -22.0 | Reachable with >= 1 mm margin: yes
- Operation: up-camera hover (4 mm above ring) | Z needed: -31.5 | Reachable with >= 1 mm margin: yes

Dynamics (PR-1..PR-4, PR-10)

Moving mass: X (bridge + head) 3.08 kg, Y (head) 0.50 kg. Profiles use catalogue kg/m (WP0 weighs the beam); motors are the stock NEMA17 40 mm, 24 V supply, required margin 2.0 on pull-out torque.

Check | Result | Pass

- Check: X drive: 3.08 kg at 3000 mm/s2, 400 mm/s (600 rpm) | Result: need 0.088 N m, pull-out 0.252 N m, margin 2.9 | Pass: yes
- Check: X step rate at 400 mm/s | Result: 32000 steps/s (board 40000) | Pass: yes
- Check: Y drive: 0.50 kg at 4000 mm/s2, 400 mm/s (600 rpm) | Result: need 0.042 N m, pull-out 0.252 N m, margin 6.1 | Pass: yes
- Check: Y step rate at 400 mm/s | Result: 32000 steps/s (board 40000) | Pass: yes
- Check: Z step rate at PR-3 60 mm/s (1/4, 400 steps/mm) | Result: 24000 steps/s, board limit gives 100 mm/s | Pass: yes
- Check: Z drive NEMA8_LIN30 travel (slide 0.14 kg, 1500 mm/s2): 1.6 N at 1800 rpm | Result: need 0.0014 N m, pull-out 0.0045 N m, margin 3.2 | Pass: yes
- Check: Z drive NEMA8_LIN30 contact (slide weight + 1.5 N compliance, slow): 2.9 N at 300 rpm | Result: need 0.0026 N m, pull-out 0.0144 N m, margin 5.5 | Pass: yes
- Check: C step rate, 90 deg in 150 ms (PR-4) | Result: 10667 steps/s peak | Pass: yes
- Check: PR-10 cycle estimate (avg lane -> up camera -> PCB centre -> back) | Result: 4.97 s/part = 724 parts/h (814 with the feed done during the place) | Pass: **NO**

Cycle breakdown: XY 1.71 s, Z 2.34 s (at 60 mm/s, from Safe Z), vision 0.25 s, vacuum 0.12 s, feed 0.55 s. OpenPnP Safe Z = -9.2 (machine Z: tip 2 mm above the tallest tooling the head passes over). Moving between pick, camera and place at Safe Z instead of Z 0 is what makes PR-10.

Interference check

No interference between moving and fixed parts at the checked poses.

Custom parts (what has to be made)

Part | Kind | Qty | Mass each g

- Part: Down camera angle, Al 3 mm, slotted for focus | Kind: machined | Qty: 1 | Mass each g: 19
- Part: Nozzle magazine bar, Al 24x12, 6 pockets along Y | Kind: machined | Qty: 1 | Mass each g: 79
- Part: Tooling plate Al 6 mm 635 x 444, PCB area + tooling strip (one piece) | Kind: machined | Qty: 1 | Mass each g: 4504
- Part: X chain tray, Al angle 50x18x2, 495 mm | Kind: machined | Qty: 1 | Mass each g: 176
- Part: Y idler bracket, machined Al (end plate + side flange + slotted cheeks) | Kind: machined | Qty: 1 | Mass each g: 140
- Part: Y motor bracket, machined Al 6 mm (end plate + motor plate + side flange + gussets) | Kind: machined | Qty: 1 | Mass each g: 225
- Part: Y tension block, steel 7x10x12, M3 tapped | Kind: machined | Qty: 2 | Mass each g: 9
- Part: Z slide, one piece: block plate + C motor arm + nut seat (CNC Al 6061) | Kind: machined | Qty: 1 | Mass each g: 25
- Part: Bridge end: foot + riser + belt clamp (printed PETG-CF, 100% infill) | Kind: printed | Qty: 2 | Mass each g: 86
- Part: Corner block front-left with X motor mount (printed PETG/ASA) | Kind: printed | Qty: 1 | Mass each g: 325

- Part: Corner block front-right (printed PETG/ASA) | Kind: printed | Qty: 1 | Mass each g: 179
- Part: Corner block rear-left (printed PETG/ASA) | Kind: printed | Qty: 1 | Mass each g: 199
- Part: Corner block rear-right (printed PETG/ASA) | Kind: printed | Qty: 1 | Mass each g: 179
- Part: Cover tape waste bin (printed PETG) | Kind: printed | Qty: 1 | Mass each g: 27
- Part: Empty-tape exit chute, 8 lanes (printed PETG) | Kind: printed | Qty: 1 | Mass each g: 48
- Part: Feeder v3.0 cover plate, bayonet rails (printed PETG) | Kind: printed | Qty: 8 | Mass each g: 1
- Part: Feeder v3.0 cover tape guide bar D2.9 with head, pushed into the post (printed PETG) | Kind: printed | Qty: 8 | Mass each g: 0
- Part: Feeder v3.0 idler gear 36T m1 with axle stub (printed PETG) | Kind: printed | Qty: 8 | Mass each g: 2
- Part: Feeder v3.0 main body (printed PETG) | Kind: printed | Qty: 8 | Mass each g: 35
- Part: Feeder v3.0 pawl, feed + detent (printed PETG) | Kind: printed | Qty: 16 | Mass each g: 0
- Part: Feeder v3.0 pinch axle D6 x 14, pressed into the yoke (printed PETG) | Kind: printed | Qty: 8 | Mass each g: 0
- Part: Feeder v3.0 pinch yoke (printed PETG) | Kind: printed | Qty: 8 | Mass each g: 0
- Part: Feeder v3.0 pinion 14T m1 with the D6 drive shaft, one part (printed PETG) | Kind: printed | Qty: 8 | Mass each g: 1
- Part: Feeder v3.0 rack m1 on the slider (printed PETG) | Kind: printed | Qty: 8 | Mass each g: 0
- Part: Feeder v3.0 roller body, drive + pinch (printed PETG) | Kind: printed | Qty: 16 | Mass each g: 1
- Part: Feeder v3.0 servo gear sector 13T m1 on the spline (printed PETG, replaces the horn) | Kind: printed | Qty: 8 | Mass each g: 0
- Part: Feeder v3.0 side plate (printed PETG) | Kind: printed | Qty: 8 | Mass each g: 13
- Part: Feeder v3.0 slider (printed PETG) | Kind: printed | Qty: 8 | Mass each g: 4
- Part: Head belt clamp + Y chain post (printed PETG-CF) | Kind: printed | Qty: 1 | Mass each g: 22
- Part: Head carriage plate: block mount + Z rail + Z motor shelf (printed PETG-CF) | Kind: printed | Qty: 1 | Mass each g: 107
- Part: PCB slide clamp (printed PETG) | Kind: printed | Qty: 1 | Mass each g: 9
- Part: Reel hub / spacer (printed PETG) | Kind: printed | Qty: 8 | Mass each g: 4
- Part: Reel rack side plate (printed PETG) | Kind: printed | Qty: 2 | Mass each g: 20
- Part: Up camera housing, one piece (printed PETG) | Kind: printed | Qty: 1 | Mass each g: 25
- Part: Waste bin leg (printed PETG) | Kind: printed | Qty: 2 | Mass each g: 3
- Part: X chain arm (printed PETG), on the Y motor bracket | Kind: printed | Qty: 1 | Mass each g: 8
- Part: X chain tray bracket (printed PETG) | Kind: printed | Qty: 3 | Mass each g: 22
- Part: X idler tension slider (printed PETG) | Kind: printed | Qty: 2 | Mass each g: 5

150 printed + 9 machined parts; 411 fasteners (see bom.csv, kind = fastener).

Printing (printed parts)

All printed parts are orange PETG (PETG-CF for the head parts). STLs in parts_stl/ are already turned to the print orientation below and lie on Z = 0. Horizontal holes over 5.2 mm diameter have a 45 deg teardrop roof (no support). Settings: 0.2 mm layers, 4 perimeters, 5 top/bottom layers, 40% gyroid, elephant-foot compensation 0.2 mm, no supports. Bed check: 220 x 220 x 250 mm.

Part | Qty | Orientation (bed face) | Size on bed mm | Bed contact mm2 | Overhang mm2 | Flat bridge mm2 | Teardrops | Fits

- Part: Corner_FrontLeft | Qty: 1 | Orientation (bed face): +Z up (as installed) | Size on bed mm: 104 × 104 × 96 | Bed contact mm2: 8557 | Overhang mm2: 0 | Flat bridge mm2: 596 | Teardrops: 29 | Fits: yes
- Part: Corner_RearLeft | Qty: 1 | Orientation (bed face): +Z up (as installed) | Size on bed mm: 104 × 68 × 90 | Bed contact mm2: 6685 | Overhang mm2: 0 | Flat bridge mm2: 596 | Teardrops: 22 | Fits: yes
- Part: Corner_FrontRight | Qty: 1 | Orientation (bed face): +Z up (as installed) | Size on bed mm: 100 × 68 × 90 | Bed contact mm2: 6413 | Overhang mm2: 0 | Flat bridge mm2: 610 | Teardrops: 15 | Fits: yes
- Part: Corner_RearRight | Qty: 1 | Orientation (bed face): +Z up (as installed) | Size on bed mm: 100 × 68 × 90 | Bed contact mm2: 6413 | Overhang mm2: 0 | Flat bridge mm2: 610 | Teardrops: 15 | Fits: yes
- Part: Xldler_Slider_Front | Qty: 2 | Orientation (bed face): +Z up (as installed) | Size on bed mm: 22 × 21 × 28 | Bed contact mm2: 161 | Overhang mm2: 0 | Flat bridge mm2: 0 | Teardrops: 0 | Fits: yes
- Part: Bridge_End_Front | Qty: 2 | Orientation (bed face): on the -X face | Size on bed mm: 91 × 64 × 44 | Bed contact mm2: 2567 | Overhang mm2: 0 | Flat bridge mm2: 0 | Teardrops: 12 | Fits: yes
- Part: XChain_Arm | Qty: 1 | Orientation (bed face): on the -Y face | Size on bed mm: 28 × 17 × 56 | Bed contact mm2: 371 | Overhang mm2: 8 | Flat bridge mm2: 0 | Teardrops: 0 | Fits: yes
- Part: Head_Carriage | Qty: 1 | Orientation (bed face): on the -X face | Size on bed mm: 114 × 128 × 44 | Bed contact mm2: 14446 | Overhang mm2: 0 | Flat bridge mm2: 0 | Teardrops: 1 | Fits: yes
- Part: Head_BeltClamp | Qty: 1 | Orientation (bed face): on the -Y face | Size on bed mm: 52 × 47 × 24 | Bed contact mm2: 1037 | Overhang mm2: 0 | Flat bridge mm2: 0 | Teardrops: 0 | Fits: yes
- Part: PCB_SlideClamp | Qty: 1 | Orientation (bed face): +Z up (as installed) | Size on bed mm: 60 × 30 × 8 | Bed contact mm2: 1683 | Overhang mm2: 0 | Flat bridge mm2: 0 | Teardrops: 0 | Fits: yes
- Part: UpCam_Housing | Qty: 1 | Orientation (bed face): +Z up (as installed) | Size on bed mm: 48 × 48 × 42 | Bed contact mm2: 2240 | Overhang mm2: 0 | Flat bridge mm2: 36 | Teardrops: 0 | Fits: yes
- Part: Lane*_Body | Qty: 8 | Orientation (bed face): on the +Y face | Size on bed mm: 127 × 66 × 20 | Bed contact mm2: 4142 | Overhang mm2: 0 | Flat bridge mm2: 88 | Teardrops: 0 | Fits: yes
- Part: Lane*_SidePlate | Qty: 8 | Orientation (bed face): on the -Y face | Size on bed mm: 127 × 61 × 16 | Bed contact mm2: 3385 | Overhang mm2: 0 | Flat bridge mm2: 149 | Teardrops: 0 | Fits: yes
- Part: Lane*_CoverPlate | Qty: 8 | Orientation (bed face): +Z up (as installed) | Size on bed mm: 62 × 18 × 10 | Bed contact mm2: 973 | Overhang mm2: 0 | Flat bridge mm2: 0 | Teardrops: 0 | Fits: yes
- Part: Lane*_Slider | Qty: 8 | Orientation (bed face): +Z up (as installed) | Size on bed mm: 85 × 14 × 17 | Bed contact mm2: 619 | Overhang mm2: 0 | Flat bridge mm2: 31 | Teardrops: 0 | Fits: yes

- Part: Lane*_Pawl_1 | Qty: 16 | Orientation (bed face): on the -Y face | Size on bed mm: 12 × 8 × 2 | Bed contact mm2: 48 | Overhang mm2: 0 | Flat bridge mm2: 0 | Teardrops: 0 | Fits: yes
- Part: Lane*_ServoGear | Qty: 8 | Orientation (bed face): +Z up (as installed) | Size on bed mm: 11 × 12 × 5 | Bed contact mm2: 68 | Overhang mm2: 0 | Flat bridge mm2: 10 | Teardrops: 0 | Fits: yes
- Part: Lane*_Rack | Qty: 8 | Orientation (bed face): on the -Y face | Size on bed mm: 15 × 12 × 3 | Bed contact mm2: 142 | Overhang mm2: 0 | Flat bridge mm2: 17 | Teardrops: 0 | Fits: yes
- Part: Lane*_Idler | Qty: 8 | Orientation (bed face): on the +Y face | Size on bed mm: 38 × 38 × 4 | Bed contact mm2: 999 | Overhang mm2: 0 | Flat bridge mm2: 0 | Teardrops: 0 | Fits: yes
- Part: Lane*_Pinion | Qty: 8 | Orientation (bed face): on the -Y face | Size on bed mm: 16 × 16 × 18 | Bed contact mm2: 136 | Overhang mm2: 0 | Flat bridge mm2: 0 | Teardrops: 0 | Fits: yes
- Part: Lane*_DriveRoller | Qty: 16 | Orientation (bed face): on the -Y face | Size on bed mm: 15 × 15 × 12 | Bed contact mm2: 102 | Overhang mm2: 0 | Flat bridge mm2: 0 | Teardrops: 0 | Fits: yes
- Part: Lane*_PinchAxle | Qty: 8 | Orientation (bed face): on the -Y face | Size on bed mm: 6 × 6 × 14 | Bed contact mm2: 28 | Overhang mm2: 0 | Flat bridge mm2: 0 | Teardrops: 0 | Fits: yes
- Part: Lane*_PinchYoke | Qty: 8 | Orientation (bed face): +Z up (as installed) | Size on bed mm: 18 × 15 × 14 | Bed contact mm2: 160 | Overhang mm2: 0 | Flat bridge mm2: 0 | Teardrops: 1 | Fits: yes
- Part: Lane*_GuideBar | Qty: 8 | Orientation (bed face): on the -Y face | Size on bed mm: 6 × 6 × 18 | Bed contact mm2: 26 | Overhang mm2: 0 | Flat bridge mm2: 0 | Teardrops: 0 | Fits: yes
- Part: Feeder_ExitChute | Qty: 1 | Orientation (bed face): on the -Y face | Size on bed mm: 21 × 17 × 178 | Bed contact mm2: 325 | Overhang mm2: 4 | Flat bridge mm2: 258 | Teardrops: 5 | Fits: yes
- Part: Feeder_WasteBin | Qty: 1 | Orientation (bed face): +Z up (as installed) | Size on bed mm: 36 × 198 × 28 | Bed contact mm2: 7011 | Overhang mm2: 0 | Flat bridge mm2: 12 | Teardrops: 0 | Fits: yes
- Part: Feeder_WasteBinLeg_1 | Qty: 2 | Orientation (bed face): on the -X face | Size on bed mm: 43 × 10 × 8 | Bed contact mm2: 430 | Overhang mm2: 0 | Flat bridge mm2: 0 | Teardrops: 0 | Fits: yes
- Part: ReelRack_Side_1 | Qty: 2 | Orientation (bed face): on the -Y face | Size on bed mm: 120 × 107 × 8 | Bed contact mm2: 2827 | Overhang mm2: 0 | Flat bridge mm2: 0 | Teardrops: 0 | Fits: yes
- Part: ReelHub_1 | Qty: 8 | Orientation (bed face): on the +Y face | Size on bed mm: 22 × 22 × 22 | Bed contact mm2: 323 | Overhang mm2: 0 | Flat bridge mm2: 0 | Teardrops: 0 | Fits: yes
- Part: XChain_TrayBracket_1 | Qty: 3 | Orientation (bed face): on the -X face | Size on bed mm: 80 × 54 × 20 | Bed contact mm2: 1258 | Overhang mm2: 0 | Flat bridge mm2: 0 | Teardrops: 1 | Fits: yes

Flat bridges left: corner-block pocket roofs (8 mm flat between 45 deg chamfers), down-facing counterbores under the feet, feeder body slider tunnel (5 / 13 mm) and pinch-yoke pocket (3.3 mm). All are within normal PETG bridging. Feeder waste bin is printed upside down; its floor bridges 34 mm.

Design notes against the TS

- D1 The stock C-beams are 480 mm apart (inner faces) in Tevo Black Widow assembly.step, not 500. The 20x80x500 top tie sits on the C-beam ends and does not fit between them; it is cut to 480. WP0 confirms.
- D2 In the project model the 40x80x550 extrusion is the bed (Y) axis beam; the stock X gantry is two 20x40x485 bars. This model uses the 40x80x550 as the Y bridge (TS 4 item 2) and the 20x40 pair, cut to 480, as the second tie (TS 7.1).
- D3 The bridge sits on 32 mm risers so the beam underside (Z 153) clears the tape bank, magazine and clamps. The head hangs in front of the beam (+X), so the nozzle reach is offset by 65.0 mm in X.
- D4 The X belts run inside the open C channels; the C-beam end caps are replaced by printed drive and idler blocks. The jackshaft sits at the left end (TS 7.1).
- D5 Head Z motor defaults to NEMA8_LIN30 to cut mass; the stock NEMA17 is the fallback named in WP4 step 6.
- D6 (rev B) The 20x80x210 legs are gone. Four printed corner blocks are end cap, pulley/idler housing, jackshaft bearing seat (608ZZ pressed in), tie corner bracket and leg in one; the front-left one also holds the X motor. Rubber feet screw into them.
- D7 (rev B) Tape lanes, up camera and nozzle magazine sit in a strip right of the bridge travel (X > 526, beam right face at most X 522), so the beam never passes over them. Tapes feed in from the right end. Magazine pockets run along Y (TS 7.5 says along X; the change is still single-axis motion).
- D8 (rev B) One tooling plate carries the PCB fixture and the whole tooling strip, so all feeder, camera and magazine coordinates share one datum. 3-point support: left tie + two posts on the right tie.
- D9 (rev B) Head = one printed head carriage (mother plate, head plate, Z motor bracket, camera bracket, switch arm, belt clamp, chain post) + one machined Z slide (Z plate, both bearing bores in one setup for FR-14, nut seat, C motor shelf). Bridge = one printed end per side (foot plate, riser, belt clamp, chain arm). The LP mother plate is no longer used.
- D10 (rev B) Vacuum panel and electronics box are not modelled for now; TS 7.4 and 8 still apply.
- D11 (rev D, current head) The head uses a NEMA8 hollow-shaft C motor and a NEMA8 external linear Z motor with integral T5x2 screw. The nozzle stack is approximately 15 g; bearing runout and loaded repeatability must be measured on the selected motor. Weigh the assembled head and verify motor torque at the actual supply voltage before setting acceleration.
- D12 (superseded feeder history) The v2.0/v2.1 lane used a flexure pawl and ratchet-based peel mechanism and is archived under legacy/feeder_v2.1/. It is not the current geometry. Current feeder details are D14 and feeder/ASSEMBLY.md.
- D13 (rev F, dynamics) (1) Aluminium profiles now carry catalogue kg/m instead of the solid-core model volume: the bridge beam was 3.6 kg in the model, ~1.3 kg in reality, so X moves 3.08 kg, not ~5.3 kg. (2) Z runs at 1/4 microstepping (400 steps/mm, 2.5 um): at 1/16 PR-3 needs 96k steps/s, over the ~40k of the 8-bit board. Configure the Z driver jumpers and Marlin DEFAULTAXISSTEPSPERUNIT to match. (3) OpenPnP Safe Z = -9.2 (tip 2 mm over the tallest tooling in reach, the magazine tips); travel at Safe Z is checked in pnp check. Together they bring the PR-10 estimate from ~610 to ~724 parts/h. (4) The torque checks assume a 24 V supply; at 12 V the X margin drops to ~1.9 (WP0 confirms).

• D14 (current v3.0 feeder) Eight independent servo lanes use a rigid spring-loaded feed pawl and detent, plus a rack/idler/pinion peel drive with two HF0612 one-way bearings per lane. The segmented cover plate is screw-locked; lane body and side plate split so PETG parts print without supports. One separate Nano controller runs the servos. This geometry is awaiting its documented physical first article; see feeder/README.md and feeder/TEST.md.

Feeder v3.0 project guide

Исходный файл: pnp/feeder/README.md

TBW-Feeder v3.0: motorised 8 mm SMD tape feeder

This folder holds the tape feeder for the TEVO-BW pick-and-place: 8 lanes at a 22 mm pitch. One servo per lane drives both the feed and the peel. The geometry is in ../model/feeder.py. The machine model uses the same code, so the two can't drift apart.

```
feeder build   # out/: one lane + bank parts as FCStd/STEP, stl/ (printed, lying on the bed), bom.csv
               #       (per lane, per bank, totals for 8 lanes)
feeder audit   # out/audit.md: overlaps, motion sweep, pawl ride-over, assembly paths, cover plate bayonet,
               #       support-free printing, walls, loads, bank vs machine
feeder all
feeder animate # out/feeder_animation.gif: one 4 mm feed cycle rendered from the model (lane, pawls, peel drive)
python orca_project.py   # out/TBW_Feeder_v3.0.3mf: OrcaSlicer project, one lane + bank parts
python print_time.py ... # slicer time + filament per part on your printer profile (see DESIGN.md)
```

orca_project.py takes the printer, process and PETG settings from the newest Adventurer 5M project in Documents (or a .3mf given as the first argument). Each part carries its own settings from ASSEMBLY.md. Run feeder build first so the STLs are current.

To view the feeder in FreeCAD with one colour per part, open it through open_gui.py. Opening the FCStd directly shows grey parts.

```
$env:PNP_FCSTD = "D:\TEVO_BW\pnp\feeder\out\tbw_feeder_lane.FCStd"
& "$env:LOCALAPPDATA\Programs\FreeCAD_1_1\bin\freecad.exe" D:\TEVO_BW\pnp\open_gui.py
```

Use feeder.cmd (cmd) or .\feeder.ps1 (PowerShell), with FREECAD_BIN set to FreeCAD's bin folder. On this PC that is %LOCALAPPDATA%\Programs\FreeCAD11\bin. audit exits with code 1 if any check fails.

Document | Contents

- Document: DESIGN.md | Contents: Why v3.0, TRIZ analysis, layout, mechanisms, numbers, print times, limits
- Document: ASSEMBLY.md | Contents: Print settings, bought parts, preparation, assembly order, calibration, tape loading
- Document: ELECTRONICS.md | Contents: Controller modules, wiring table, power budget, flashing
- Document: OPENPNP.md | Contents: Driver, actuators, feeder setup
- Document: TEST.md | Contents: First-article and acceptance tests
- Document: `firmware/tbw_feeder_fw/` | Contents: Controller firmware (Arduino Nano), G-code M600–M620, settings in EEPROM (μs)

The feeder bank's power branch is part of the machine panel design in ../electronics/POWERCONTROLDESIGN.md (../electronics/POWERCONTROLDESIGN.md). Keep the Nano on USB serial and feed servo current from the separate fused 5 V rail described in ELECTRONICS.md (ELECTRONICS.md).

The v2.1 sources and outputs are kept in ../legacy/feeder_v2.1/.

How it works

An MG90S servo hangs upside down in the tower above the tape. A printed gear on its spline drives a rack on the slider, which runs in a tunnel under the tape: 4.2 mm stroke = 37° of servo rotation.

- Feed: a rigid printed pawl on a screw pivot, pressed up by a steel spring, pushes the tape exactly one

sprocket hole (4.0 mm); a half stroke gives 2 mm. The pawl's pivot sits below and behind its tooth, so the push load drives the tooth deeper into the hole.

- Detent: the same pawl, mounted in the body, blocks back-drag.
- Peel: the same stroke peels the cover tape. A rack fixed to the slider turns a 36-tooth idler, and the

idler turns a 14-tooth pinion printed with its shaft. An HF0612 one-way bearing in the drive roller passes the motion on only in the feed direction, giving 4.8 mm of cover tape per 4 mm fed. A second HF0612 in the pinch roller keeps the cover tape taut on the retract. The sprung nip slips at about 1.2 N, which limits the tension.

- Cover plate: it sits in segmented rails and is locked by 2 countersunk M2 screws. Take them out, slide it

6.8 mm and lift it off to load a tape.

- Body: a main body plus a side plate. Every part is laid in from the split face, and every printed part

prints flat in PETG on a 0.4 mm nozzle without supports.

Status (2026-10-22)

Design revision 3.0, ready for the first article. It replaces v2.1 after that lane's first article: weak, slipping feed; ratchet slipping or jamming; printed springs and the snap cap not holding; 10 h of printing per lane with supports on a 0.25 mm nozzle.

- feeder audit: see out/audit.md.
- The firmware (3.0.0) compiles for the Nano.

What remains needs hardware. Build one lane plus the controller and pass TEST.md, including A0 (print without supports and compare the times) and the tests marked as v2.1 issues. Only then print the other seven lanes. The lane mounting holes are the same as for v2.1, so the tooling plate doesn't change. The waste bin tray is new: it is 3 mm lower and 2.5 mm further back.

The technical specification (TS 2.2, 7.7, WP8) still describes head-actuated lever lanes. Decisions D12 and D14 in ../out/report.md record the change to this feeder.

Feeder design, mechanisms and limits

Исходный файл: pnp/feeder/DESIGN.md

TBW-Feeder v3.0: design notes

All numbers come from ../model/feeder.py and are checked by feeder audit (out/audit.md). Heights are measured from the tooling plate top. X runs along the tape: the tape moves toward −X and the pick is at the inner end. Y is measured from the lane centre line.

Why v3.0: what the v2.1 first article showed

Problem on the v2.1 lane | Cause in the v2.1 design

- Problem on the v2.1 lane: Weak feed that's hard to adjust; the tape slips | Cause in the v2.1 design: The pawl was a 0.9 mm PETG flexure with 0.26 N of spring force, so the tooth lifted out of the hole under load. The servo positions were whole degrees, 0.11 mm steps at the pawl.
- Problem on the v2.1 lane: The "gear" slips or doesn't move | Cause in the v2.1 design: The peel ratchet was Ø12 with 18 teeth only 0.6 mm deep, driven by a 0.8 mm flexure finger. Even with a 0.25 mm nozzle the teeth print rounded.
- Problem on the v2.1 lane: The springs don't hold | Cause in the v2.1 design: Every spring was printed PETG (pawl, lifter finger, hold pawl, detent tongue), and PETG creeps.
- Problem on the v2.1 lane: The cover latch doesn't hold firmly | Cause in the v2.1 design: The cover plate was held only at its ends (2 M2 thumb screws and 2 hinge tabs 58 mm apart), so it bowed over the pawl. The cheek cap snapped on with 0.85 mm legs.
- Problem on the v2.1 lane: Printing took long, and the supports were hard to remove | Cause in the v2.1 design: One body full of closed tunnels and chambers needed supports inside. The fine features forced a 0.25 mm nozzle: 10.0 h per lane, 8.3 h of it the body.

The servo-driven swing feed stays: one SG90 per lane, the same controller and the same wiring. The mechanism around it is new.

TRIZ analysis

Ideal final result: *the tape itself holds the pawl in the hole, the cover tape stays taut by itself, and the lane prints without supports in one material.* Each contradiction below was resolved with existing resources (the push load, the servo stroke, the cover tape tension, standard bought parts) instead of with extra parts.

 # | Contradiction | Principles | v3.0 solution

- #: 1 | Contradiction: Physical: the pawl must be **stiff** to push the tape and **compliant** to ride back under it | Principles: 1 Segmentation, 40 Composite, 25 Self-service | v3.0 solution: Rigid printed lever with a separate steel spring on an M2 set screw pivot. The pivot is 5.9 mm below and behind the tooth, so the push load turns the tooth deeper into the hole (self-locking) against a positive up-stop. The spring only has to seat the tooth: 1.25–1.57 N, 5× v2.1.
- #: 2 | Contradiction: The detent needs the same function as the pawl in the other direction | Principles: 6 Universality, 33 Homogeneity | v3.0 solution: It is the same part, pivot screw and spring as the feed pawl, mounted in the body. Same self-locking geometry, play 0.15 mm.
- #: 3 | Contradiction: Technical: a reliable one-way peel drive (27) against manufacturability (32): fine ratchet teeth can't be printed. Then: a second moving member (the lifter) against simplicity (36) | Principles: 2 Taking out, 14 Curvature, 5 Merging, 6 Universality | v3.0 solution: The one-way function moves out of the printed parts into a bought HF0612 needle clutch. The rack sits on the slider, which already moves: a 36T idler passes the motion to the 14T pinion on the roller shaft and sets the direction. There is no second moving member. Teeth are 2.25 mm deep instead of 0.6.
- #: 4 | Contradiction: Hold the cover tape taut between feeds without a hold pawl | Principles: 5 Merging, 25 Self-service | v3.0 solution: A second HF0612 goes in the pinch roller that already exists. It can't turn back, so the nip holds the cover tape on the retract.
- #: 5 | Contradiction: Limit the peel tension without a sensor | Principles: 11 Beforehand cushioning | v3.0 solution: Kept from v2.1: the sprung nip (now on NBR O-rings) slips at 1.2 N and takes the 20 % surplus.
- #: 6 | Contradiction: Physical: the cover plate must be **fixed** firmly and **free** for tape loading without tools | Principles: 1 Segmentation, 15 Dynamics, 28 Mechanics substitution | v3.0 solution: Segmented rails: 3 lips per side and 2 rear tongues hold the plate down every 14 mm. Slide it 6.8 mm toward the pick (bayonet) and lift it off. Two countersunk M2 screws through its upstand into the tower lock it home instead of snaps or magnets. Deflection over the pawl drops from 2.2 mm to 0.03 mm.
- #: 7 | Contradiction: Technical: manufacturability (32, no supports, larger nozzle) against device complexity (36, tunnels and pockets) | Principles: 1 Segmentation, 17 Another dimension, 7 Nested doll, 2 Taking out | v3.0 solution: The body is split into a main body and a side plate. Every pocket opens to the split face, so both halves print flat on their outer faces and all moving parts are laid in from one side. Shafts, axles and the guide bar are printed PETG; the drive shaft is printed in one piece with the pinion (5 Merging).
- #: 8 | Contradiction: A few features still face down in print | Principles: 3 Local quality, 4 Asymmetry | v3.0 solution: A 45° chamfer goes only on the one edge that faces down. The strip beside the pawl slot and the riser wall belong to the side plate, where they print standing up. Bridging layers sit in the counterbores.
- #: 9 | Contradiction: Fine pick adjustment (accuracy, 29) without extra hardware (36) | Principles: 35 Parameter change, 23 Feedback | v3.0 solution: Servo positions are in microseconds instead of degrees: 0.01 mm per step instead of 0.11. Calibrated with the down camera.

Result: there are 13 printed parts per lane (11 different), all PETG, none needing supports on a 0.4 mm nozzle. Every spring and axle is steel, and every one-way function is a bought needle clutch.

Layout (one lane, 127 × 20.5 × 68 mm)

Zone | X (frame) | Height | Contents

- Zone: Low section | X (frame): 526–592 | Height: ≤ 16.3 (lips 18.3) | Contents: Tape channel, cover plate on segmented rails, detent under the hole row, slider tunnel under the tape
- Zone: Tower front | X (frame): 592–600 | Height: ≤ 52 | Contents: Post: cover tape riser, printed Ø2.9 guide bar, 2 M2 pilot holes for the cover plate screws, slots for the rear tongues of the plate
- Zone: Servo | X (frame): 600–633 | Height: ≤ 48.2 | Contents: SG90 hanging upside down, tabs captured in a slot between the halves
- Zone: Peel head | X (frame): 612–652 | Height: ≤ 67 | Contents: Drive roller (O-rings) over the pinch roller. On the −Y side: take-up rack on the slider, 36T idler beside the servo, 14T pinion on the roller shaft

The body is split at y = −4.85 into a main body (prints on its +Y face) and a side plate (prints on its −Y face), held together by 4 M3 screws into inserts. All moving parts lie in pockets that open to the split face. The lane mounting holes and the tooling plate are the same as for v2.1. The rail lips are 18.3 mm high, still under the 20 mm the head clears, and the parts above 20 mm are beyond X 588, which the head never reaches (pnp check).

Tape path

• Channel 8.2 × 1.2 mm (paper carriers up to 1.1 mm, embossed 0.2–0.4 mm). A relief groove 4.0 × 2.1 mm

sits under the pockets for embossed tape (2.4 mm pockets clear). Its sprocket-side edge has a 45° chamfer.

• The slider tunnel roof (11.8) ties the sprocket-side tape floor to the main body. In v2.1 that strip was held

only by the solid side wall, which is now the side plate.

• Pick pocket at X 540, Y lane + 1.25. Sprocket holes at Y −2.25, with a hole centre at X 542.

• Empty tape leaves at X 526 through the chute. Reels stand on the rack behind the machine.

Advance mechanism

• Servo upside down (MG90S, SG90 size), output axis at X 611. The model includes the round boss around the

output gear (Ø11.8 × 4 mm under the case face) and the spline (Ø4.8 × 2.8). The case face is at 27.1 mm, and the servo top at 49.8 mm, under the cover tape path. SG90/MG90S drawings: body 22.2–22.8 × 11.8–12.2 mm, tab span 32.2–32.5, tab top 18.5 above the base, 28.4–28.5 high without the horn. Check the delivered servo (TEST A7).

• A printed servo gear, a 13T m1 sector (teeth only within ±55° of the rack side, the rest is hub, so it clears

the idler), replaces horn, crank and pin. It is pressed onto the spline (the metal spline cuts its teeth into the bore) and held by the servo's screw from below.

• It drives a rack moulded on the slider: an arm rises beside the tape on two short necks, and its teeth face

the gear at Y + 6.5. The stroke is 4.2 mm = 37° of servo rotation, linear in the angle (the scotch yoke was not), so 1 µs ≈ 0.01 mm everywhere in the stroke. The slider prints standing on its bottom, so the rack teeth lie in the print plane.

• Feed pawl: a rigid 2 mm PETG lever on an M2 × 5 set screw (pivot, threaded into the slider walls) in the slider

fork. A steel spring (OD 3 × 0.3 × L0 8, installed 5.5 mm) presses it up. The tooth is 0.8 mm wide (flush with the face it prints on), with a vertical hook face 0.75 mm into the tape, a 0.3 mm flat and a 45° ramp. At the tape underside it is 1.05 mm long, so it fits a hole at both stroke ends with 0.2 mm free travel.

• Self-locking: the tooth contact is 5.9 mm above the pivot, behind the tooth. The push load therefore turns the

tooth deeper, against the up-stop (the tail on the fork floor). It can't climb out under load. Riding back under the tape it turns 5.8° down (checked for clearance in the audit).

• Detent: the same pawl, in a housing in the body at the hole next to the pick pocket. Its hook face sits 0.15

mm behind the hole edge and blocks back-drag. The tape pushes it down over its ramp on the feed stroke.

• Feed per stroke is 4.0 mm, and a half stroke gives 2.0 mm. Positions are in µs: forward 1709, half 1510,

retract 1291 (108.9 / 90.9 / 71.1°), calibrated per lane in steps of about 0.01 mm.

Load case | Value | Limit / margin

• Load case: Pawl spring, seated / riding over | Value: 1.25 / 1.57 N | Limit / margin: v2.1 flexure 0.26 N
• Load case: Tape push (detent ramp + friction + reel drag) | Value: 3.8 N | Limit / margin: —
• Load case: Servo torque, tape only (gear R 6.5) | Value: 0.024 N m | Limit / margin: Stall 0.18 N m (7×)
• Load case: Pawl tooth root bending (prints lying, in-plane) | Value: 10.7 MPa | Limit / margin: PETG ≈ 45 MPa
• Load case: Servo positioning resolution | Value: 1 µs ≈ 0.01 mm | Limit / margin: v2.1: 1° ≈ 0.11 mm

Peel mechanism (gears + one-way bearings)

• Path of the cover tape: peel slot in the plate (X 546–548), back over the plate, up the riser in the post,

over the printed Ø2.9 guide bar, horizontally above the servo at 50.9 mm, into the nip under the drive roller, and out rearward over the waste bin lip.

• Drive:

• A printed rack (m1, teeth up, pitch line 16.8 mm) is fixed to the slider tail with two M2 × 6 screws.

It moves with the slider, 4.2 mm per stroke, with no extra member and no cam.

• The rack turns a 36T m1 idler (Ø36, axis at X 631.7 / Z 34.8). Its Ø5 axle stub is printed on and runs

in a hole through the side plate. It bridges the 42 mm up to the roller shaft.

• The idler turns a 14T m1 pinion, printed in one piece with its Ø6 drive shaft (no press fit to slip).

The shaft runs in the body at +Y; its −Y end runs on a Ø2.9 journal pin printed on the side plate.

• Rack → idler → pinion is one gear more than rack → pinion, which gives the right winding direction (the nip

pulls the cover tape rearward on the feed stroke). The idler's size doesn't change the ratio.

• The shaft is the inner race of an HF0612 needle clutch in the drive roller (PETG body, two NBR O-rings

12 × 1.5, Ø16). The clutch drives the roller on the feed stroke and freewheels on the retract.

- Per full stroke the roller surface moves 4.2 × 8 / 7 = 4.8 mm per 4 mm of tape (20 % surplus). A half

stroke gives 2.4 mm per 2 mm, so every 2 mm feed now peels its own share.

- Hold on the idle (return) stroke: the rack goes back and turns the idler, pinion and shaft back. The drive roller's

HF0612 freewheels, so the roller isn't taken back. The pinch roller is the same roller body (O-rings) on a second HF0612, on a fixed printed Ø6 × 14 axle pressed into the single yoke arm (+Y side, the idler is on −Y), and it can't turn back. With the cover tape clamped between the two rollers at 1.5 N, the drive roller and the cover tape stand still: the nip holds 1.2 N, the same as the most it can pull, so nothing slips back. The carrier tape is held by the detent at the same time.

- Nip: a steel spring (OD 5 × 0.4 × L0 9, installed 7 mm) pushes the yoke up: 1.5 N. On NBR (μ ≈ 0.8) the nip

slips at 1.2 N, which limits the peel tension (EIA-481 peel force 0.1–1.3 N). To thread the cover tape, push the yoke tab down 3 mm.

Peel load case | Value | Limit / margin

- Peel load case: Rack force at nip slip (1.2 N at R 8 / R 7) | Value: 1.4 N | Limit / margin: —
- Peel load case: Slider force for the peel (two spur meshes) | Value: 1.5 N | Limit / margin: v3.0 lifter slot: 3.1 N
- Peel load case: Servo torque, tape + peel | Value: 0.036 N m | Limit / margin: Stall 0.18 N m (5×)
- Peel load case: Rack / idler / pinion tooth bending (Lewis, face 1.8) | Value: 2.8 MPa | Limit / margin: PETG fatigue ≈ 10 MPa
- Peel load case: Pinion on its shaft | Value: one printed part | Limit / margin: no press fit
- Peel load case: HF0612 torque | Value: 9.6 N mm | Limit / margin: Rated ≈ 1.8 N m
- Peel load case: Cover tape tension held on the idle stroke | Value: 1.2 N | Limit / margin: = nip slip tension

Cover plate

- Printed PETG 1.2 mm, tape side down. It has the pick window (8 × 5.2), the peel slot, rail tongues at both

edges (3 segments per side), 2 rear tongues into the post, and an upstand at the rear with 2 countersunk holes: 2 × M2 × 8 countersunk screws (DIN 965) go through it along X into the tower.

- To load tape: take out the 2 screws, slide it 6.8 mm toward the pick and lift it off. At home it can't lift (checked in the audit).
- Deflection under the pawl spring, rails every 14 mm: 0.03 mm. With the v2.1 supports (58 mm apart) the same

load would bow it 2.2 mm.

- Hold: the screws are a positive lock (≈ 40 N pull-out each, self-tapped into PETG) against 0.9 N of tape

drag on the feed stroke.

Manufacturing

Per lane there are 15 printed parts (Body, SidePlate, CoverPlate, Slider, 2 × Pawl, Rack, Idler, Pinion with its shaft, 2 × roller body, PinchYoke, PinchAxle, GuideBar, ServoGear), all PETG on one 0.4 mm nozzle at 0.20 mm layers. The audit checks each one in its print orientation:

- longest bridge ≤ 10 mm
- no flat ceiling that hangs from one side by more than 1 mm
- no overhang steeper than 45°
- walls ≥ 0.8 mm

Print time and filament, sliced by OrcaSlicer on the AD5M profile (print_time.py):

Version | Nozzle / layers | Supports | Body (+ side plate) | Small parts | Per lane | 8 lanes

- Version: v2.1, as printed | Nozzle / layers: 0.25 / 0.12 | Supports: tree, in the body | Body (+ side plate): 498 min | Small parts: 103 min | Per lane: **10.0 h, 56 g** | 8 lanes: 80 h
- Version: v2.1 on a 0.4 nozzle (reference only: its 0.6 mm ratchet teeth don't print at 0.4) | Nozzle / layers: 0.4 / 0.2 (0.12 small parts) | Supports: tree | Body (+ side plate): 147 min | Small parts: 38 min | Per lane: 3.1 h, 62 g | 8 lanes: 25 h
- Version: **v3.0** | Nozzle / layers: **0.4 / 0.2** | Supports: **none** | Body (+ side plate): 134 min (incl. side plate, yoke) | Small parts: 25 min | Per lane: **2.7 h, 66 g** | 8 lanes: **22 h**

v3.0 prints 3.7× faster than v2.1 as printed. It also needs no support removal, and nothing inside the tunnels has to be cleaned out.

Bought per lane: the SG90 and the following standard parts (ASSEMBLY.md):

- 2 HF0612 one-way bearings
- 4 O-rings
- 3 springs
- 6 M3 inserts, 6 M3 screws, 2 M2 set screws, 2 M2 screws and 2 M2 countersunk screws (plus the servo's own

screw)

Every shaft and axle is printed (or a screw): no steel pins, no magnets, no TPU, no laser-cut plate.

Electronics and firmware

One controller per bank: an Arduino Nano on an IO shield with a 5 V 4 A supply, servos only (ELECTRONICS.md). Firmware 3.0.0 (firmware/tbwfeederfw) stores the positions in µs (EEPROM layout 3) and compiles for the Nano: 6.4 kB flash, 401 B RAM.

Known limits

- 8 mm tape with 4 mm sprocket pitch only. Part pitch 2 or 4 mm, or any multiple of 2 up to 24 mm.
- Feed time is about 0.55 s per 4 mm (3 servo moves, 180 ms settle each); it can be tuned down per lane.
- The HF0612 is made to run on a hardened shaft (HRC 58+). On the printed PETG shafts the needles lock on a

softer surface: the clutch can slip or wear the shaft. Print the pinion shaft and the pinch axle solid (100 % infill) and check them in TEST.md A10. If they wear, the drop-in fix is a hardened Ø6 dowel (6 × 20 for the shaft, then the pinion needs its old Ø6 bore; 6 × 14 for the pinch axle).

- Over the rack (X 619–643, under the servo, 80 mm from the pick) the tape channel has no −Y wall above the

tape floor, so the side plate can pass the rack. The tape can move 0.85 mm sideways there. It is guided everywhere else.

- The servo dimensions are SG90/MG90S catalogue typicals. Check the delivered parts (TEST.md A7).
- The audit draws the springs at their installed length. The pawl ride-over is checked by rotating the pawl,

and the spring forces by calculation.

Feeder printing and assembly instructions

Исходный файл: pnp/feeder/ASSEMBLY.md

TBW-Feeder v3.0: manufacturing and assembly

Quantities are per lane unless marked "per bank". out/bom.csv has the full list with totals for 8 lanes. Run feeder build first; it writes every file named below.

1. Make the parts

Printed (STLs in out/stl/, already lying in print orientation)

One material (PETG) and one 0.4 mm nozzle for everything. Use 0.20 mm layers and no supports; every part lies flat on a face, and every pocket opens upward. python orcaproject.py writes out/TBWFeeder_v3.0.3mf with the three lane plates and the bank plates already set up.

File | Qty | Settings

- File: `Body.stl` | Qty: 1 | Settings: 4 walls, 40 % gyroid. The main body prints on its +Y outer face.
- File: `SidePlate.stl` | Qty: 1 | Settings: 4 walls, 40 % gyroid, on its outer face. Each of the 3 counterbores ends in a 0.2 mm bridging layer; pierce it with a 3 mm drill by hand.
- File: `CoverPlate.stl` | Qty: 1 | Settings: 100 %. It prints tape side down, so the smooth bed side touches the tape.
- File: `PinchYoke.stl` | Qty: 1 | Settings: 4 walls, 40 %, bridge down
- File: `Slider.stl`, `Rack.stl`, `Idler.stl`, `Pinion.stl` | Qty: 1 each | Settings: 100 %, flat as exported
- File: `DriveRoller.stl` | Qty: 2 | Settings: 100 %. The same roller body serves as the drive roller and the pinch roller.
- File: `ServoGear.stl` | Qty: 1 | Settings: 100 %, flat as exported. It replaces the stock horn.
- File: `Pawl.stl` | Qty: 2 | Settings: 100 %. One part serves as both the feed pawl and the detent.
- File: `Feeder_ExitChute.stl` | Qty: 1 per bank | Settings: 3 walls, 30 %, standing on its end
- File: `Feeder_WasteBin.stl`, `Feeder_WasteBinLeg.stl` | Qty: 1 + 2 per bank | Settings: 3 walls, 20 %
- File: `ReelRack_Side.stl` | Qty: 2 per bank | Settings: 4 walls, 30 %
- File: `ReelHub.stl` | Qty: 8 per bank | Settings: 3 walls, 30 %

Check the printed parts:

- The tape channel passes an 8.0 mm gauge strip.
- The slider lies in its tunnel and slides over 4.2 mm without force.
- Both pawls turn freely on an M2 screw.
- python print_time.py gives the slicer's time per part on your printer profile (see the table in DESIGN.md).

Purchased

All standard parts. Order them per 8 lanes from out/bom.csv.

Part | Qty | Use

- Part: Servo MG90S (metal gears and a metal spline, SG90 size) | Qty: 1 | Use: Advance. An SG90 fits, but its plastic spline grips the gear less.
- Part: Set screw M2 × 5 | Qty: 2 | Use: Pawl pivots, threaded into the printed holes
- Part: M2 × 6 SHCS | Qty: 2 | Use: Take-up rack to slider
- Part: M2 × 8 countersunk, DIN 965 | Qty: 2 | Use: Cover plate upstand to the tower (self-tapping into PETG)
- Part: One-way needle bearing HF0612 (6 × 10 × 12) | Qty: 2 | Use: Drive roller, pinch roller
- Part: O-ring 12 × 1.5, NBR 70 | Qty: 4 | Use: Tyres on both rollers
- Part: Compression spring OD 3 × 0.3 wire × L0 8 (≈ 0.5 N/mm) | Qty: 2 | Use: Pawls
- Part: Compression spring OD 5 × 0.4 wire × L0 9 (≈ 0.75 N/mm) | Qty: 1 | Use: Pinch
- Part: M3 heat-set insert | Qty: 6 | Use: 3 in the lane bottom, 3 in the split face
- Part: M3 × 8 SHCS | Qty: 3 | Use: Tooling plate to lane, from below
- Part: M3 × 10 SHCS | Qty: 3 | Use: Side plate to main body
- Part: (servo kit screw) | Qty: 1 | Use: Servo gear on the spline

There are no metal pins: the drive shaft (on the pinion), the pinch axle and the guide bar are printed PETG, the pawl pivots are screws. There is no TPU, no laser-cut part, no thumb screw and no stock horn.

2. Prepare

For a trial assembly before the bearings arrive, print plate 4 of the Orca project: 2 × TestFit_HF0612, a plain Ø9.8 × 12 tube with a 6.3 bore. It slides into the rollers and onto the shafts but has no clutch, so with it the peel doesn't work; swap in the real HF0612 before running tape. out/peel_exploded.png shows the order.

1. Inserts. Use a soldering iron at 220 °C. Put 3 M3 inserts in the lane bottom and 3 in the holes in the split face of the main body (6 per lane, matching out/bom.csv). Each insert finishes flush.
2. Servo gear. Set the servo to 90° (M604 N<lane> S90). Push the gear sector onto the spline with its teeth

along +Y; the metal spline cuts its teeth into the 4.8 mm bore. Fit the servo's screw from below. Try to turn the gear against the servo by hand: there must be no play.

3. One-way bearings. Press one HF0612 into each of the two roller bodies, flush, in a vise, and roll two O-rings onto each roller. The direction matters:

- Drive roller, on its shaft: the roller must turn freely when its top moves toward the tower front (the pick side) and lock the other way.
- Pinch roller, on its axle: it must turn freely when its top moves toward the rear (the waste bin) and lock the other way.
- If a direction is wrong, press the bearing out and turn it around.

4. Drive group. The pinion and the drive shaft are one printed part. Slide one roller onto the shaft from its free end, up to the pinion. If the HF0612 is tight on the printed shaft, sand the shaft lightly; it must not wobble.

5. Pinch group. Push the printed 6 × 14 axle through the pinch roller into the yoke arm (the yoke has one arm, on +Y). The axle is a press fit in the yoke; add a drop of CA glue if it is loose.

6. Cover plate screws. Turn the 2 M2 × 8 countersunk screws once into the pilot holes in the tower face (side plate) to cut the threads, then take them out again.

7. Pawls. Put a spring into the bore under each pawl seat, lay the pawl on it (tooth toward the tape), and turn the M2 × 5 set screw through both walls. The pawl must spring back onto its stop by itself.

3. Assemble one lane

Lay the main body on its +Y outer face, split face up. Every part goes in from above, in this order. `feeder audit` checks each path.

1. Slider. Fit the feed pawl (step 2.7). Screw the take-up rack to the slider tail with 2 × M2 × 6, teeth up. Lay the slider into its tunnel; its rack arm goes into the passage beside the tape channel.
2. Servo. With the gear fitted (step 2.2) at 90° and the slider at mid stroke, lay the servo into its pocket. The gear teeth come into mesh with the slider rack. The tabs go into the tab slot. No screws: the side plate captures the tabs.
3. Detent. Put the second pawl with its spring into its housing in front of the slider; it goes in through the opening in the tape floor. Turn its M2 × 5 set screw in from the split face.
4. Guide bar. It is fitted after the side plate (step 8).
5. Drive group. Put the shaft into the hole in the +Y cheek, with the pinion on the split side.
6. Pinch group. Put the pinch spring into its pocket in the tower top. Press the yoke down onto it and lay it in with the roller under the drive roller.
7. Idler. Lay the 36T idler (axle stub toward you) onto the side of the servo, so it meshes with the rack below and the pinion above. Turn the drive roller a little until the teeth slide in.
8. Side plate. Lay it on; the idler's stub goes into its hole and the journal pin into the end of the pinion shaft. Fit 3 × M3 × 10, then push the printed guide bar (head first in the counterbore) through the side plate, across the riser, into the post. The idler, the pawl screws, the drive shaft and the servo tabs are now captured.
9. Function check. Step the servo with M604 N<lane> U1291 and U1709:
   - The slider and the rack move 4.2 mm; the idler turns about 13° back and forth.
   - The roller turns only on the feed stroke (U1291 → U1709) and stands still on the retract.
10. Cover plate. Lower it with its rear end about 7 mm in front of the post, so the edge tongues drop into the gaps between the rail lips. Slide it toward the tower until the upstand touches it, then fit the 2 M2 × 8 countersunk screws. Lifting it at home must not be possible.
11. Cable. Run the servo lead out at the rear and label it with the lane number.

4. Mount the bank

1. Tooling plate. Tap M3 into the six chute holes and the two waste-bin holes. The lane holes and the empty-tape slot are machined from the plate drawing (../out/partsstep/ToolingPlate.step). They are the same as for v2.1.
2. Lanes. Set each lane on the plate and fix it from below with 3 × M3 × 8. The heads sit in the 3.2 mm counterbores. Lane 1 is at Y 150 and the pitch is 22 mm.
3. Exit chute. Fix it with 6 × M3 × 20. Put a box at least 200 × 150 mm on the bench under the slot.
4. Waste bin. Stand the two legs behind the lanes and set the tray on top. Fix it with 2 × M3 × 50 through the tray floor, the legs and into the plate. In v3.0 the tray is 3 mm lower than in v2.1 and starts 2.5 mm further back, under the cover tape exit.

5. Reel rack. Fit the two side plates and the two M8 rods with the 8 hubs between them (flange toward +Y).

Stand the rack on the bench with the axle 760 mm from the machine origin X.

6. Controller. Wire it as in ELECTRONICS.md and flash firmware 3.0.0. Send M502 and M500 to store

the defaults: settings from v2.1 are not carried over.

5. Calibrate each lane (once, with a tape loaded)

The positions are servo pulse widths in microseconds. 1 µs is about 0.01 mm at the pawl, so the forward end can be set to a hundredth of a millimetre (v2.1 had whole degrees, 0.11 mm per step).

1. M604 N<n> U1709 moves the pawl forward and U1291 moves it back. If the slider moves the wrong way, the

servo turns the other way on this unit. Mirror the values around 1500: M603 N<n> A1291 B1490 C1709.

2. Load a tape (section 6). Send M600 N<n> repeatedly and watch the pocket in the pick window with the down

camera.

3. Adjust the forward end with M603 N<n> A<us> in steps of 5 to 10 µs until the pocket is centred.
4. Set the retract end C about 20 µs past the point where the tooth clicks into the next hole.
5. Set B so that a half stroke moves the tape 2.0 mm (for 2 mm tapes). The default is 1510.
6. Peel check: run 20 × M600 N<n>. The roller turns on every feed, the cover tape stays taut and clear of

the pick window. There is nothing to set; the nip slips the surplus.

7. M500 to save. Re-teach the pick location in OpenPnP (OPENPNP.md).

6. Load a tape

1. Take out the 2 cover plate screws, slide the plate 7 mm toward the pick end and lift it off.
2. Peel about 150 mm of cover tape back by hand. Push the carrier into the inlet at the rear, with the sprocket

holes on the −Y side (the side plate side) and the pockets up, until it is under the pick window. The pawl and detent ramps let it pass.

3. Pass the free cover tape up through the peel slot of the plate from below. Lay the plate on, 7 mm short of

the post, slide it home and fit the 2 screws.

4. Lead the cover tape up the riser in the post, over the guide bar and along the top of the servo. Push the

tab of the pinch yoke down (about 3 mm), slide the cover tape between the rollers and let go. Lay the end into the waste bin.

5. Pull the cover tape rearward by hand until it is taut. It stays taut, because the pinch roller cannot turn

back. Then send M600 N<n> twice; the first pocket is now at the pick window.

Empty the waste bin and the empty-tape box at every reel change.

Feeder controller and servo wiring

Исходный файл: pnp/feeder/ELECTRONICS.md

TBW-Feeder v3.0: electronics

One Arduino Nano controls one bank of eight lanes. It connects to the machine host over USB serial. The servo rail is supplied from the machine's dedicated 24-to-5 V DC/DC converter described in ../electronics/POWERCONTROLDESIGN.md (../electronics/POWERCONTROLDESIGN.md).

Per bank

Qty | Part | Note

- Qty: 1 | Part: Arduino Nano (ATmega328P, CH340 or FT232) | Note: `firmware/tbw_feeder_fw`
- Qty: 1 | Part: Nano IO shield with GVS headers and external servo-power terminal | Note: Set jumper to external servo rail
- Qty: 1 | Part: 5 V DC/DC supply, 8 A design rating for the whole bank | Note: Dedicated servo branch; validate with actual servos
- Qty: 1 | Part: 1000 uF / 10 V electrolytic capacitor | Note: At the servo rail distribution point; observe polarity
- Qty: 8 | Part: 3-wire servo extension, 300 mm | Note: Route away from tape and moving gantry
- Qty: 1 | Part: USB cable matching Nano connector | Note: Host serial and Nano logic power

Wiring

```
Machine 24 V fused branch -> DC/DC set to 5.0 V -> IO shield external servo V+
                                                -> capacitor + at the distribution point
DC/DC 0 V ---------------------------------------> IO shield GND
                                                -> capacitor -
Nano USB ----------------------------------------> Raspberry Pi / host USB
Servo signal pins -------------------------------> D2..D9 (lane N = N+1)
```

The USB cable powers the Nano logic. The external 5 V rail powers servos only. Configure the shield jumper so external V+ does not feed USB 5 V or the Nano 5 V pin. Join signal ground and servo supply ground once at the shield. Confirm continuity and jumper wiring on the exact shield revision before connecting the Nano or servos. Do not connect 24 V to the Nano.

Lane index | Physical lane | Servo signal

- Lane index: 0 | Physical lane: 1, front lane | Servo signal: D2
- Lane index: 1 | Physical lane: 2 | Servo signal: D3
- Lane index: 2 | Physical lane: 3 | Servo signal: D4
- Lane index: 3 | Physical lane: 4 | Servo signal: D5
- Lane index: 4 | Physical lane: 5 | Servo signal: D6
- Lane index: 5 | Physical lane: 6 | Servo signal: D7
- Lane index: 6 | Physical lane: 7 | Servo signal: D8
- Lane index: 7 | Physical lane: 8, rear lane | Servo signal: D9

Servo colours are typically brown/black = GND, red = +5 V, orange/yellow = signal; verify the actual cable. D10-D13 and A0-A7 remain unused by this firmware.

Power budget and validation

Previous project estimate: one moving servo 0.15-0.7 A peak, seven servos lightly holding 0.1-0.35 A, Nano/LEDs about 0.05 A, total roughly 1.1 A. This is not a worst-case stall rating. Select the converter using the real servo stall current and measured simultaneous load. Use a converter with thermal headroom and check 5 V at the farthest servo during repeated feeds. If 5 V drops or any servo resets, increase wire gauge/branch capacity or reduce rail length; do not solve it by powering servos from USB.

The master design proposes an 8 A converter rating as a capacity target, not a measured load. Verify the converter can dissipate its losses inside the electronics enclosure. Branch protection must be selected from measured load, conductor rating and converter data. Keep servo power cables away from motor phase wires.

Firmware and bring-up

```
arduino-cli lib install Servo
arduino-cli compile --fqbn arduino:avr:nano:cpu=atmega328 firmware/tbw_feeder_fw
arduino-cli upload  --fqbn arduino:avr:nano:cpu=atmega328 -p COMx firmware/tbw_feeder_fw
```

Some Nano clones need cpu=atmega328old. After flash, verify start TBW-Feeder 3.0.0, M115, M620, then send M502 and M500 once if EEPROM layout 3 is new. First power the Nano without servos; then test one servo at low force; finally connect the bank and verify all eight channels and USB serial recovery.

The machine host may be Raspberry Pi 5 only after the ARM64 OpenPnP/camera acceptance gate in ../electronics/POWERCONTROLDESIGN.md (../electronics/POWERCONTROLDESIGN.md). The feeder Nano itself remains a separate USB serial controller.

Feeder setup in OpenPnP

Исходный файл: pnp/feeder/OPENPNP.md

TBW-Feeder v3.0: OpenPnP setup

The feeder controller is a second G-code device next to the motion board. Field names are from OpenPnP 2.x. If your version labels a field differently, the values stay the same.

Driver

Machine Setup → Drivers → add a GcodeDriver named Feeders:

Field | Value

- Field: Communications | Value: Serial, the Nano's COM port, 115200 8N1, no flow control
- Field: `CONNECT_COMMAND` | Value: `M115`
- Field: `COMMAND_CONFIRM_REGEX` | Value: `^ok.*`
- Field: `COMMAND_ERROR_REGEX` | Value: `^error.*`
- Field: Timeout | Value: 5000 ms (a 24 mm feed takes about 3.5 s)
- Field: Axes | Value: none

Actuators

Add two actuators on the Feeders driver, value type Double:

Actuator | `ACTUATE_DOUBLE_COMMAND` | Used for

- Actuator: `Feed4` | `ACTUATE_DOUBLE_COMMAND`: `M600 N{IntegerValue} F4` | Used for: 4 mm part pitch (0603 / 0805 / 1206 paper, SOT-23 / SOD-123 embossed)
- Actuator: `Feed2` | `ACTUATE_DOUBLE_COMMAND`: `M600 N{IntegerValue} F2` | Used for: 2 mm part pitch (0402 paper)

Feeders

Add one ReferenceAutoFeeder per lane:

Field | Value

- Field: Feed actuator | Value: `Feed4` or `Feed2`, depending on the part pitch
- Field: Feed actuator value | Value: lane index 0–7 (lane 1 = 0, the front lane)
- Field: Post-pick actuator | Value: none
- Field: Location | Value: `pick_xyz` of that lane from `../out/openpnp_coords.json`, then taught with the down camera on the first pocket
- Field: Rotation in tape | Value: 0° for parts as they sit in the tape, measured along the feed direction (−X)

Each feed moves the next pocket to the same pick location, so 2 mm tapes need no second location. Use bottom vision on every part. The servo-defined stop repeats within about 0.05 mm, and bottom vision corrects the rest. The forward end is set per lane in microseconds (M603 N<lane> A<us>, ASSEMBLY.md §5), about 0.01 mm per step.

Head Safe Z

Set the nozzle's Safe Z to safez from ../out/openpnpcoords.json. The generated value is derived from the current model and includes clearance above the tallest tooling the head can pass over. Do not copy a value from an older report or machine profile: regenerate the coordinates after changing tooling heights or the machine origin. pnp check checks modeled travel at that height; verify physical clearance at low speed before enabling automatic travel.

First run

1. Connect the driver. The log should show FIRMWARENAME:TBW-Feeder FIRMWAREVERSION:3.0.0 LANES:8.
2. Feeders tab → select a lane → Feed. One pocket moves into the window.
3. Teach the location on that pocket (camera), then Pick to check the height. The pick Z is the tape top

minus 0.6 mm (pick_xyz Z).

Feeder verification and acceptance procedure

Исходный файл: pnp/feeder/TEST.md

TBW-Feeder v3.0: first-article and acceptance tests

v3.0 is verified in CAD (feeder audit: geometry, motion, assembly paths, support-free printing, loads; pnp check), and the firmware compiles for the target. Build one lane plus the controller and pass these tests before printing the other seven lanes. Record the results in a copy of the table at the end. The tests marked (v2.1 issue) check the problems found on the v2.1 first article.

A. First article: parts against the model

 # | Check | Pass

- #: A0 | Check: Print every lane part on a 0.4 mm nozzle, 0.20 mm layers, **supports off**; note the printer's time per plate *(v2.1 issue)* | Pass: No part needs support or cleanup beyond piercing the 4 counterbore membranes; the times are within 15 % of `print_time.py`
- #: A1 | Check: The tape channel takes an 8.0 mm gauge strip along its full length, with no binding at the inlet flare | Pass: It slides under its own weight with the body tilted 30°
- #: A2 | Check: The slider lies in its tunnel; with the side plate fitted (no servo) it moves over 4.2 mm | Pass: It moves when pushed with ≤ 0.2 N
- #: A3 | Check: The SG90 tabs slide into the tab slot; with the side plate fitted, the servo can't move | Pass: No play you can feel up or down
- #: A4 | Check: Peel drive: step the servo between `U1291` and `U1709` by hand commands *(v2.1 issue)* | Pass: Rack, idler and pinion run without binding or skipping. The roller turns forward on every feed stroke and stands still on every retract. With the shaft held, the drive roller locks in one direction. The pinion doesn't slip on the shaft at 50 N mm (500 g at 10 mm).
- #: A5 | Check: Pinch: push the yoke tab down 3 mm and let go. Then step the servo back (idle stroke) with the cover tape loaded and 100 g hung on it on the peel side | Pass: It springs back onto the roller. During the idle stroke the roller and the cover tape don't move back.
- #: A6 | Check: Cover plate: slide it home; try to lift it; pull it toward the pick end *(v2.1 issue)* | Pass: With the 2 screws in, it can't be lifted or slid. Without them it slides out freely. It lies flat: a 0.1 mm feeler doesn't enter between the plate and the body anywhere.
- #: A7 | Check: The servo matches the size used in the model: case 22.5 × 12 × 22.7, tab span 32.2, tab top 4.2 below the output-side face, output boss Ø11.8 × 4.0, spline Ø4.8 × 2.8 | Pass: Within ±0.2 mm, otherwise update `model/feeder.py` and reprint
- #: A9 | Check: Servo gear: pressed on the spline, try to turn it against the held servo; run the slider over its stroke by hand commands *(new)* | Pass: No play you can feel; gear and slider rack run without binding or skipping over 37°.
- #: A8 | Check: Pawls: both turn freely on their pins; the spring returns each to its stop; the tooth stands 0.75 ± 0.1 mm above the tape floor *(v2.1 issue)* | Pass: Pushing the tooth down 0.85 mm needs 1.2–1.8 N (120–180 g), and it returns fully
- #: A10 | Check: Printed shafts in the HF0612: hold the pinion and turn the drive roller (and the pinch roller on its axle) in the locking direction with 50 N mm (50 g on a 10 cm lever arm, 5× the 9.6 N mm in use); then run 2000 feed strokes and repeat | Pass: Both clutches lock with no slip, before and after the 2000 strokes; no visible flats or dents on the shafts. If one slips, fit a hardened Ø6 dowel there (see DESIGN.md, Known limits).

B. Function, one lane on the bench

 # | Test | Pass

- #: B1 | Test: Calibrate as in ASSEMBLY.md §5 | Pass: The forward end centres the pocket. The tooth drops into the next hole with ≥ 20 µs margin on the retract.
- #: B2 | Test: 4 mm feed accuracy: 200 × `M600 F4` on 0805 paper tape, measuring the pocket centre with the down camera after each feed *(v2.1 issue)* | Pass: 3σ ≤ 0.10 mm along X, no lost or double feeds
- #: B3 | Test: 2 mm feed accuracy: 200 × `M600 F2` on 0402 paper tape | Pass: 3σ ≤ 0.10 mm, every pocket presented
- #: B4 | Test: Embossed tape (SOT-23): 100 × `M600 F4` | Pass: No missed feeds; the pockets clear the groove; the detent catches every hole
- #: B5 | Test: Peel: 200 × `M600 F4`, then 200 × `M600 F2` *(v2.1 issue)* | Pass: The cover tape is always clear of the pick pocket and taut between feeds. No tape tear. The servo never stalls.
- #: B6 | Test: Back-drag: after a feed, pull the inlet tape rearward with 1 N; repeat over 200 feeds | Pass: The tape doesn't move by more than 0.15 mm
- #: B7 | Test: Endurance: 20 000 × `M600 F4` on a tape loop (≈ 3 h) | Pass: Pawl teeth, rack and pinion show wear within 0.1 mm. The pins and springs are unchanged. The one-way bearings still lock (A4). B2 and A5 still pass afterwards.
- #: B8 | Test: Push the slider back by hand with the servo detached (`M610 S0`) | Pass: The detent holds the tape. The slider moves back, and the pawl rides out of the hole without lifting the plate.

C. In the machine

 # | Check | Pass

- #: C1 | Check: `pnp check` with the as-built heights (measure the tower and update the model if different) | Pass: 0 overlaps
- #: C2 | Check: Dry run: the head picks from every lane with vacuum off, Z raised by 5 mm | Pass: No contact with the cover plate, its rail lips (top 18.3 mm), the tower or the cover tape
- #: C3 | Check: 100 picks of 0805 per lane with vacuum and bottom vision | Pass: Lost-part rate < 2 % (PR-11)
- #: C4 | Check: Empty tape runs through the chute into the box; used cover tape lands in the waste bin over a full reel | Pass: No jam; nothing reaches the reel path

Record

Test | Date | Lane | Result | Note

- Test: | Date: | Lane: | Result: | Note:

Current feeder geometry and design audit

Исходный файл: pnp/feeder/out/audit.md

TBW-Feeder v3.0 audit

Generated by feeder audit on 2026-10-24 17:59.

Area | Check | Result | Status | Note

• Area: geometry | Check: static overlaps, all 30 lane parts + tape (model pose = forward end) | Result: none | Status: OK | Note: press fits are modelled line-on-line

• Area: geometry | Check: feed pawl and detent teeth sit in sprocket holes at the forward end | Result: 0.000 / 0.000 mm3 | Status: OK | Note: detent hook face 0.15 mm behind its hole edge (back-drag play)

• Area: advance | Check: servo sweep 71.5..108.5 deg (9 poses): servo gear, slider, pawl, rack, idler, pinion vs every part | Result: free (39 pairs) | Status: OK | Note: rack, idler and pinion in mesh over the whole stroke; the pawl riding under the tape is checked below

• Area: advance | Check: feed tooth drops into the next hole at the retract end | Result: tooth/tape 0.000 mm3 | Status: OK | Note: stroke 4.2 = 4.0 feed + 0.2 free travel

• Area: peel | Check: rack/idler and idler/pinion engaged at both stroke ends (moved 0.6 mm into mesh they overlap) | Result: 2.11 / 1.51, 1.02 / 1.17 mm3 | Status: OK | Note:

• Area: advance | Check: feed pawl rides over: tooth 0.85 mm down (5.8 deg about the pin) / up-stop | Result: free (0.000 mm3) / stop hit (0.11 mm3 at +0.6 deg) | Status: OK | Note: the tail sits on the fork floor, so the push load (self-locking) cannot lift the tooth further

• Area: advance | Check: detent rides over: tooth 0.85 mm down (5.8 deg about the pin) / up-stop | Result: free (0.000 mm3) / stop hit (0.11 mm3 at +0.6 deg) | Status: OK | Note: the tail sits on the fork floor, so the push load (self-locking) cannot lift the tooth further

• Area: peel | Check: pinch yoke pushed down 3 mm by its rear tab (thread the cover tape) | Result: free | Status: OK | Note:

• Area: assembly | Check: lay-in assembly paths (8), reverse order of ASSEMBLY.md (screws go in along their axis) | Result: all free | Status: OK | Note: one direction for everything; then 3 side plate screws, the guide bar and the 2 cover plate screws

• Area: cover | Check: bayonet cover plate: held at home / slides 6.8 mm to the pick side / lifts off | Result: held / free / free | Status: OK | Note: 3 rail segments per side + 2 rear tongues; 2 M2 screws lock it

• Area: printing | Check: Body on the +Y face: bridge / one-sided ceiling / steep overhang | Result: 8.8 mm / 1.0 mm / 0.0 mm2 | Status: OK | Note: no supports: limits 10 / 1 / 3; 127 × 66 × 21 mm, bed contact 4142 mm2

• Area: printing | Check: Body thinnest wall (X/Y line sampling) | Result: 0.90 mm at ('Y', 535.25, 9.3, 16.9) | Status: OK | Note:

• Area: printing | Check: SidePlate on the -Y face: bridge / one-sided ceiling / steep overhang | Result: 6.3 mm / 1.0 mm / 0.0 mm2 | Status: OK | Note: no supports: limits 10 / 1 / 3; 127 × 61 × 16 mm, bed contact 3384 mm2

• Area: printing | Check: SidePlate thinnest wall (X/Y line sampling) | Result: 0.85 mm at ('Y', 602.25, 4.7, 14.4) | Status: OK | Note:

• Area: printing | Check: CoverPlate +Z up (as installed): bridge / one-sided ceiling / steep overhang | Result: 0.0 mm / 0.0 mm / 0.0 mm2 | Status: OK | Note: no supports: limits 10 / 1 / 3; 62 × 18 × 10 mm, bed contact 973 mm2

• Area: printing | Check: CoverPlate thinnest wall (X/Y line sampling) | Result: 1.20 mm at ('Y', 588.25, -9.3, 24.2) | Status: OK | Note:

• Area: printing | Check: Slider +Z up (as installed): bridge / one-sided ceiling / steep overhang | Result: 3.4 mm / 0.9 mm / 0.0 mm2 | Status: OK | Note: no supports: limits 10 / 1 / 3; 85 × 14 × 17 mm, bed contact 619 mm2

• Area: printing | Check: Slider thinnest wall (X/Y line sampling) | Result: 0.80 mm at ('X', 556.3, 147.55, 8.6) | Status: OK | Note:

• Area: printing | Check: Pawl on the -Y face: bridge / one-sided ceiling / steep overhang | Result: 0.0 mm / 0.0 mm / 0.0 mm2 | Status: OK | Note: no supports: limits 10 / 1 / 3; 12 × 8 × 2 mm, bed contact 48 mm2

• Area: printing | Check: Pawl thinnest wall (X/Y line sampling) | Result: 0.84 mm at ('X', 557.4, 148.1, 15.3) | Status: OK | Note:

• Area: printing | Check: ServoGear +Z up (as installed): bridge / one-sided ceiling / steep overhang | Result: 3.6 mm / 1.0 mm / 0.0 mm2 | Status: OK | Note: no supports: limits 10 / 1 / 3; 11 × 12 × 5 mm, bed contact 68 mm2

• Area: printing | Check: ServoGear thinnest wall (X/Y line sampling) | Result: 2.11 mm at ('X', 606.5, 149.75, 21.7) | Status: OK | Note:

• Area: printing | Check: Rack on the -Y face: bridge / one-sided ceiling / steep overhang | Result: 0.0 mm / 0.6 mm / 0.0 mm2 | Status: OK | Note: no supports: limits 10 / 1 / 3; 15 × 12 × 3 mm, bed contact 142 mm2

• Area: printing | Check: Rack thinnest wall (X/Y line sampling) | Result: 1.13 mm at ('X', 626.4, 144.05, 8.7) | Status: OK | Note:

• Area: printing | Check: Idler on the +Y face: bridge / one-sided ceiling / steep overhang | Result: 0.0 mm / 0.0 mm / 0.0 mm2 | Status: OK | Note: no supports: limits 10 / 1 / 3; 38 × 38 × 4 mm, bed contact 999 mm2

• Area: printing | Check: Idler thinnest wall (X/Y line sampling) | Result: 1.03 mm at ('X', 631.2, 140.55, 37.2) | Status: OK | Note:

• Area: printing | Check: Pinion on the -Y face: bridge / one-sided ceiling / steep overhang | Result: 0.0 mm / 0.0 mm / 0.0 mm2 | Status: OK | Note: no supports: limits 10 / 1 / 3; 16 × 16 × 18 mm, bed contact 136 mm2

• Area: printing | Check: Pinion thinnest wall (X/Y line sampling) | Result: 1.35 mm at ('X', 635.0, 144.05, 58.9) | Status: OK | Note:

• Area: printing | Check: DriveRoller on the -Y face: bridge / one-sided ceiling / steep overhang | Result: 0.0 mm / 0.0 mm / 0.0 mm2 | Status: OK | Note: no supports: limits 10 / 1 / 3; 15 × 15 × 12 mm, bed contact 103 mm2

• Area: printing | Check: DriveRoller thinnest wall (X/Y line sampling) | Result: 0.83 mm at ('Y', 635.65, 1.3, 51.8) | Status: OK | Note:

• Area: printing | Check: PinchAxle on the -Y face: bridge / one-sided ceiling / steep overhang | Result: 0.0 mm / 0.0 mm / 0.0 mm2 | Status: OK | Note: no supports: limits 10 / 1 / 3; 6 × 6 × 14 mm, bed contact 28 mm2

• Area: printing | Check: PinchAxle thinnest wall (X/Y line sampling) | Result: 2.99 mm at ('X', 636.5, 144.65, 40.3) | Status: OK | Note:

• Area: printing | Check: PinchYoke +Z up (as installed): bridge / one-sided ceiling / steep overhang | Result: 0.0 mm / 0.0 mm / 0.0 mm2 | Status: OK | Note: no supports: limits 10 / 1 / 3; 18 × 15 × 14 mm, bed contact 160 mm2

• Area: printing | Check: PinchYoke thinnest wall (X/Y line sampling) | Result: 1.25 mm at ('X', 633.8, 156.25, 42.4) | Status: OK | Note:

• Area: printing | Check: GuideBar on the -Y face: bridge / one-sided ceiling / steep overhang | Result: 0.0 mm / 0.0 mm / 0.0 mm2 | Status: OK | Note: no supports: limits 10 / 1 / 3; 6 × 6 × 18 mm, bed contact 26 mm2

• Area: printing | Check: GuideBar thinnest wall (X/Y line sampling) | Result: 2.10 mm at ('X', 595.2, 143.2, 48.4) | Status: OK | Note:

• Area: printing | Check: printed lane parts: material / count / filament | Result: PETG / 15 parts (13 different) / 59 g | Status: OK | Note: one material, one 0.4 mm nozzle; no TPU (O-rings are bought)

• Area: loads | Check: pawl spring (steel, OD 3 x 0.3 x L0 8, ~0.5 N/mm): seated / riding over | Result: 1.25 / 1.56 N | Status: OK | Note: solid length ~3 mm; the v2.1 flexure pawl gave 0.26 N

• Area: loads | Check: pawl pivot below the tooth: the push load turns the tooth deeper (self-locking) | Result: contact 5.9 mm above the pivot | Status: OK | Note: positive up-stop: the tail on the fork floor

• Area: loads | Check: tape push (detent ramp + pawl friction + reel drag) / servo torque vs SG90 stall 0.18 N m | Result: 3.8 N / 0.0244 N m | Status: OK | Note:

• Area: loads | Check: pawl tooth root bending (0.8 x 1.15 root, load 0.5 above it; prints lying, in-plane) | Result: 10.7 MPa | Status: OK | Note: PETG ~45 MPa in the layer plane

• Area: loads | Check: pinch spring / nip slip tension (NBR O-rings, mu ~0.8) | Result: 1.50 / 1.20 N | Status: OK | Note: EIA-481 peel force 0.1..1.3 N

• Area: peel | Check: cover tape tension held on the idle stroke (pinch roller with O-rings, locked by its HF0812) | Result: 1.20 N | Status: OK | Note: = nip slip tension, so the cover tape does not slip back on the retract

• Area: peel | Check: drive roller surface per full stroke (4 mm fed) / half stroke (2 mm fed) | Result: 4.80 / 2.40 mm | Status: OK | Note: rack 4.2 x R 8 / R 7 (the 38T idler doesn't change it), less 0.31 mm backlash + clutch take-up; the surplus slips in the nip

• Area: loads | Check: servo torque, tape + peel at nip slip vs stall 0.18 N m | Result: 0.0343 N m (peel adds 1.5 N at the slider) | Status: OK | Note: the v3.0 lifter slot added 3.1 N

• Area: loads | Check: pinion / idler / rack tooth bending (Lewis, 14T m1, face 1.8) | Result: 2.8 MPa | Status: OK | Note: the v2.1 ratchet teeth were 0.6 mm deep

• Area: loads | Check: pinion shaft (printed with the pinion, D6, bored D3.3 at the -y end) in torsion | Result: 10 N mm, 0.25 MPa | Status: OK | Note: PETG creep limit ~5 MPa; no press fit left to slip

• Area: loads | Check: HF0612 one-way bearings: torque / catalogue rating | Result: 10 / ~1800 N mm | Status: OK | Note: on printed PETG shafts (the catalogue assumes HRC 58): TEST A10 checks the lock

• Area: cover | Check: cover plate deflection under the pawl spring: rails every 14 mm vs v2.1 (screws + hinge 58 mm apart) | Result: 0.031 mm vs 2.2 mm | Status: OK | Note: the v2.1 plate bowed enough to let the pawl lift the tape out of the hole

• Area: cover | Check: 2 x M2 countersunk screws (self-tapped into PETG, ~40 N pull-out each) lock the plate home vs tape drag on the feed stroke | Result: 80 N vs 0.94 N | Status: OK | Note: the bayonet cannot slide out while they are in: positive lock, no magnets

• Area: bank | Check: chute, waste bin, reel rack vs frame, plate, lanes | Result: none | Status: OK | Note:

• Area: bank | Check: empty tape drop path below the plate slot to the bench | Result: clear | Status: OK | Note:

• Area: bank | Check: carrier tape path reel -> lane inlet | Result: clear | Status: OK | Note:

• Area: bank | Check: used cover tape leaves the nip over the waste bin lip | Result: clear | Status: OK | Note:

54 OK, 0 FAIL

Machine power and control architecture

Исходный файл: pnp/electronics/POWER_CONTROL_DESIGN.md

TEVO-BW — силовой блок и компьютер управления (концепт A)

Редакция 0.1 · 31.10.2026 · компоновка для текущей модели машины и feeder v3.0

Решение

- Основной DC источник: Mean Well LRS-350-24, 24 V / 14.6 A / 350.4 W, только после сверки фактических токов драйверов, насоса и клапанов. Текущий CAD-проект считает динамику XY для 24 V; это не доказывает, что имеющаяся donor-плата или моторы рассчитаны на такую шину.
- Компьютер OpenPnP: x86-64 mini-PC с Linux 64-bit — рекомендуемый производственный host. Официальные OpenPnP Linux packages указывают x86/x64, не ARM64.
- Raspberry Pi 5: можно установить Raspberry Pi OS 64-bit Desktop и оставить Pi как вспомогательный компьютер/логгер. Запуск OpenPnP на ARM64 считать экспериментом: он требует сборки/проверки Java-приложения, нативного захвата камер, двух UVC -потоков и USB serial. Orange Pi имеет ту же ARM64 оговорку.
- Host получает отдельное штатное питание. Для Pi 5 это официальный USB-C БП 27 W (5 V / 5 A); не подключать 24 V к GPIO/USB-C и не питать SBC через неизвестный buck.
- Feeder: штатный Arduino Nano по USB как отдельный USB/serial-контроллер; сервошина 5 V от собственного DC/DC 24→5 V, не от Pi/Nano/USB. Общий GND логики и сервоприводов соединить одной точкой у feeder controller; исключить обратную подачу питания в USB перемычкой IO-shield.
- Драйверы движения: существующая donor-плата и её драйверы, на отдельной защищённой 24 V ветви, если это подтверждено паспортами. Сигналы шаг/направление и E-stop согласовать с фактической платой.

Корпус предусматривает место для Pi 5 как вспомогательной платы, но OpenPnP в производственной конфигурации размещается на x86-64 host. Перенос OpenPnP на Pi/Orange Pi допускается после успешной сборки ARM64 и теста обеих камер, USB-serial motion board/feeder, стабильности изображения и задержки управления.

Однолинейная схема

```
230 VAC* → вводной разъём/выключатель → вводная защита → распределительная колодка
                                                  ├─ AC → LRS-350-24 → DC распределитель 24 V
                                                  │                    ├─ предохранитель → motion board / drivers / motors
                                                  │                    ├─ предохранитель → pump (если 24 V) / vacuum valve
                                                  │                    ├─ DC/DC 24→5 V, ≥8 A → servo rail, 8 lanes
                                                  │                    └─ DC/DC 24→5 V, ≥2 A → Nano logic* / sensors (если не USB)
                                                  └─ AC → официальный 27 W USB-C PSU → Raspberry Pi 5

PE ──────────────── корпус/дверь, DIN rail (если проводящий), PE stud БП
E-stop (2 NC контакта) → safety relay/contactor: разрыв 24 V приводов + аппаратный DRIVER_EN
Pi остаётся запитан для логирования; остановка движения не должна зависеть от Linux/OpenPnP.
```

*** 230 VAC показано как проектное допущение. Проверить местное напряжение, класс защиты источников, разъём, защиту и правила монтажа. Не проводить сетевую часть на печатной плате и не оставлять открытые клеммы. Сетевые работы поручить квалифицированному электрику. PE не коммутировать предохранителем или E-stop. Номинал ввода, автомата, кабеля, контактора, клемм и ответвительных предохранителей выбрать по паспортам нагрузки, способу прокладки и местным нормам; этот документ намеренно не назначает их на глаз.**

DC ветви и расчётный бюджет

Ветвь | Проектное решение | Проверка до заказа

- Ветвь: 24 V motion | Проектное решение: Отдельный fuse holder → фактическая плата и драйверы | Проверка до заказа: Допуск платы/драйверов по входному напряжению; ток и суммарная мощность; фактическая PSU машины (в проекте WP0)
- Ветвь: 24 V vacuum | Проектное решение: Отдельная ветвь для насоса и нормально закрытого клапана | Проверка до заказа: Паспортный пусковой ток насоса, duty cycle клапана, диод/TVS катушки по её паспорту
- Ветвь: 5 V servo | Проектное решение: DC/DC 5 V, не менее 8 A, с тепловым запасом; звезда к шине servo | Проверка до заказа: Измерить пиковый ток одного серво и худший одновременный сценарий 8 дорожек; проверить просадку на дальнем servo
- Ветвь: 5 V feeder logic | Проектное решение: Nano питается по USB во время настройки; при автономном питании — выделенный защищённый DC/DC | Проверка до заказа: На IO shield не соединять USB 5 V с внешней сервошиной; сверить схему конкретного shield
- Ветвь: Pi 5 | Проектное решение: Производитель рекомендует питание 5 V / 5 A и адаптер 27 W; штатный USB-C PSU | Проверка до заказа: Две камеры, SSD, fan и USB hub: проверить сообщения undervoltage, ток и пропускную способность

LRS-350-24 даёт 24×14.6 = 350.4 W на выходе по паспорту; это предельный источник, не утверждение о потреблении машины. Сложить измеренные токи всех одновременно работающих ветвей, учесть пуск насоса и запас, после чего подобрать защиту каждой ветви и сечение. Предусмотреть не менее 20% резерва по длительной нагрузке как проектную цель; подтвердить тепловым тестом в закрытом корпусе. В текущем проекте feeder-заметка оценивает сервошину приблизительно до 1.1 A, но это прежняя оценка для одного активного серво и лёгкого удержания, а не гарантированный all-servo stall current.

Корпус и компоновка

Предлагаемый монтажный щит 300×240×120 mm (Ш×В×Г), IP20 внутри сухого закрытого основания станка. Перед изготовлением проверить реальные габариты LRS-350, доступ к винтам/клеммам, радиус изгиба и выделение тепла по паспорту конкретной партии. Металлический корпус предпочтителен как защитный экран; отдельная PE шпилька с шайбами-зубцами, провод двери/крышки. Пластиковый корпус требует закрытых клемм и отдельного решения по защите от прикосновения и пожара.

Зоны: сверху/слева — ввод AC и 24 V PSU; справа — защищённые DC клеммы и DC/DC; низ — PE и вводы кабеля; Pi в отдельном вентилируемом низковольтном отсеке, удалённом от 230 V и силовых моторов/драйверов. Сервошина и USB-кабели вести отдельно от моторных фаз и насоса; пересекать силовые провода под 90°. Поставить strain relief на вводах. Не ставить Pi поверх PSU/драйвера и не закрывать радиатор/активный cooler.

В комплекте находятся векторные эскизы powerwiring.svg и enclosurelayout.svg, их PNG-экспорты и редактируемый источник render_concept.ps1. Это концептуальные схемы/рендер компоновки, не производственный чертёж с допусками. Корпус не смоделирован в исходном FreeCAD assembly.

Подробная клеммная схема AC/PE/24 V, аварийного останова и DC/DC дана в WIRINGDETAILS.md (WIRINGDETAILS.md) и wiringdetail.svg / wiringdetail.png. USB, host, камеры и распиновка Nano/servo показаны в hostiodetail.svg / hostiodetail.png. Процедура установки ОС и совместимость ПО вынесены в PI5SOFTWARE.md (PI5SOFTWARE.md).

Пуск и приёмка

1. Записать фото шильдиков фактической платы, драйверов, PSU, насосов, клапанов, световых колец и камер. Сверить 12/24 V до подключения.
2. С отключённой нагрузкой настроить DC/DC; измерить 24.0 V и 5.0 V на выводах нагрузки. Ограничить ток и проверить полярность.
3. Подключить только motion board; проверить E-stop аппаратно: DRIVER_EN снимается, силовая ветвь приводов отключается, Pi остаётся online.
4. Проверить Nano/servo отдельно: внешний 5 V, правильный jumper, просадку при feed всех lane, нагрев buck и кабелей.
5. На Pi установить целевой OpenPnP и провести длительный тест обеих камер, motion serial, feeder serial, USB и логирования. Записать версию ОС, OpenPnP, camera driver, latency и ошибки.
6. 30 минут при максимальном расчётном цикле: измерить температуры PSU/DC/DC/контактора, напряжения на дальнем servo и SBC; любой brownout, перегрев или сброс означает остановку и пересмотр проекта.

Что ещё не подтверждено

- Идентичность и входное напряжение donor motherboard/stepper drivers.
- Точные токи моторов, вакуумного насоса, клапанов, кольцевого света, камер и всех сервоприводов.
- Совместимость выбранного SBC/ОС с OpenPnP и захватом обеих камер.
- Размещение корпуса и крепёж к реальной раме, вентиляция/температура, EMC и электрическая безопасность.
- Уставки защит, номиналы проводов/предохранителей, пригодность E-stop и safety relay.

Источники

- Mean Well LRS-350 datasheet (https://www.meanwell.com/Upload/PDF/LRS-350/LRS-350-SPEC.PDF) — выходные параметры и требования к охлаждению/монтажу.
- Raspberry Pi 5 product brief (https://datasheets.raspberrypi.com/rpi5/raspberry-pi-5-product-brief.pdf) и официальные рекомендации по питанию (https://www.raspberrypi.com/documentation/computers/getting-started.html).
- OpenPnP camera setup (https://github.com/openpnp/openpnp/wiki/Setup-and-Calibration_General-Camera-Setup) — проверить конкретную сборку и драйвер на целевой ARM64 ОС.

Terminal schedule and detailed wiring

Исходный файл: pnp/electronics/WIRING_DETAILS.md

Detailed wiring sheets — concept A

This sheet defines the terminal-to-terminal intent for the proposed panel. It is deliberately explicit about interfaces while leaving fuse ratings, wire sizes and exact motherboard pins unresolved until the physical parts are identified.

1. AC and protective earth

From | To | Conductors | Notes

- From: X1.1 L | To: Q1 input L | Conductors: Brown / local code | Notes: AC input connector; voltage per site
- From: X1.2 N | To: Q1 input N | Conductors: Blue / local code | Notes: Switch both poles where required by the chosen device and code
- From: X1.3 PE | To: PE bar, chassis stud, door bond, PSU FG/PE | Conductors: Green-yellow | Notes: PE is unswitched and unfused
- From: Q1 output L/N | To: F0 and protected AC distribution | Conductors: L/N | Notes: F0/input protection is selected from supply datasheets and installation code
- From: AC distribution | To: PS1 L/N | Conductors: L/N | Notes: 24 V PSU AC input; respect its input range/selector
- From: AC distribution | To: PS2 input | Conductors: L/N | Notes: Dedicated certified Pi USB-C PSU; can be outside the low-voltage enclosure

Never use the CAD drawing to choose AC conductor size, fuse or contactor rating. Cover live terminals; keep the AC path physically segregated from all SELV DC terminals; bond any conductive door with a flexible PE braid.

2. 24 V DC and safety shutdown

```
PS1 V+ -> TB24+ -> F1 -> K1 power contact -> X-MOTION (+24 V) -> board VIN / motor drivers (only if rated)
                  -> F2 --------------------> X-VAC (+24 V) -> pump / valve branch (by nameplate)
                  -> F3 --------------------> DC/DC1 IN+ -> DC/DC1 OUT+ (5.0 V) -> F4 -> TB5-SERVO V+
                  -> F5 --------------------> X-AUX (spare, fuse fitted only for a defined load)
PS1 V- -> TB0 star -> X-MOTION 0 V; X-VAC 0 V; DC/DC1 IN-; controller signal reference as required
S0 E-stop, dual NC channels -> SR1 safety relay -> K1 coil + isolated DRIVER_EN path
K1 drops the motion branch. The Pi host stays powered for fault logging; servo/vacuum stop behavior must be validated.
```

F1-F5 are identifiers, not fuse ratings. Measure the loads; select each fuse from maximum continuous/inrush current, conductor ampacity, DC interrupt rating, PSU behavior and protection coordination. Keep the pump and motion return wiring separate back to TB0; do not route motor return current through the Nano/Pi ground.

3. Feeder controller and servo connector schedule

The Nano IO shield uses GVS signal connectors. Verify the shield's external power jumper with a meter before insertion. USB powers Nano logic; the isolated external 5 V branch powers only servo V+. Join grounds at the shield star point so signal levels have a reference. Never tie external servo V+ to USB VBUS.

X-SERVO connector | Lane | Signal | V+ | GND

- X-SERVO connector: X3.1 | Lane: Front / 1 | Signal: Nano D2 | V+: DC/DC1 OUT+ via F4 | GND: TB0 / DC/DC1 OUT-
- X-SERVO connector: X3.2 | Lane: 2 | Signal: Nano D3 | V+: same | GND: same
- X-SERVO connector: X3.3 | Lane: 3 | Signal: Nano D4 | V+: same | GND: same
- X-SERVO connector: X3.4 | Lane: 4 | Signal: Nano D5 | V+: same | GND: same
- X-SERVO connector: X3.5 | Lane: 5 | Signal: Nano D6 | V+: same | GND: same
- X-SERVO connector: X3.6 | Lane: 6 | Signal: Nano D7 | V+: same | GND: same
- X-SERVO connector: X3.7 | Lane: 7 | Signal: Nano D8 | V+: same | GND: same
- X-SERVO connector: X3.8 | Lane: Rear / 8 | Signal: Nano D9 | V+: same | GND: same

Typical servo cable: brown/black = GND, red = +5 V, orange/yellow = signal. Check the actual servo lead, connector orientation, voltage range and locked-rotor current. Add the existing 1000 uF / 10 V capacitor across servo V+/GND at the distribution point, correct polarity. Keep the Nano's USB serial connection physically distinct from the servo rail.

4. Host I/O and camera wiring

Host port | Device | Cable/role | Routing

- Host port: USB 3.x | Device: Powered USB hub (if needed) | Cable/role: Camera data only; hub must have a certified separate supply | Routing: Keep away from stepper phase bundles
- Host port: Hub port 1 | Device: Down-looking camera | Cable/role: UVC video | Routing: Fix device identity in OpenPnP, not only enumeration index
- Host port: Hub port 2 | Device: Up-looking camera | Cable/role: UVC video | Routing: Prefer MJPEG if supported to reduce USB bandwidth
- Host port: USB 2.x | Device: Motion board | Cable/role: USB serial / control | Routing: Add strain relief; verify actual board protocol/baud
- Host port: USB 2.x | Device: Feeder Nano | Cable/role: USB serial / feeder commands | Routing: Separate serial port identity in OpenPnP
- Host port: Ethernet | Device: Machine LAN | Cable/role: Setup, logs, remote access | Routing: Do not expose an unauthenticated remote desktop on an untrusted network

Use a powered hub only after its current rating is verified. Measure both camera streams concurrently under exposure/lighting settings. If Pi 5 is selected, OpenPnP/camera support is a commissioning gate; the official OpenPnP Linux package does not currently list ARM64 as a packaged platform. For reliable production use, host OpenPnP on x86-64 and use Pi as an optional auxiliary computer until an ARM build is demonstrated.

5. Connector IDs and wire labels

ID | Function | Label at both ends

- ID: X1 | Function: AC input L/N/PE | Label at both ends: `AC-L`, `AC-N`, `PE`
- ID: X2 | Function: 24 V PSU output | Label at both ends: `24V+`, `0V`
- ID: X3 | Function: Eight feeder servo ports | Label at both ends: `SV1..SV8`, each `SIG/V+/GND`
- ID: X4 | Function: Motion branch after K1 | Label at both ends: `MOTION+`, `MOTION-`, `DRIVER_EN`
- ID: X5 | Function: Vacuum pump/valve branch | Label at both ends: `VAC+`, `VAC-`, `VALVE+`, `VALVE-`
- ID: X6 | Function: SBC host and camera USB | Label at both ends: `HOST`, `CAM-DOWN`, `CAM-UP`

Do not connect any connector based only on color. Check continuity, polarity and protective earth before fitting the boards. Record actual pin numbers and wire gauges in the as-built copy of this table.

SBC operating system and OpenPnP software preparation

Исходный файл: pnp/electronics/PI5_SOFTWARE.md

SBC software preparation — Raspberry Pi 5 candidate

Date checked: 2026-10-31. This is a preparation guide; no Pi board or microSD card is attached to this workstation, so the image cannot be flashed and no software can be installed on hardware here.

Compatibility decision

Raspberry Pi OS 64-bit supports Pi 5 and is available on Debian Trixie. Choose the Desktop edition because OpenPnP is a desktop application. Raspberry Pi Imager is the preferred installer.

OpenPnP's official Linux download exists, but its developer guide lists packaged Linux binaries for x86/x64 and does not list ARM64. The OpenPnP OpenCV Java package has an ARMv8 build; that alone does not provide the missing OpenPnP application packaging or prove camera capture. Therefore:

- Recommended machine host for dependable control: x86-64 mini-PC running the official OpenPnP Linux package.
- Optional Raspberry Pi 5: use as auxiliary machine computer/logger, or as an experimental OpenPnP ARM port only after the build and dual-camera acceptance test below.
- Orange Pi: same OpenPnP ARM64 packaging caveat; it does not avoid the application compatibility issue.

Official downloads

Item | Official source | Selection

- Item: Raspberry Pi Imager | Official source: https://www.raspberrypi.com/software/ | Selection: Windows Imager; use it to fetch and write the OS to a new microSD/SSD
- Item: Raspberry Pi OS | Official source: https://www.raspberrypi.com/software/operating-systems/ | Selection: Raspberry Pi OS (64-bit), Desktop, Debian Trixie, Pi 5
- Item: Direct OS image | Official source: https://downloads.raspberrypi.com/raspios_arm64/images/raspios_arm64-2026-06-19/2026-06-18-raspios-trixie-arm64.img.xz | Selection: Published size 1,282 MB; SHA-256 `123287c05f27b0eebd8f65456f6369b8f6635fa50a3d440a4f9f6223bf58c8e2`
- Item: OpenPnP Linux | Official source: https://openpnp.org/downloads/ | Selection: Official Linux packages; verify target CPU architecture before installation
- Item: OpenPnP source/release | Official source: https://github.com/openpnp/openpnp/releases | Selection: Stable release currently shown as 2.6
- Item: Capture library source | Official source: https://github.com/openpnp/openpnp-capture | Selection: V4L2 camera capture library; ARM64 application packaging remains unverified

The current OS image is about 1.3 GB compressed and requires a separate boot device. Raspberry Pi Imager downloads the image directly and performs its integrity checks; prefer that workflow. Do not overwrite an existing SD card without first backing it up.

Setup procedure when hardware is available

1. Use a new high-endurance 64 GB microSD card or a supported USB SSD. Back up any card that will be erased.
2. Install Raspberry Pi Imager on the setup PC. Select Raspberry Pi 5 → Raspberry Pi OS (64-bit) Desktop → new storage.
3. In Imager settings, set device name, user/password, locale/timezone, Wi-Fi only if needed, and enable SSH only if remote maintenance is required. Prefer wired Ethernet for setup.
4. Boot Pi 5 with active cooling and its official 27 W USB-C supply. Update using the supported OS updater; record OS/kernel and firmware versions.
5. For Pi diagnostics / source-build experiments, install basic tools: sudo apt update && sudo apt install -y v4l-utils usbutils openjdk-17-jre git cmake build-essential pkg-config libusb-1.0-0-dev.
6. Connect one UVC camera and verify it with v4l2-ctl --list-devices; repeat for the second camera. Check MJPEG/YUYV modes and bandwidth.
7. Confirm that USB serial adapters / motion board / Nano are enumerated consistently. Use stable device paths/identifiers; do not rely only on /dev/ttyUSB0 ordering.
8. For production, install the official Linux package on the x86-64 host from https://openpnp.org/downloads/ and complete OpenPnP's quick-start camera/controller setup.
9. For an ARM OpenPnP experiment, build from the official OpenPnP source with an ARM64 JDK, build/install OpenPnP Capture for AArch64/V4L2, and follow the project build instructions. Start with one camera; then validate both cameras, image processing, serial motion and feeder communication for at least a 30-minute run.
10. If application launch, native camera capture, serial latency or dual-camera operation fails, move OpenPnP to an x86-64 host; retain Pi only for auxiliary monitoring.

Software test record

Test | Expected record

- Test: OS boot/update | Expected record: OS release, kernel, boot media, supply voltage, temperature
- Test: Camera 1/2 | Expected record: UVC device ID, resolutions/fps, pixel formats, dropped frames and exposure control
- Test: Motion serial | Expected record: Controller name, VID/PID, baud, command/reply and safe-stop behavior
- Test: Feeder serial | Expected record: Nano VID/PID, firmware 3.0.0, `M115`, `M620`, one-lane feed test

- Test: OpenPnP | Expected record: Version/commit, CPU architecture, Java version, camera driver and 30-minute run result

Network limitation in this workspace

Direct binary retrieval was attempted from the provided workstation but HTTPS download failed at Windows TLS credential negotiation, and the web download endpoint denied the image fetch. No incomplete OS image or unsupported OpenPnP binary was saved. Official links and the published image checksum are recorded above so the OS can be obtained with Raspberry Pi Imager on the board setup PC.

Engineering progress and remaining gates

Исходный файл: pnp/electronics/PROGRESS.md

Power/control design progress

Date: 2026-10-31

Delivered in this revision

- Proposed 24 V distribution architecture with independently protected motion, vacuum and servo branches.
- Selected Raspberry Pi 5 as the first SBC candidate with a separate official USB-C PSU; documented the ARM64/OpenPnP/dual-camera gate and x86-64 fallback.
- Specified an enclosure concept envelope of 300 x 240 x 120 mm and separated AC, motor power, servo DC and SBC/camera zones.
- Added editable vector wiring and enclosure diagrams, PNG concept renders, preliminary BOM and machine-readable power/control notes.
- Replaced feeder electronics note with v3.0 wiring, explicit USB backfeed prevention and revised power validation instructions.
- Added power/control pages to the combined technical dossier and clarified which feeder render is current.
- Regenerated feeder.png, feedermech.png, feedersheet.png and peel_exploded.png from the current FreeCAD v3.0 model; corrected FreeCAD install-path discovery in the PowerShell wrappers.
- Regenerated the feeder cycle animation from the current model (38 frames).
- Added terminal-level AC/PE/DC/E-stop wiring and detailed host/camera/USB/Nano topology diagrams in editable SVG and PNG.
- Added Raspberry Pi OS/OpenPnP preparation notes with official download locations and the ARM64 packaging limitation.

Limits and next engineering gates

The donor board/drivers, PSU voltage, pump and valve, motors, servo stall currents, camera models and physical enclosure mounting have not been measured or identified. Protection ratings and cable gauges therefore remain TBD. The diagrams are concepts, not mains wiring or manufacturing drawings. Do not apply power until the actual module ratings, protective earth, E-stop behavior, branch protection and DC/DC outputs have been checked.

Next: identify all donor electrical components; perform an OpenPnP ARM64/camera bring-up; confirm the 24 V and 5 V load budgets; choose branch fuse/cable/contactors; check component clearances and ventilation; then produce a dimensioned mechanical panel drawing.

Source/render status

The current feeder model source is pnp/model/feeder.py; all four current feeder renders were regenerated with FreeCAD 1.1 on 2026-10-31. pnp/out/render/revD.png remains a historical render. The electrical enclosure renders are concept illustrations generated from editable SVG/source scripts.

Version control

Commit requested by the user. The supplied directory does not currently contain a .git repository (git status reports “not a git repository”), so a commit cannot be created in this workspace state. Preserve this progress record and commit the changes after restoring/initializing the intended repository.

Render revision manifest

Исходный файл: pnp/out/render/RENDER_STATUS.md

Render manifest

Current feeder geometry comes from ../../model/feeder.py and ../../feeder/tbw_feeder.py (TBW-Feeder v3.0).

- feeder.png — regenerated from the current machine CAD scene on 2026-10-31; shows the eight-lane bank in context.
- feeder_mech.png — regenerated from the current machine CAD scene on 2026-10-31; shows the v3 lane kinematics.
- feeder_sheet.png — regenerated composite of the bank, mechanism and peel-head exploded view on 2026-10-31.
- ../../feeder/out/peel_exploded.png — regenerated from feeder v3.0 source on 2026-10-31.
- ../../feeder/out/feeder_animation.gif — regenerated from the current feeder cycle on 2026-10-31, 38 frames.
- revD.png remains a historical render and is not the current assembly.

Rebuild the composite with ../../feeder/makefeedersheet.ps1. ../../pnp.ps1 render --views feeder,feeder_mech discovers both common FreeCAD Windows install folders. Other main machine views remain from October 2026; rerender them after future geometry changes.

Main machine parts list

Исходный CSV: pnp/out/bom.csv · строк: 132

Колонки: kind | item | qty | systems | mass_g_total | note

1. kind: stock | item: C-Beam 40x80x550 (stock Z column), laid horizontal | qty: 2 | systems: frame | mass_g_total: 2640.0
2. kind: stock | item: Drag chain 10x15 (stock), X run | qty: 1 | systems: chains | mass_g_total: 0.0
3. kind: stock | item: Drag chain 10x15 (stock), Y run | qty: 1 | systems: chains | mass_g_total: 0.0
4. kind: stock | item: Endstop microswitch (stock) | qty: 2 | systems: head/y_axis | mass_g_total: 6.0
5. kind: stock | item: Endstop microswitch with long lever (stock) | qty: 1 | systems: x_axis | mass_g_total: 4.0
6. kind: stock | item: NEMA17 (stock X motor) -> Y | qty: 1 | systems: y_axis | mass_g_total: 280.0
7. kind: stock | item: NEMA17 (stock bed motor) -> X | qty: 1 | systems: x_axis | mass_g_total: 280.0
8. kind: stock | item: V-slot 20x40x485 (stock X gantry bar) cut to 480, tie | qty: 1 | systems: frame | mass_g_total: 384.0
9. kind: stock | item: V-slot 20x80x500 (stock top tie) cut to 480, tie | qty: 1 | systems: frame | mass_g_total: 720.0
10. kind: stock | item: V-slot 40x80x550 (stock), Y bridge | qty: 1 | systems: y_axis | mass_g_total: 1320.0
11. kind: buy | item: Bearing 608ZZ 8x22x7 | qty: 4 | systems: x_axis | mass_g_total: 48.0
12. kind: buy | item: Camera board OV9281 global shutter 1280x800 mono, USB UVC, M12 mount, 38x38 mm (down + up) | qty: 2 | systems: head/table | mass_g_total: 20.0
13. kind: buy | item: Compliance spring + sleeve on the hollow shaft, ~1 N, 1.0-1.5 mm stroke | qty: 1 | systems: head | mass_g_total: 2.7
14. kind: buy | item: Compression spring OD 5 x 0.4 wire x L0 9, ~0.75 N/mm (pinch, 1.5 N) | qty: 8 | systems: feeders | mass_g_total: 0.8
15. kind: buy | item: Corner bracket 40x40 cast Al (tie -> C-beam) | qty: 2 | systems: frame | mass_g_total: 80.0
16. kind: buy | item: Coupler 5-8 mm | qty: 1 | systems: x_axis | mass_g_total: 18.0
17. kind: buy | item: Datum pin 3 mm hardened, press fit | qty: 2 | systems: table | mass_g_total: 1.3
18. kind: buy | item: Dowel pin ISO 8734 2 x 18 m6 (horn pin) | qty: 8 | systems: feeders | mass_g_total: 3.2
19. kind: buy | item: GT2 belt 9 mm, open, 1300 mm (ends clamped in the bridge end) | qty: 2 | systems: x_axis | mass_g_total: 196.6
20. kind: buy | item: GT2 idler 20T, 5 mm bore, 9 mm belt | qty: 2 | systems: x_axis | mass_g_total: 7.4
21. kind: buy | item: GT2 pulley 20T, 8 mm bore, 9 mm belt | qty: 2 | systems: x_axis | mass_g_total: 10.6
22. kind: buy | item: Idler axle M5x25 + nyloc | qty: 2 | systems: x_axis | mass_g_total: 6.5
23. kind: buy | item: Idler axle M5x30 + nyloc | qty: 1 | systems: y_axis | mass_g_total: 4.6
24. kind: buy | item: Jackshaft 8 mm ground rod, 592 mm | qty: 1 | systems: x_axis | mass_g_total: 233.6
25. kind: buy | item: Juki-compatible nozzle holder (CP40/500 pattern), 5 mm shaft bore | qty: 1 | systems: head | mass_g_total: 3.3
26. kind: buy | item: Juki-compatible tip (set of 6, see TS 7.4) | qty: 1 | systems: magazine | mass_g_total: 1.5
27. kind: buy | item: LED ring 12/24 V, diffuse (camera light) | qty: 1 | systems: head | mass_g_total: 6.0
28. kind: buy | item: LED ring diffuse (up camera silhouette light) | qty: 1 | systems: table | mass_g_total: 8.0
29. kind: buy | item: M12 lens f = 10 mm, for 1/4" sensor or larger, low distortion, with lock nut (down camera: FOV ~18.5 x 11.6 mm at the PCB, 0.014 mm/px) | qty: 1 | systems: head | mass_g_total: 8.0
30. kind: buy | item: M12 lens f = 6 mm, for 1/4" sensor or larger, low distortion, with lock nut (up camera: FOV ~21 x 13 mm at the hovering part, 0.017 mm/px) | qty: 1 | systems: table | mass_g_total: 8.0
31. kind: buy | item: MGN12 rail 500 mm | qty: 1 | systems: y_axis | mass_g_total: 325.0
32. kind: buy | item: MGN12 rail 550 mm | qty: 2 | systems: x_axis | mass_g_total: 715.0
33. kind: buy | item: MGN12H block | qty: 4 | systems: x_axis/y_axis | mass_g_total: 148.0
34. kind: buy | item: MGN9 rail 90 mm | qty: 1 | systems: head | mass_g_total: 34.2
35. kind: buy | item: MGN9H block | qty: 1 | systems: head | mass_g_total: 26.0
36. kind: buy | item: Magazine retaining leaf spring | qty: 1 | systems: magazine | mass_g_total: 0.7
37. kind: buy | item: NEMA8 external linear stepper 20x20x30, T5x2 screw 95 mm (head Z) | qty: 1 | systems: head | mass_g_total: 60.0
38. kind: buy | item: NEMA8 hollow-shaft stepper 20x20x30, shaft OD5/ID3, runout <= 0.02 mm (nozzle C) | qty: 1 | systems: head | mass_g_total: 60.0
39. kind: buy | item: Rotary vacuum swivel M5 for 5 mm shaft, 4 mm push-in | qty: 1 | systems: head | mass_g_total: 10.0
40. kind: buy | item: Servo SG90 9 g, 180 deg, analog (tape advance) | qty: 8 | systems: feeders | mass_g_total: 72.0
41. kind: buy | item: Spring plunger X stop block | qty: 1 | systems: table | mass_g_total: 5.9
42. kind: buy | item: Table support: M5 levelling screw + lock nut | qty: 1 | systems: table | mass_g_total: 3.6
43. kind: buy | item: Table support: M5 levelling screw + lock nut on Al post D16 | qty: 2 | systems: table | mass_g_total: 208.1
44. kind: buy | item: Threaded rod M8 x 212 + 2 nuts + 2 washers (reel rack) | qty: 2 | systems: feeders | mass_g_total: 167.5
45. kind: buy | item: Up camera cover glass D29 | qty: 1 | systems: table | mass_g_total: 1.0
46. kind: make | item: Down camera angle, Al 3 mm, slotted for focus | qty: 1 | systems: head | mass_g_total: 18.8
47. kind: make | item: Nozzle magazine bar, Al 24x12, 6 pockets along Y | qty: 1 | systems: magazine | mass_g_total: 78.9
48. kind: make | item: Tooling plate Al 6 mm 635 x 444, PCB area + tooling strip (one piece) | qty: 1 | systems: table | mass_g_total: 4504.5
49. kind: make | item: X chain tray, Al angle 50x18x2, 495 mm | qty: 1 | systems: chains | mass_g_total: 176.3
50. kind: make | item: Y idler bracket, machined Al (end plate + side flange + slotted cheeks) | qty: 1 | systems: y_axis | mass_g_total: 140.3
51. kind: make | item: Y motor bracket, machined Al 6 mm (end plate + motor plate + side flange + gussets) | qty: 1 | systems: y_axis | mass_g_total: 224.6
52. kind: make | item: Y tension block, steel 7x10x12, M3 tapped | qty: 2 | systems: y_axis | mass_g_total: 17.6
53. kind: make | item: Z slide, one piece: block plate + C motor arm + nut seat (CNC Al 6061) | qty: 1 | systems: head | mass_g_total: 25.1
54. kind: print | item: Bridge end: foot + riser + belt clamp (printed PETG-CF, 100% infill) | qty: 2 | systems: x_axis | mass_g_total: 172.9
55. kind: print | item: Cheek cap, snap-on, with lifter stop (printed PETG) | qty: 8 | systems: feeders | mass_g_total: 4.2

BOM · Main machine parts list

56. kind: print | item: Corner block front-left with X motor mount (printed PETG/ASA) | qty: 1 | systems: frame | mass_g_total: 325.2
57. kind: print | item: Corner block front-right (printed PETG/ASA) | qty: 1 | systems: frame | mass_g_total: 178.6
58. kind: print | item: Corner block rear-left (printed PETG/ASA) | qty: 1 | systems: frame | mass_g_total: 198.9
59. kind: print | item: Corner block rear-right (printed PETG/ASA) | qty: 1 | systems: frame | mass_g_total: 178.6
60. kind: print | item: Cover tape waste bin (printed PETG) | qty: 1 | systems: feeders | mass_g_total: 27.5
61. kind: print | item: Drive roller D14 x 6, D-bore (printed TPU 95A) | qty: 8 | systems: feeders | mass_g_total: 5.2
62. kind: print | item: Empty-tape exit chute, 8 lanes (printed PETG) | qty: 1 | systems: feeders | mass_g_total: 48.2
63. kind: print | item: Feeder v2.1 cover plate 1.2 mm with detent tongue (printed PETG, upside down) | qty: 8 | systems: feeders | mass_g_total: 8.8
64. kind: print | item: Feeder v2.1 hold pawl (printed PETG) | qty: 8 | systems: feeders | mass_g_total: 0.9
65. kind: print | item: Feeder v2.1 lane body (printed PETG) | qty: 8 | systems: feeders | mass_g_total: 386.4
66. kind: print | item: Feeder v2.1 peel lifter with flexure pawl (printed PETG) | qty: 8 | systems: feeders | mass_g_total: 4.5
67. kind: print | item: Feeder v2.1 ratchet 18T + hub (printed PETG) | qty: 8 | systems: feeders | mass_g_total: 3.7
68. kind: print | item: Feeder v2.1 slider with flexure pawl (printed PETG) | qty: 8 | systems: feeders | mass_g_total: 21.9
69. kind: print | item: Head belt clamp + Y chain post (printed PETG-CF) | qty: 1 | systems: head | mass_g_total: 21.7
70. kind: print | item: Head carriage plate: block mount + Z rail + Z motor shelf (printed PETG-CF) | qty: 1 | systems: head | mass_g_total: 106.6
71. kind: print | item: PCB slide clamp (printed PETG) | qty: 1 | systems: table | mass_g_total: 8.5
72. kind: print | item: Pinch axle D3 x 10 (printed PETG, lying) | qty: 8 | systems: feeders | mass_g_total: 0.5
73. kind: print | item: Pinch roller D7 x 3, runs on its axle (printed PETG) | qty: 8 | systems: feeders | mass_g_total: 0.6
74. kind: print | item: Pinch yoke (printed PETG) | qty: 8 | systems: feeders | mass_g_total: 3.8
75. kind: print | item: Reel hub / spacer (printed PETG) | qty: 8 | systems: feeders | mass_g_total: 28.3
76. kind: print | item: Reel rack side plate (printed PETG) | qty: 2 | systems: feeders | mass_g_total: 40.2
77. kind: print | item: Roller axle D3 x 20, stationary (printed PETG, lying) | qty: 8 | systems: feeders | mass_g_total: 1.0
78. kind: print | item: Slider peg D3 x 8 (printed PETG, lying) | qty: 8 | systems: feeders | mass_g_total: 0.4
79. kind: print | item: Up camera housing, one piece (printed PETG) | qty: 1 | systems: table | mass_g_total: 24.6
80. kind: print | item: Waste bin leg (printed PETG) | qty: 2 | systems: feeders | mass_g_total: 5.8
81. kind: print | item: X chain arm (printed PETG), on the Y motor bracket | qty: 1 | systems: chains | mass_g_total: 7.8
82. kind: print | item: X chain tray bracket (printed PETG) | qty: 3 | systems: chains | mass_g_total: 66.5
83. kind: print | item: X idler tension slider (printed PETG) | qty: 2 | systems: x_axis | mass_g_total: 10.0
84. kind: fastener | item: Grub screw M4 (pulley) | qty: 4 | systems: x_axis | mass_g_total: 0.0
85. kind: fastener | item: M2 heat-set insert | qty: 32 | systems: feeders | mass_g_total: 0.0
86. kind: fastener | item: M2x10 screw (X home switch) | qty: 2 | systems: frame | mass_g_total: 0.0
87. kind: fastener | item: M2x10 screw (microswitch) | qty: 2 | systems: head | mass_g_total: 0.0
88. kind: fastener | item: M2x10 screw + M3 T-nut adapter (microswitch) | qty: 2 | systems: y_axis | mass_g_total: 0.0
89. kind: fastener | item: M2x5 screw (leaf spring) | qty: 2 | systems: magazine | mass_g_total: 0.0
90. kind: fastener | item: M2x6 SHCS (NEMA8 hollow, from below) | qty: 4 | systems: head | mass_g_total: 0.0
91. kind: fastener | item: M2x6 SHCS (servo tabs) | qty: 16 | systems: feeders | mass_g_total: 0.0
92. kind: fastener | item: M2x6 knurled thumb screw (cover plate) | qty: 16 | systems: feeders | mass_g_total: 0.0
93. kind: fastener | item: M2x6 screw (camera board) | qty: 4 | systems: table | mass_g_total: 0.0
94. kind: fastener | item: M2x8 SHCS (NEMA8) | qty: 4 | systems: head | mass_g_total: 0.0
95. kind: fastener | item: M2x8 SHCS + nut (flange nut) | qty: 2 | systems: head | mass_g_total: 0.0
96. kind: fastener | item: M2x8 screw + 3 mm spacer (camera board) | qty: 4 | systems: head | mass_g_total: 0.0
97. kind: fastener | item: M3 heat-set insert | qty: 2 | systems: head | mass_g_total: 0.0
98. kind: fastener | item: M3 heat-set insert (lane bottom) | qty: 24 | systems: feeders | mass_g_total: 0.0
99. kind: fastener | item: M3x10 SHCS (NEMA) | qty: 4 | systems: y_axis | mass_g_total: 0.0
100. kind: fastener | item: M3x10 SHCS (NEMA, from the inner face, 2.5 mm ball-end key 100 mm) | qty: 4 | systems: frame | mass_g_total: 0.0
101. kind: fastener | item: M3x10 SHCS (into bracket, tapped) | qty: 2 | systems: chains | mass_g_total: 0.0
102. kind: fastener | item: M3x10 SHCS + washer (clamp slots) | qty: 2 | systems: table | mass_g_total: 0.0
103. kind: fastener | item: M3x12 SHCS (MGN block) | qty: 8 | systems: x_axis | mass_g_total: 0.0
104. kind: fastener | item: M3x12 SHCS (tooling plate -> lane, from below) | qty: 24 | systems: feeders | mass_g_total: 0.0
105. kind: fastener | item: M3x12 SHCS + nut (belt clamp) | qty: 4 | systems: x_axis | mass_g_total: 0.0
106. kind: fastener | item: M3x14 SHCS (plate -> belt clamp insert) | qty: 2 | systems: head | mass_g_total: 0.0
107. kind: fastener | item: M3x20 SHCS (chute -> plate) | qty: 6 | systems: feeders | mass_g_total: 0.0
108. kind: fastener | item: M3x30 SHCS (Y belt tension) | qty: 2 | systems: y_axis | mass_g_total: 0.0
109. kind: fastener | item: M3x45 SHCS (housing -> plate) | qty: 4 | systems: table | mass_g_total: 0.0
110. kind: fastener | item: M3x55 SHCS + washer (bin + leg -> plate) | qty: 2 | systems: feeders | mass_g_total: 0.0
111. kind: fastener | item: M3x6 low-head screw (MGN9H block) | qty: 4 | systems: head | mass_g_total: 0.0
112. kind: fastener | item: M3x6 set screw + heat-set insert (belt clamp, presses belt end) | qty: 2 | systems: head | mass_g_total: 0.0
113. kind: fastener | item: M3x8 SHCS | qty: 2 | systems: table | mass_g_total: 0.0
114. kind: fastener | item: M3x8 SHCS (MGN block) | qty: 8 | systems: head | mass_g_total: 0.0
115. kind: fastener | item: M3x8 SHCS + M3 T-nut (MGN rail) | qty: 64 | systems: x_axis/y_axis | mass_g_total: 0.0
116. kind: fastener | item: M3x8 SHCS + heat-set insert (MGN9 rail) | qty: 4 | systems: head | mass_g_total: 0.0
117. kind: fastener | item: M3x8 SHCS + heat-set insert (chain end) | qty: 4 | systems: chains/head | mass_g_total: 0.0

118. kind: fastener | item: M3x8 SHCS + heat-set insert (tray) | qty: 3 | systems: chains | mass_g_total: 0.0
119. kind: fastener | item: M3x8 SHCS + washer + heat-set insert (camera angle) | qty: 2 | systems: head | mass_g_total: 0.0
120. kind: fastener | item: M4 nut (in slider trap) | qty: 2 | systems: x_axis | mass_g_total: 0.0
121. kind: fastener | item: M4x20 SHCS + nut (idler tension) | qty: 2 | systems: frame | mass_g_total: 0.0
122. kind: fastener | item: M5x10 SHCS + M5 T-nut | qty: 42 | systems: frame | mass_g_total: 0.0
123. kind: fastener | item: M5x10 SHCS + M5 T-nut (C-beam lower slot) | qty: 3 | systems: chains | mass_g_total: 0.0
124. kind: fastener | item: M5x10 SHCS + M5 T-nut (bridge beam) | qty: 8 | systems: x_axis | mass_g_total: 0.0
125. kind: fastener | item: M5x12 SHCS (tie end tap -> C-beam slot) | qty: 6 | systems: frame | mass_g_total: 0.0
126. kind: fastener | item: M5x12 SHCS + M5 T-nut (beam side slot) | qty: 8 | systems: y_axis | mass_g_total: 0.0
127. kind: fastener | item: M5x12 SHCS + T-nut (support base) | qty: 3 | systems: table | mass_g_total: 0.0
128. kind: fastener | item: M5x16 SHCS (beam end tap) | qty: 12 | systems: y_axis | mass_g_total: 0.0
129. kind: fastener | item: M5x16 SHCS (magazine -> plate) | qty: 2 | systems: magazine | mass_g_total: 0.0
130. kind: fastener | item: M5x45 SHCS (corner block -> C-beam end tap) | qty: 16 | systems: frame | mass_g_total: 0.0
131. kind: fastener | item: Rubber foot M6 stud, D24x10 | qty: 4 | systems: frame | mass_g_total: 0.0
132. kind: fastener | item: Shaft collar 8 mm | qty: 2 | systems: x_axis | mass_g_total: 0.0

TBW-Feeder v3.0 parts list

Исходный CSV: pnp/feeder/out/bom.csv · строк: 33

Колонки: kind | item | qty_per_lane | qty_per_bank | total_8_lanes | mass_g_each

1. kind: buy | item: Compression spring OD 3 x 0.3 wire x L0 8, ~0.5 N/mm (pawls) | qty_per_lane: 2 | qty_per_bank: 0 | total_8_lanes: 16 | mass_g_each: 0.1
2. kind: buy | item: Compression spring OD 5 x 0.4 wire x L0 9, ~0.75 N/mm (pinch, 1.5 N) | qty_per_lane: 1 | qty_per_bank: 0 | total_8_lanes: 8 | mass_g_each: 0.1
3. kind: buy | item: O-ring 12 x 1.5 NBR 70 (roller tyres) | qty_per_lane: 4 | qty_per_bank: 0 | total_8_lanes: 32 | mass_g_each: 0.1
4. kind: buy | item: One-way needle bearing HF0612 (6 x 10 x 12), drive roller + pinch roller | qty_per_lane: 2 | qty_per_bank: 0 | total_8_lanes: 16 | mass_g_each: 2.5
5. kind: buy | item: Servo MG90S (metal gears + spline, SG90 size), 180 deg (tape advance) | qty_per_lane: 1 | qty_per_bank: 0 | total_8_lanes: 8 | mass_g_each: 9.0
6. kind: buy | item: Set screw M2 x 5 (pawl pivots) | qty_per_lane: 2 | qty_per_bank: 0 | total_8_lanes: 16 | mass_g_each: 0.1
7. kind: buy | item: Threaded rod M8 x 212 + 2 nuts + 2 washers (reel rack) | qty_per_lane: 0 | qty_per_bank: 2 | total_8_lanes: 2 | mass_g_each: 83.8
8. kind: print | item: Cover tape waste bin (printed PETG) | qty_per_lane: 0 | qty_per_bank: 1 | total_8_lanes: 1 | mass_g_each: 26.5
9. kind: print | item: Empty-tape exit chute, 8 lanes (printed PETG) | qty_per_lane: 0 | qty_per_bank: 1 | total_8_lanes: 1 | mass_g_each: 48.2
10. kind: print | item: Feeder v3.0 cover plate, bayonet rails (printed PETG) | qty_per_lane: 1 | qty_per_bank: 0 | total_8_lanes: 8 | mass_g_each: 1.2
11. kind: print | item: Feeder v3.0 cover tape guide bar D2.9 with head, pushed into the post (printed PETG) | qty_per_lane: 1 | qty_per_bank: 0 | total_8_lanes: 8 | mass_g_each: 0.2
12. kind: print | item: Feeder v3.0 idler gear 36T m1 with axle stub (printed PETG) | qty_per_lane: 1 | qty_per_bank: 0 | total_8_lanes: 8 | mass_g_each: 1.6
13. kind: print | item: Feeder v3.0 main body (printed PETG) | qty_per_lane: 1 | qty_per_bank: 0 | total_8_lanes: 8 | mass_g_each: 35.0
14. kind: print | item: Feeder v3.0 pawl, feed + detent (printed PETG) | qty_per_lane: 2 | qty_per_bank: 0 | total_8_lanes: 16 | mass_g_each: 0.1
15. kind: print | item: Feeder v3.0 pinch axle D6 x 14, pressed into the yoke (printed PETG) | qty_per_lane: 1 | qty_per_bank: 0 | total_8_lanes: 8 | mass_g_each: 0.4
16. kind: print | item: Feeder v3.0 pinch yoke (printed PETG) | qty_per_lane: 1 | qty_per_bank: 0 | total_8_lanes: 8 | mass_g_each: 0.4
17. kind: print | item: Feeder v3.0 pinion 14T m1 with the D6 drive shaft, one part (printed PETG) | qty_per_lane: 1 | qty_per_bank: 0 | total_8_lanes: 8 | mass_g_each: 0.6
18. kind: print | item: Feeder v3.0 rack m1 on the slider (printed PETG) | qty_per_lane: 1 | qty_per_bank: 0 | total_8_lanes: 8 | mass_g_each: 0.3
19. kind: print | item: Feeder v3.0 roller body, drive + pinch (printed PETG) | qty_per_lane: 2 | qty_per_bank: 0 | total_8_lanes: 16 | mass_g_each: 0.9
20. kind: print | item: Feeder v3.0 servo gear sector 13T m1 on the spline (printed PETG, replaces the horn) | qty_per_lane: 1 | qty_per_bank: 0 | total_8_lanes: 8 | mass_g_each: 0.3
21. kind: print | item: Feeder v3.0 side plate (printed PETG) | qty_per_lane: 1 | qty_per_bank: 0 | total_8_lanes: 8 | mass_g_each: 13.1
22. kind: print | item: Feeder v3.0 slider (printed PETG) | qty_per_lane: 1 | qty_per_bank: 0 | total_8_lanes: 8 | mass_g_each: 3.8
23. kind: print | item: Reel hub / spacer (printed PETG) | qty_per_lane: 0 | qty_per_bank: 8 | total_8_lanes: 8 | mass_g_each: 3.5
24. kind: print | item: Reel rack side plate (printed PETG) | qty_per_lane: 0 | qty_per_bank: 2 | total_8_lanes: 2 | mass_g_each: 20.1
25. kind: print | item: Waste bin leg (printed PETG) | qty_per_lane: 0 | qty_per_bank: 2 | total_8_lanes: 2 | mass_g_each: 2.7
26. kind: fastener | item: M2x6 SHCS (take-up rack -> slider) | qty_per_lane: 2 | qty_per_bank: 0 | total_8_lanes: 16 | mass_g_each: 0.0
27. kind: fastener | item: M2x6 screw (servo gear on the spline, from the servo kit) | qty_per_lane: 1 | qty_per_bank: 0 | total_8_lanes: 8 | mass_g_each: 0.0
28. kind: fastener | item: M2x8 countersunk DIN 965 (cover plate upstand -> tower) | qty_per_lane: 2 | qty_per_bank: 0 | total_8_lanes: 16 | mass_g_each: 0.0
29. kind: fastener | item: M3 heat-set insert (lane bottom + side plate joint) | qty_per_lane: 6 | qty_per_bank: 0 | total_8_lanes: 48 | mass_g_each: 0.0
30. kind: fastener | item: M3x10 SHCS (side plate -> body) | qty_per_lane: 3 | qty_per_bank: 0 | total_8_lanes: 24 | mass_g_each: 0.0
31. kind: fastener | item: M3x20 SHCS (chute -> plate) | qty_per_lane: 0 | qty_per_bank: 6 | total_8_lanes: 6 | mass_g_each: 0.0
32. kind: fastener | item: M3x50 SHCS + washer (bin + leg -> plate) | qty_per_lane: 0 | qty_per_bank: 2 | total_8_lanes: 2 | mass_g_each: 0.0
33. kind: fastener | item: M3x8 SHCS (tooling plate -> lane, from below) | qty_per_lane: 3 | qty_per_bank: 0 | total_8_lanes: 24 | mass_g_each: 0.0

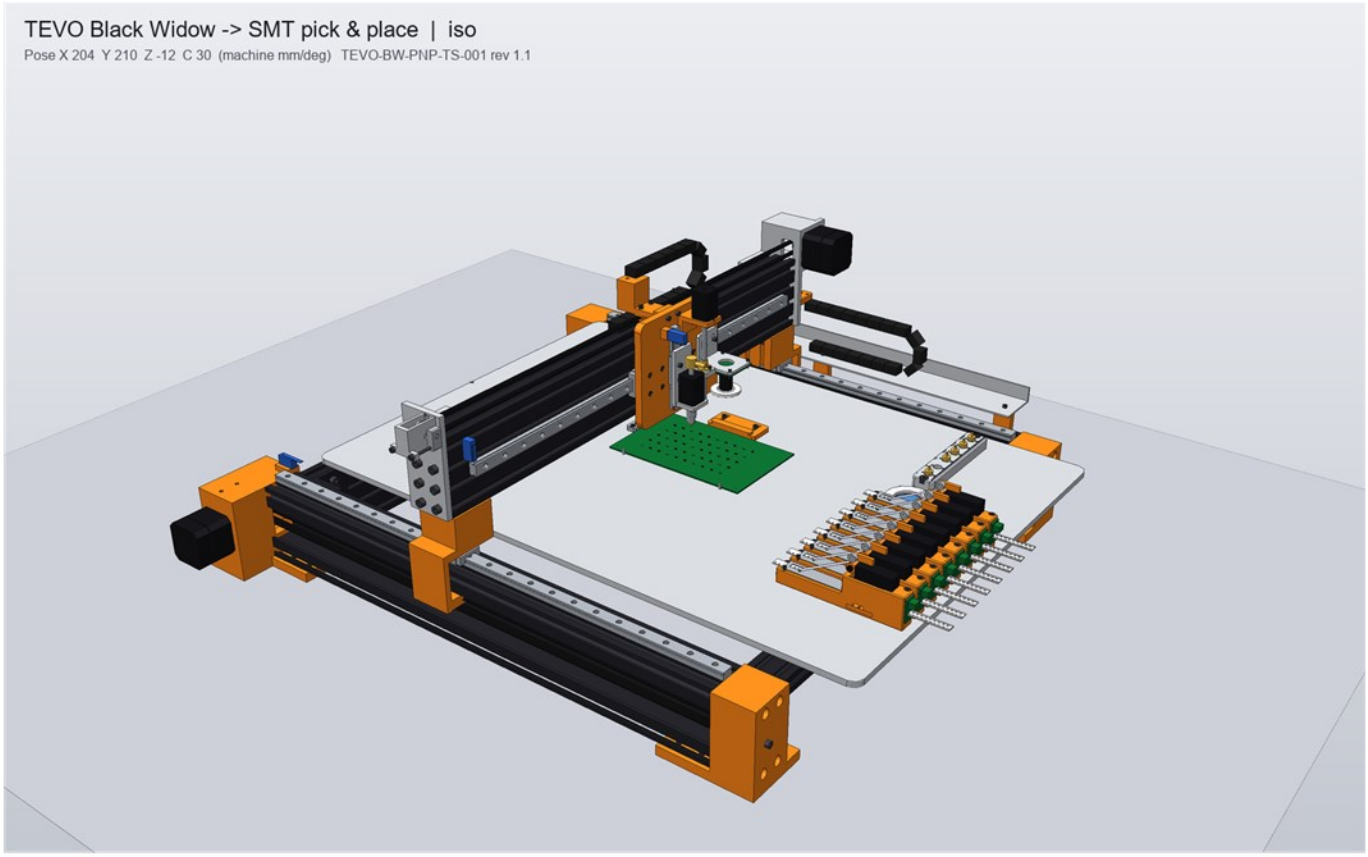
Power and control panel preliminary BOM

Исходный CSV: pnp/electronics/BOM.csv · строк: 14

Колонки: Item | Qty | Specification | Status | Notes

1. Item: Main 24V PSU | Qty: 1 | Specification: Mean Well LRS-350-24 24V 14.6A 350.4W | Status: PROPOSED | Notes: Confirm measured load and actual donor board rating
2. Item: OpenPnP host | Qty: 1 | Specification: x86-64 mini-PC Linux 64-bit | Status: RECOMMENDED | Notes: Official OpenPnP Linux packages list x86/x64
3. Item: Aux SBC | Qty: 1 | Specification: Raspberry Pi 5 preferably 8GB with active cooling | Status: OPTIONAL | Notes: Auxiliary/logger by default; ARM64 OpenPnP is experimental
4. Item: Aux SBC PSU | Qty: 1 | Specification: Official Raspberry Pi 27W USB-C supply 5V 5A | Status: OPTIONAL | Notes: Separate AC branch; do not feed 24V into Pi
5. Item: Servo DC/DC | Qty: 1 | Specification: 24V input 5.0V output rated >=8A | Status: PROPOSED | Notes: Choose exact converter after servo peak/stall measurements
6. Item: Motion branch protection | Qty: 1 | Specification: DC fuse/holder sized to motion board and conductors | Status: TBD | Notes: Actual board/drivers and inrush unknown
7. Item: Vacuum branch protection | Qty: 1 | Specification: DC fuse/holder sized to pump/valve | Status: TBD | Notes: Pump/valve datasheets and inrush unknown
8. Item: Servo branch protection | Qty: 1 | Specification: DC fuse/holder sized to converter and wiring | Status: TBD | Notes: Coordinate with converter rating and wire ampacity
9. Item: Emergency stop | Qty: 1 | Specification: 2 NC contacts with suitable contactor/safety relay | Status: PROPOSED | Notes: Verify safety category and DC interrupt rating
10. Item: Feeder logic | Qty: 1 | Specification: Arduino Nano + IO shield | Status: EXISTING DESIGN | Notes: Check external servo power jumper before connection
11. Item: Servo bulk capacitor | Qty: 1 | Specification: 1000uF 10V electrolytic | Status: EXISTING DESIGN | Notes: Place at servo distribution point with correct polarity
12. Item: Enclosure | Qty: 1 | Specification: Metal cabinet 300x240x120mm concept | Status: PROPOSED | Notes: Final size and thermal design require component fit check
13. Item: DIN rail terminals | Qty: As required | Specification: Finger-safe PE and DC distribution terminals | Status: PROPOSED | Notes: AC/DC zones separated
14. Item: Cable glands | Qty: As required | Specification: Strain relief matched to cables | Status: PROPOSED | Notes: Separate motor power and USB/camera routing

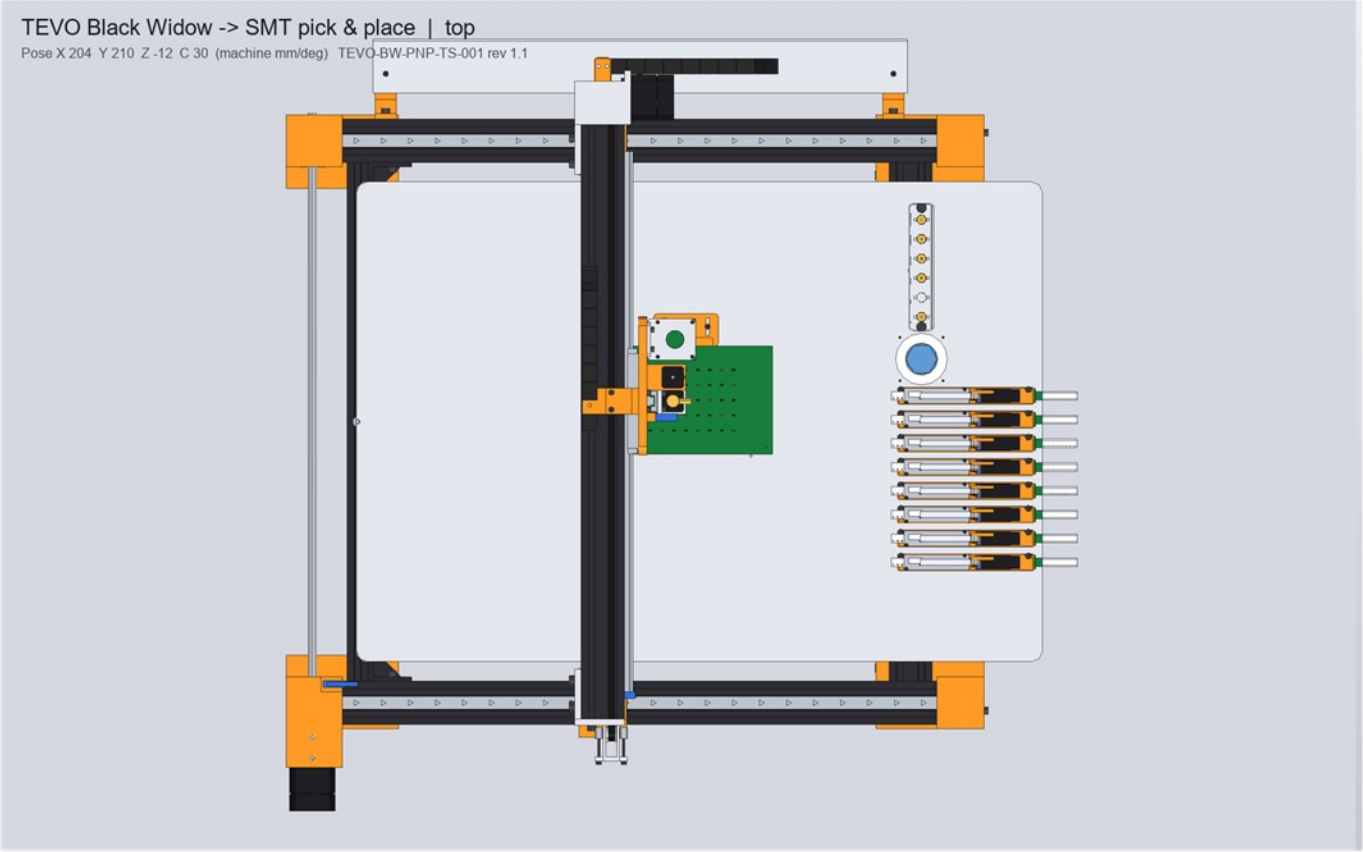
Machine · isometric



Current project CAD overview; compare physical donor hardware before fabrication.

Источник: pnp/out/render/iso.png

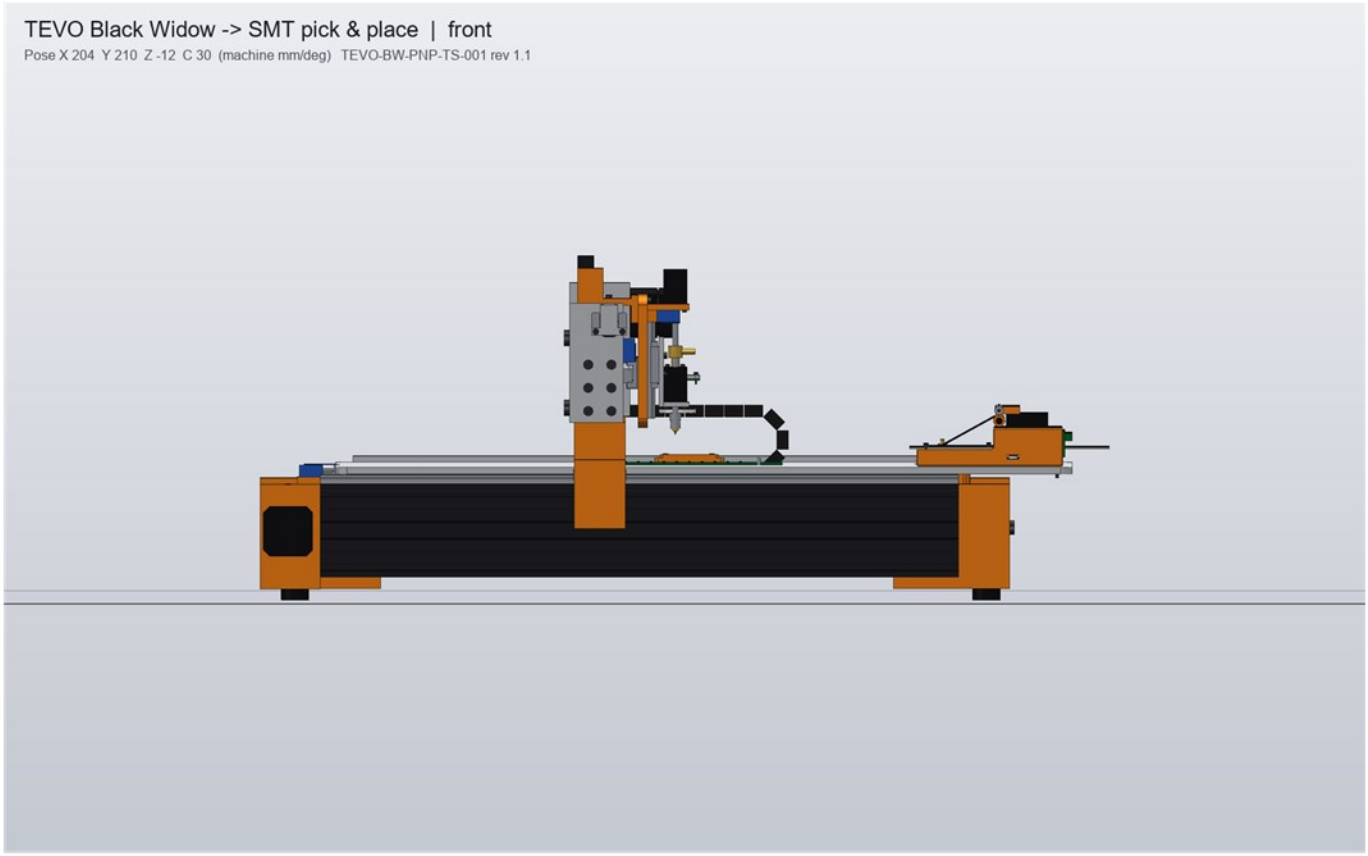
Machine · top view



Working envelope and fixed tooling locations.

Источник: pnp/out/render/top.png

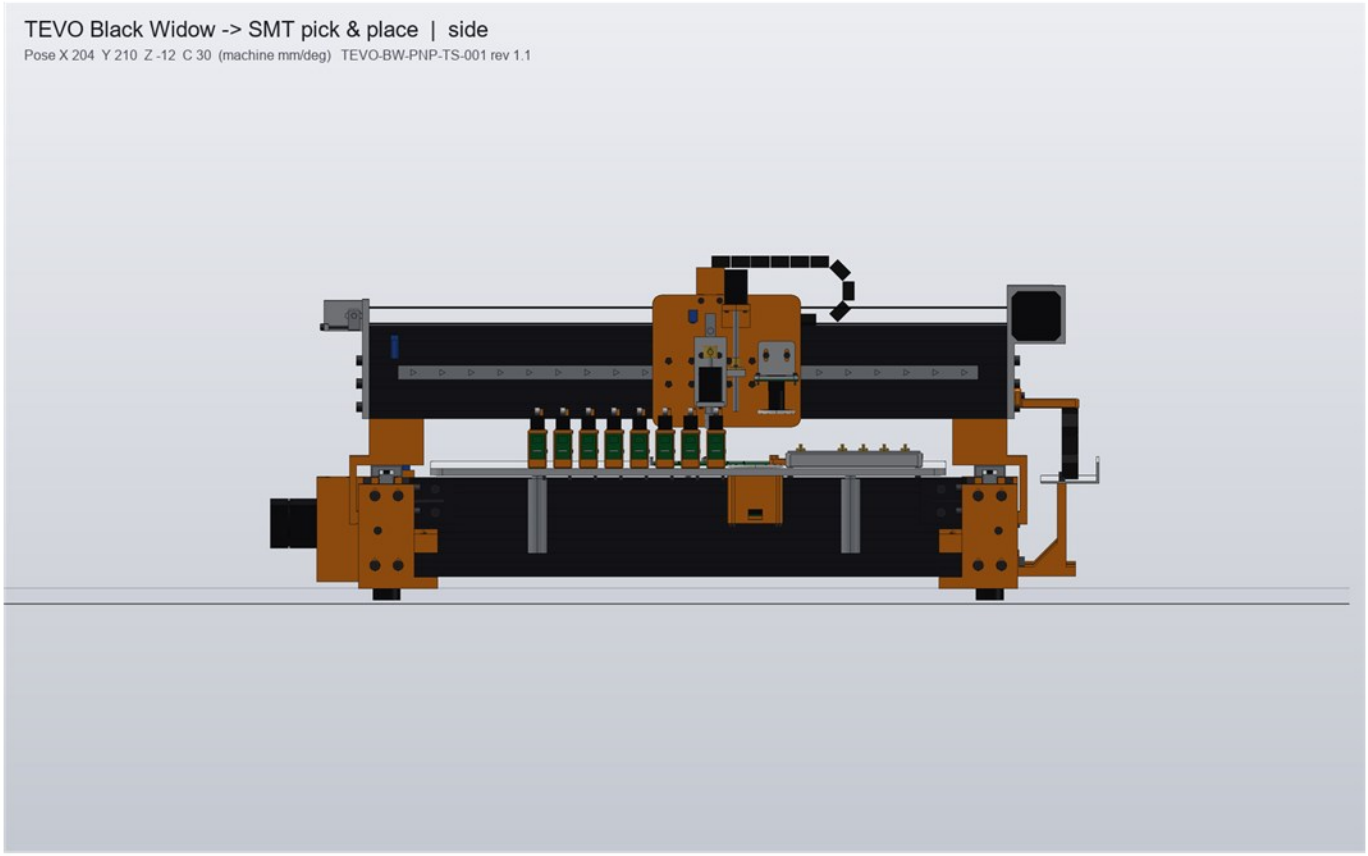
Machine · front view



Frame, gantry and vertical stack.

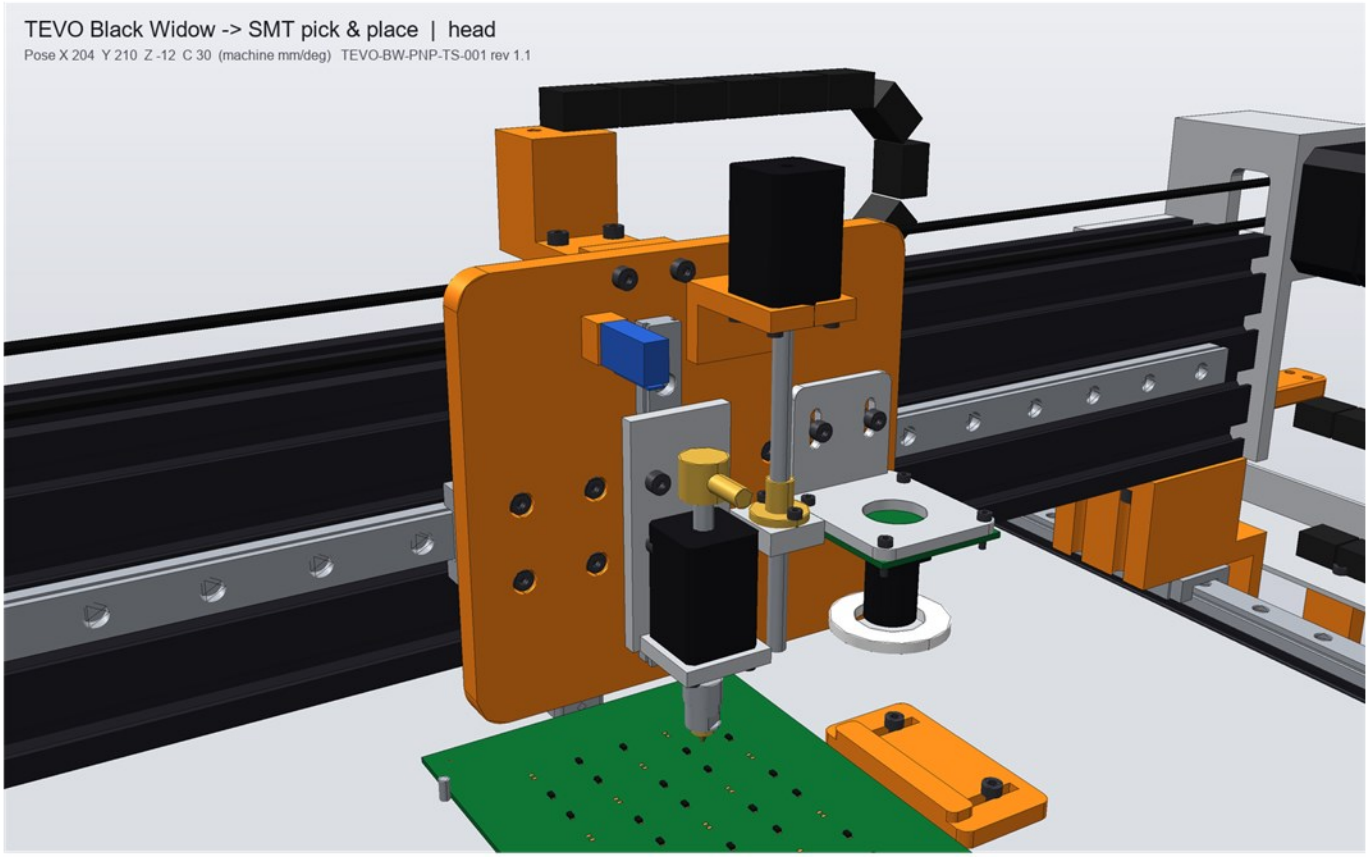
Источник: pnp/out/render/front.png

Machine · side view



Side clearance and motion arrangement.

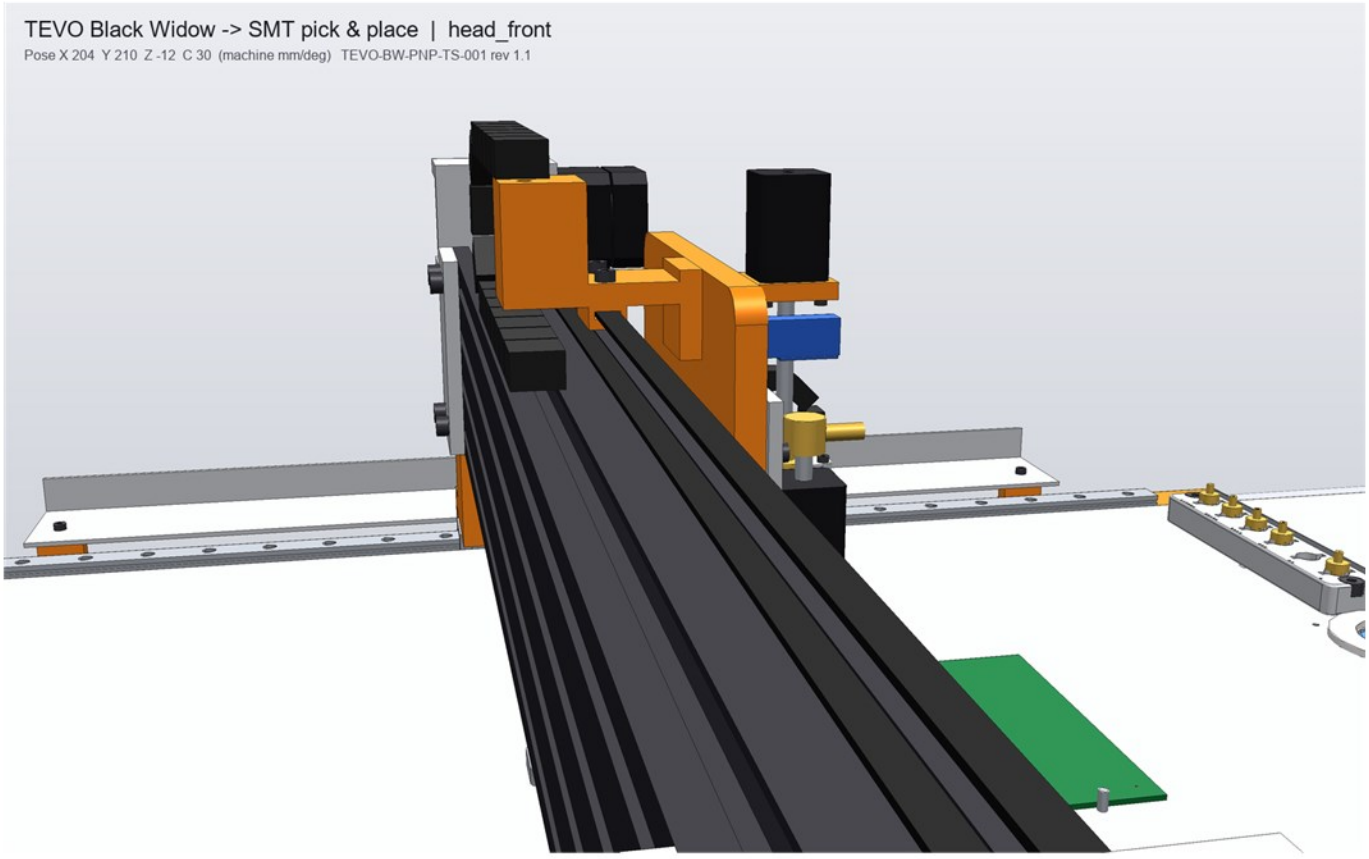
Источник: pnp/out/render/side.png



Current nozzle head and camera geometry.

Источник: pnp/out/render/head.png

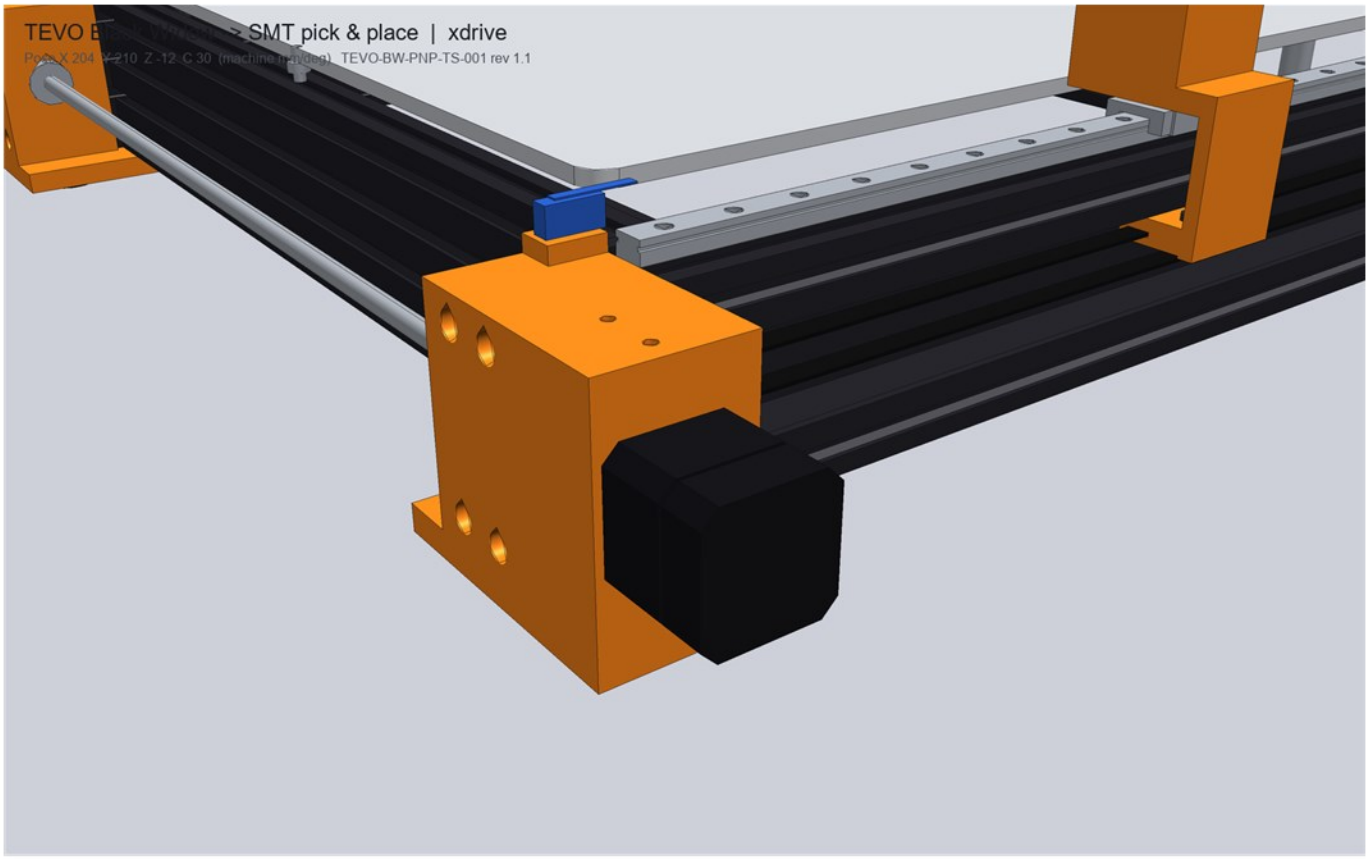
Head · front detail



Front detail of Z/C head arrangement.

Источник: pnp/out/render/head_front.png

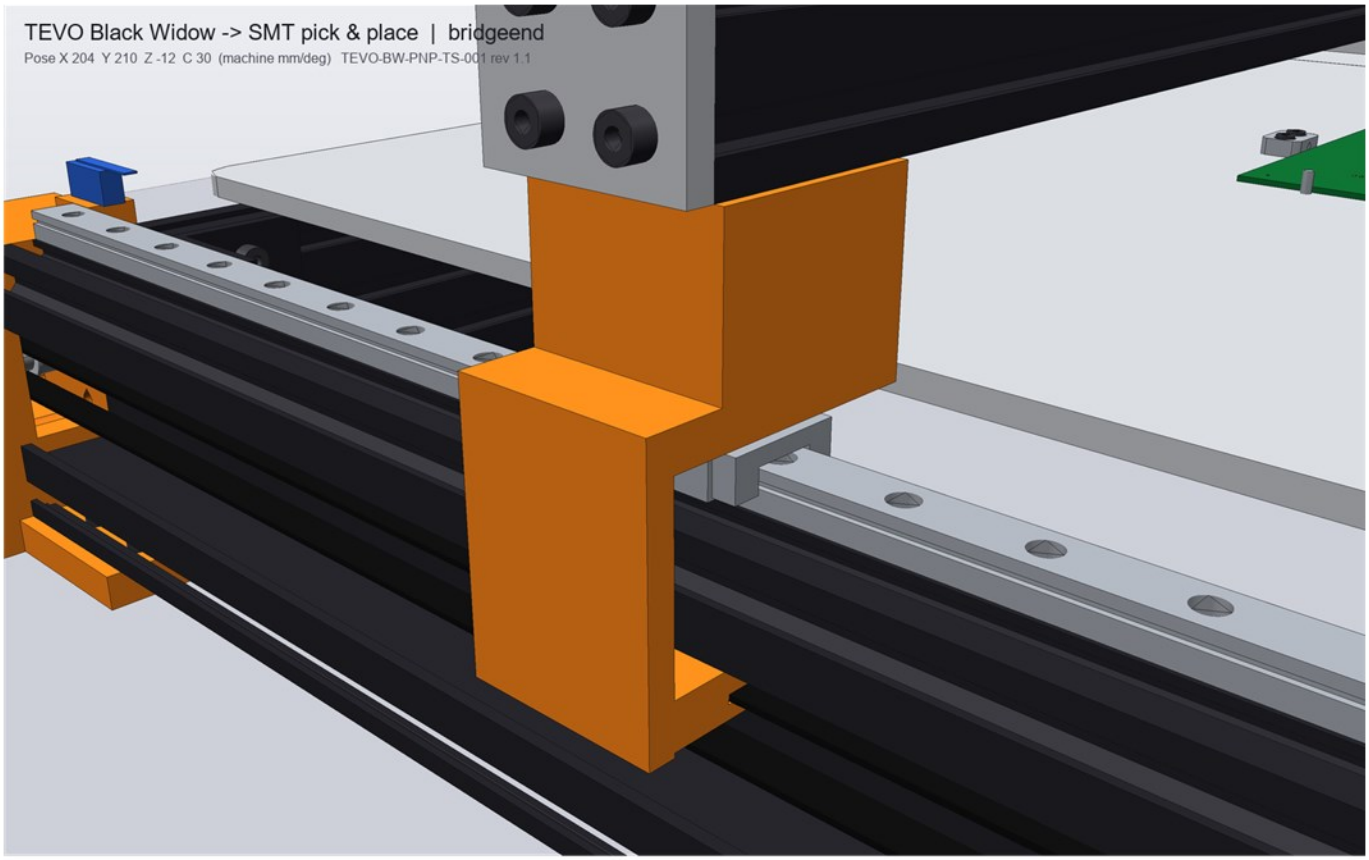
X drive · common shaft



Synchronized dual-belt X drive.

Источник: pnp/out/render/xdrive.png

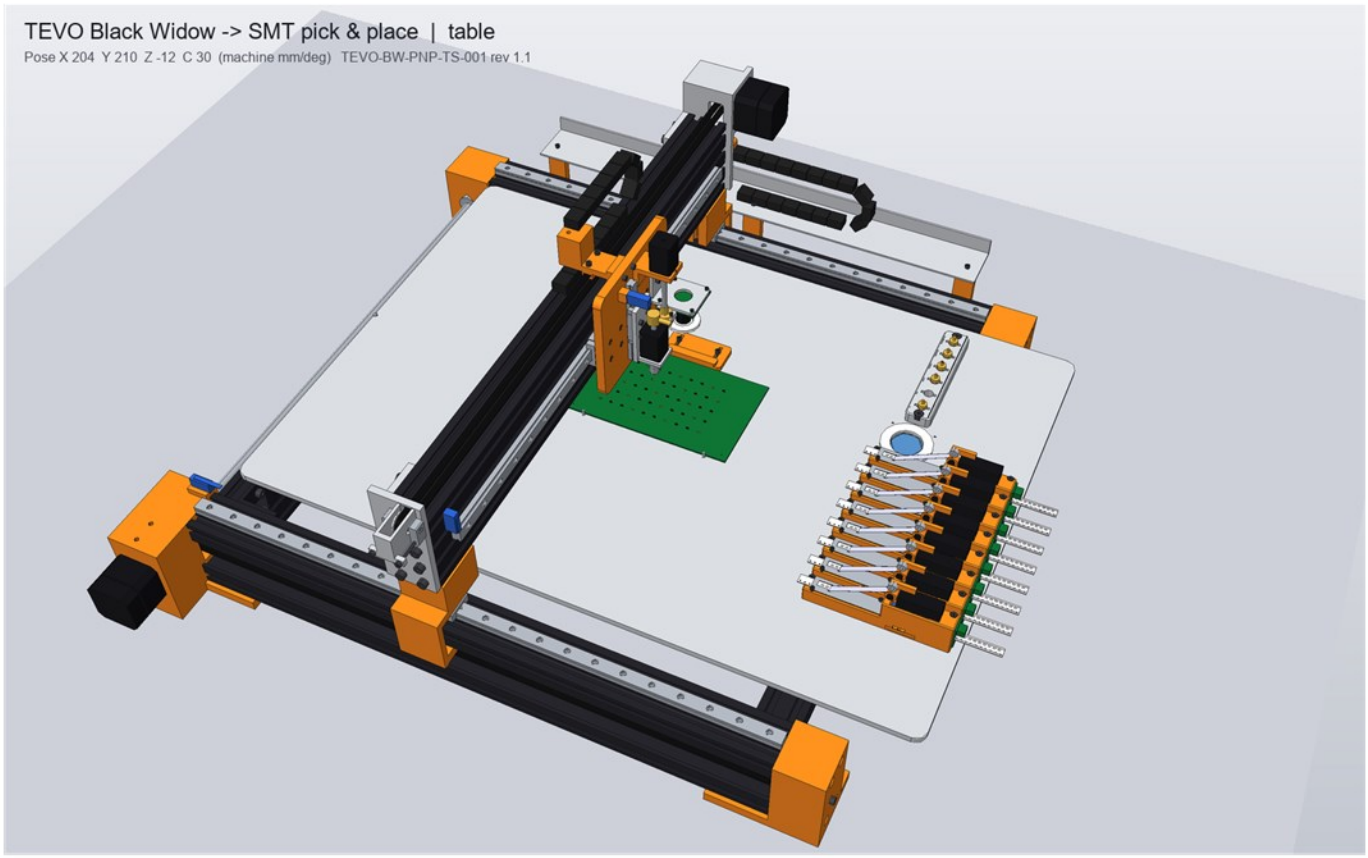
Bridge end · rail block



Bridge end and linear guide interface.

Источник: pnp/out/render/bridgeend.png

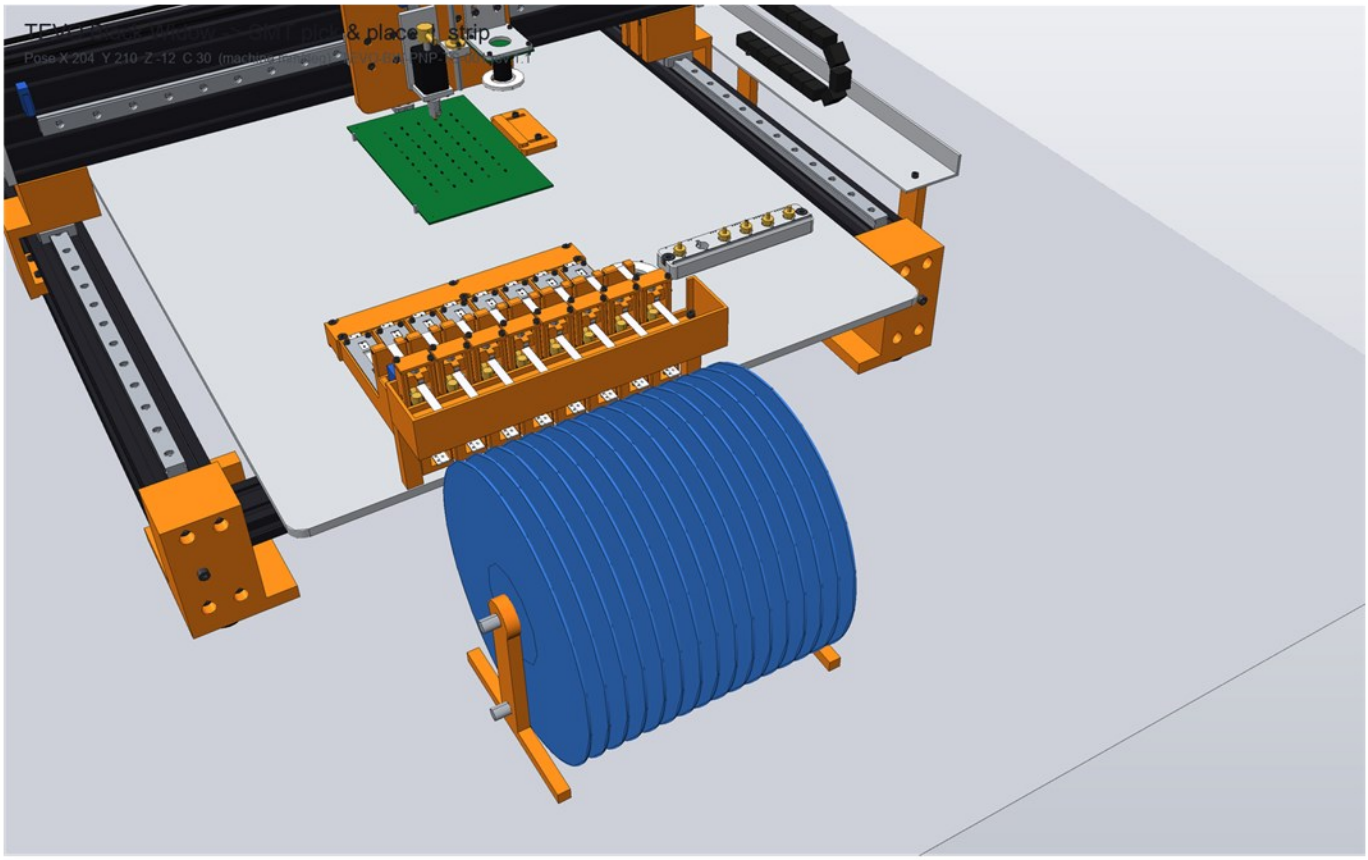
Tooling plate



Work plate, datums and support arrangement.

Источник: pnp/out/render/table.png

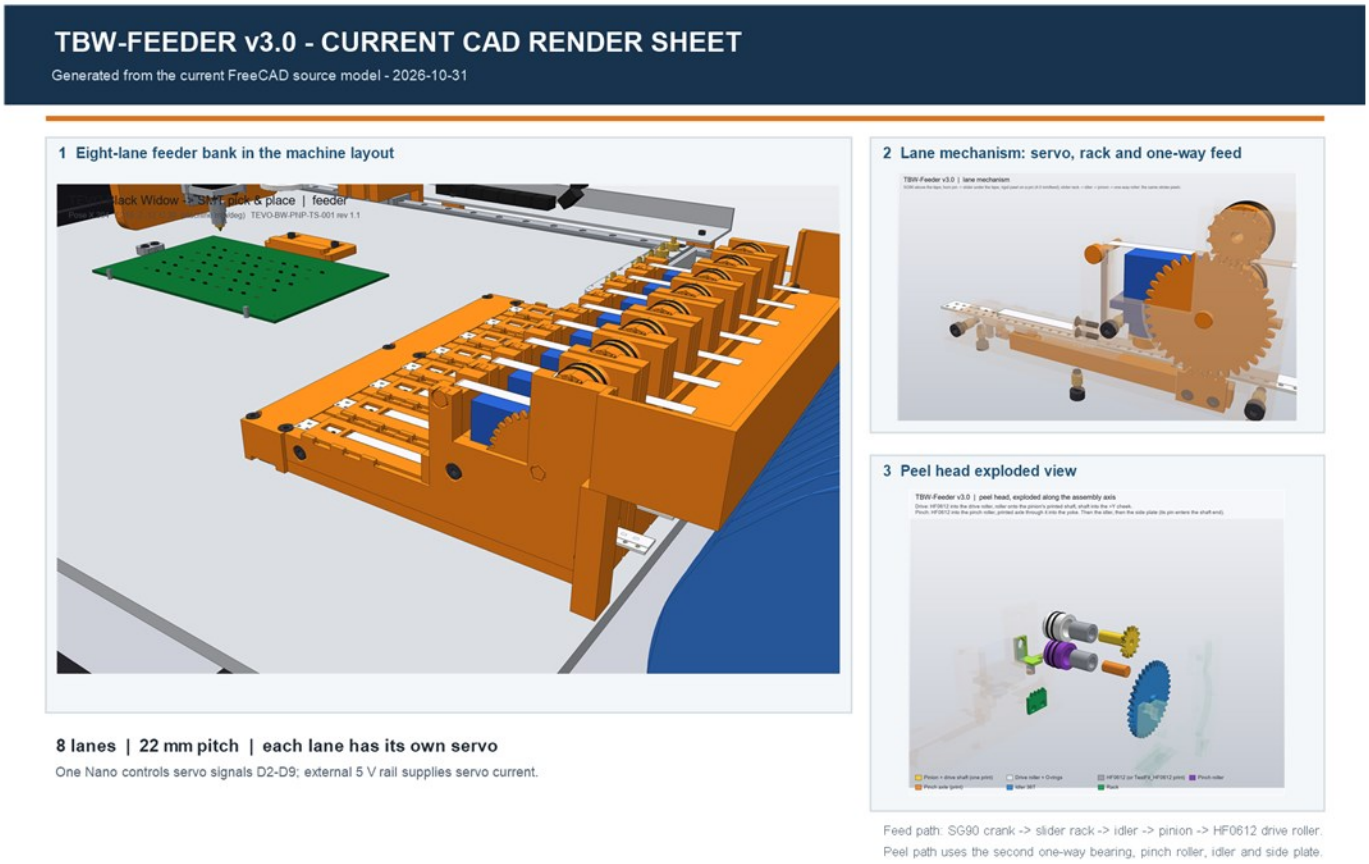
Tooling strip



Fixed feeder/tooling strip and lane arrangement.

Источник: pnp/out/render/strip.png

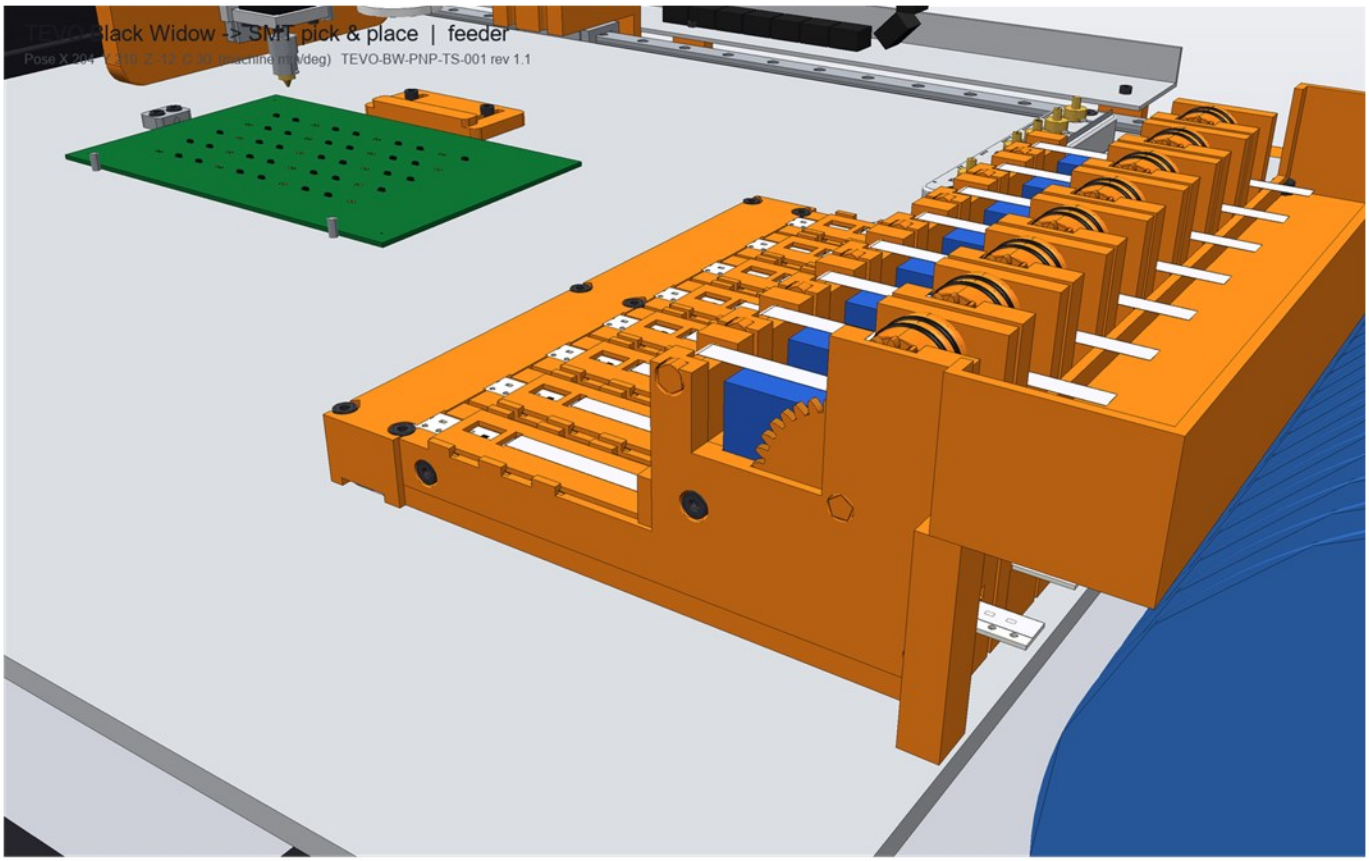
Feeder v3.0 · current render sheet



Current eight-lane bank, lane mechanism and exploded peel head.

Источник: pnp/out/render/feeder_sheet.png

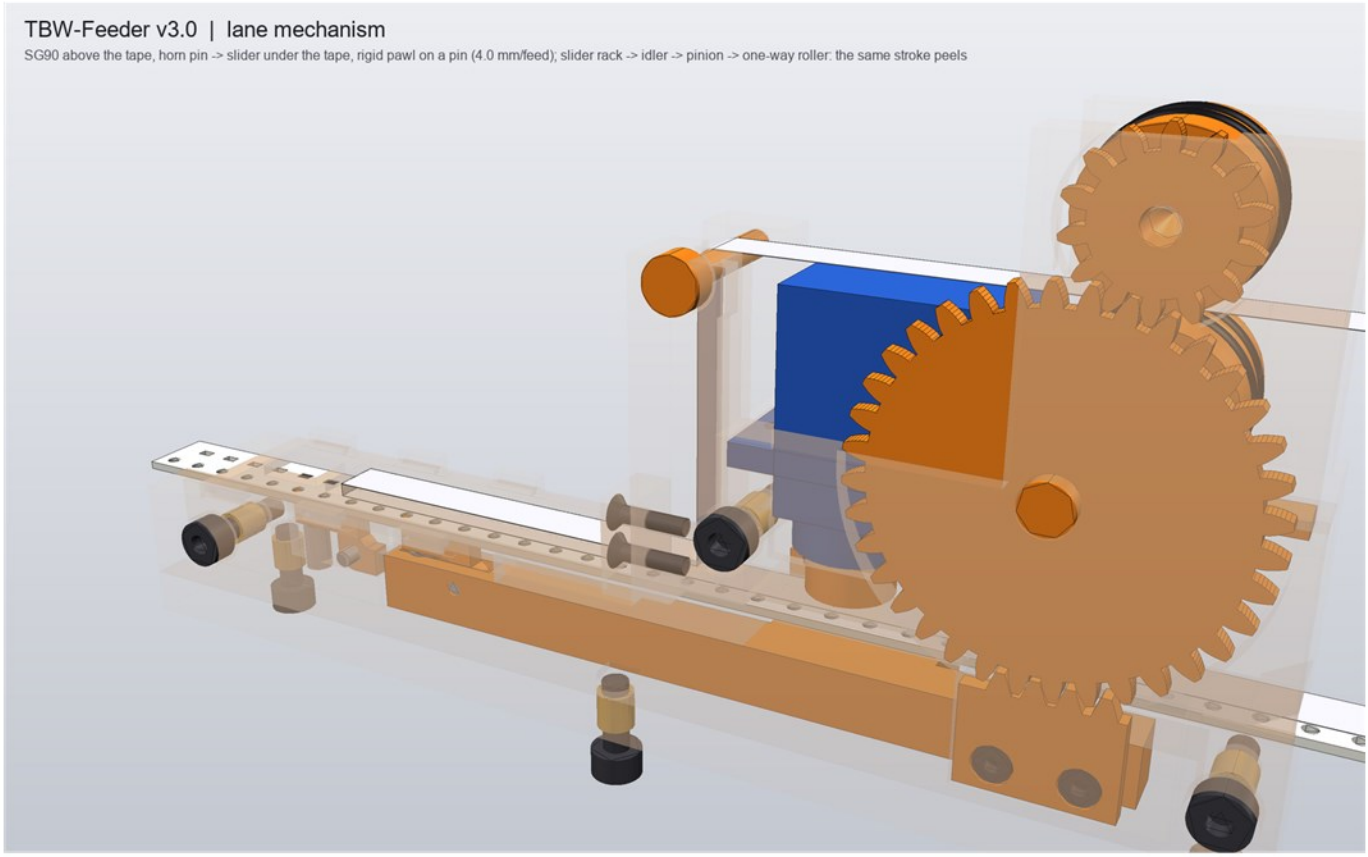
Feeder v3.0 · bank in machine



Feeder bank generated from the current machine CAD scene.

Источник: pnp/out/render/feeder.png

Feeder v3.0 · lane mechanism



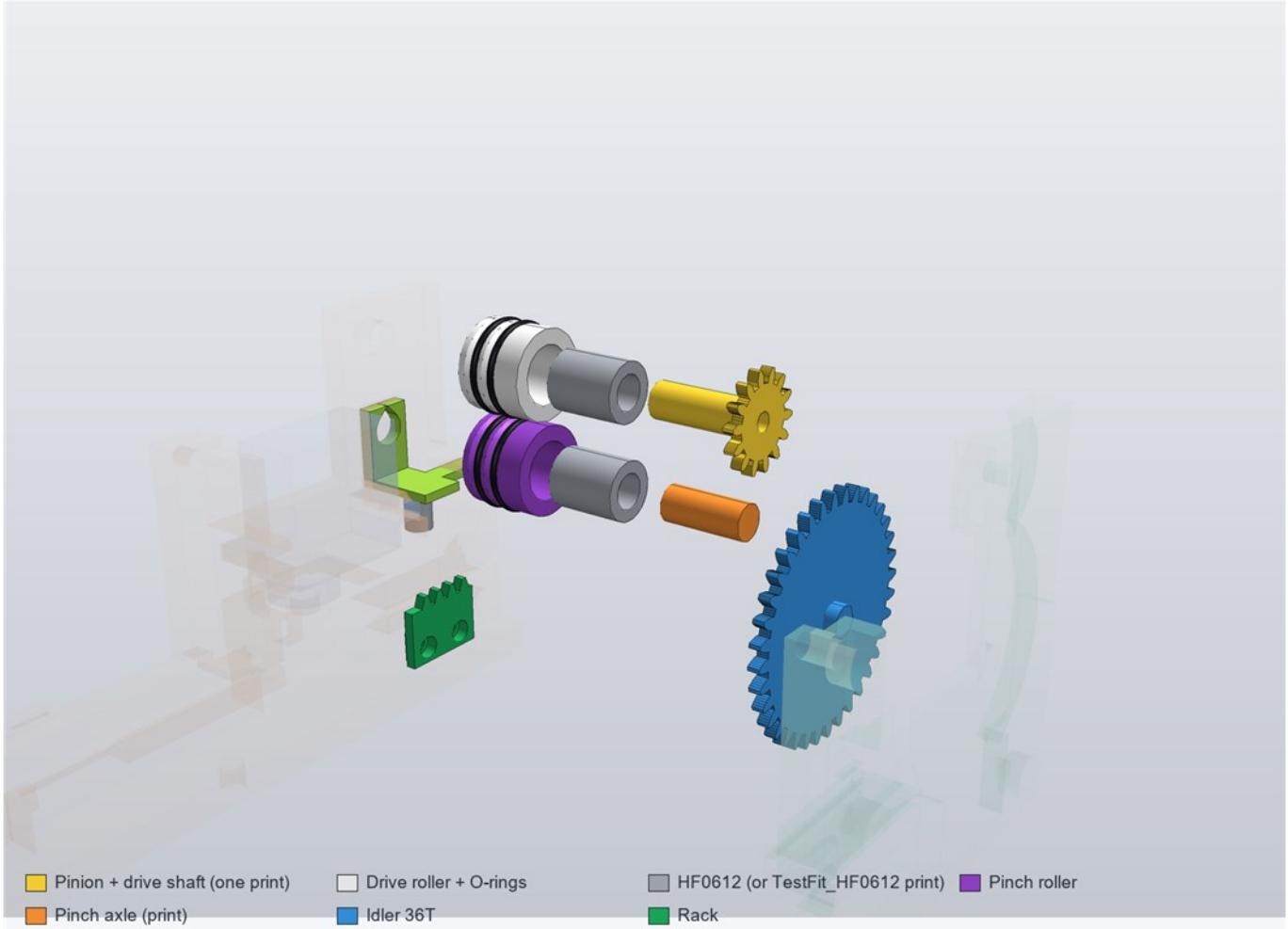
Servo, rack, idler and one-way roller drive.

Источник: pnp/out/render/feeder_mech.png

Feeder v3.0 · peel head exploded

TBW-Feeder v3.0 | peel head, exploded along the assembly axis

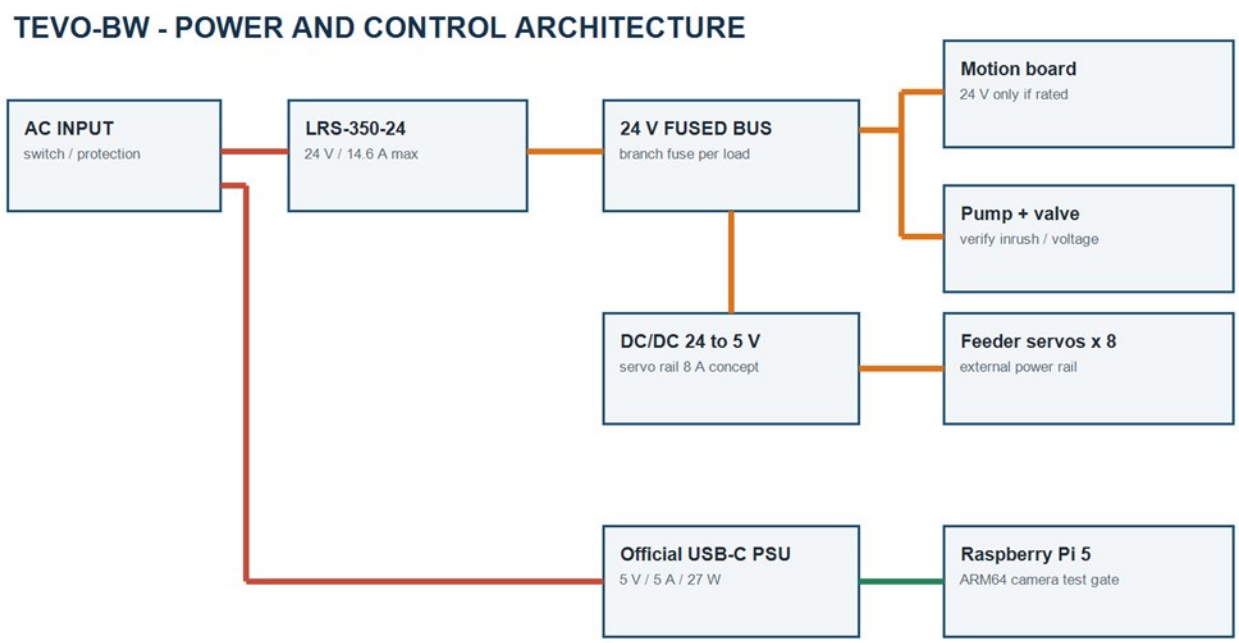
Drive: HF0612 into the drive roller, roller onto the pinion's printed shaft, shaft into the +Y cheek.
Pinch: HF0612 into the pinch roller, printed axle through it into the yoke. Then the idler, then the side plate (its pin enters the shaft end).



Current exploded peel head and assembly axis.

Источник: pnp/feeder/out/peel_exploded.png

Power architecture

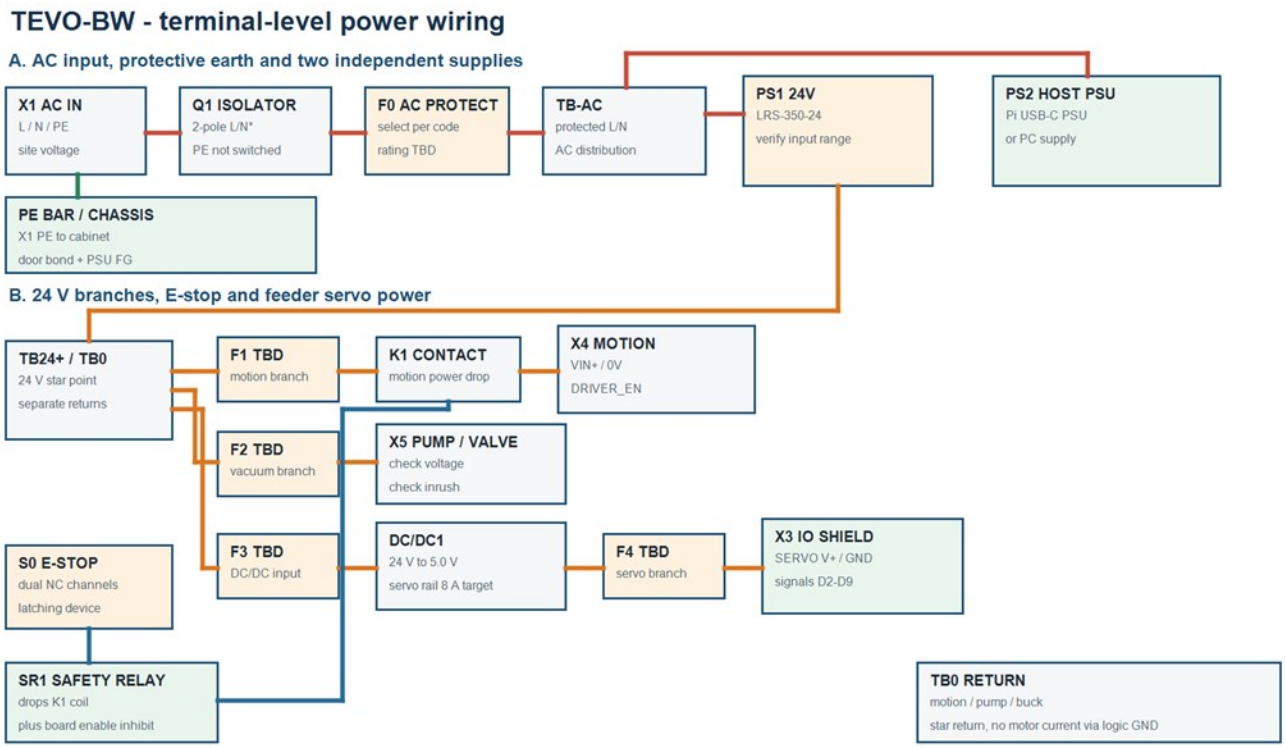


PE bonds enclosure, door and PSU. E-stop drops drive power/enable; Pi remains powered for logging.
Concept: rate branch fuses, cables and AC protection from measured loads and local code.

Proposed 24 V branches, servo DC/DC and SBC power.

Источник: pnp/electronics/power_wiring.png

Detailed terminal wiring



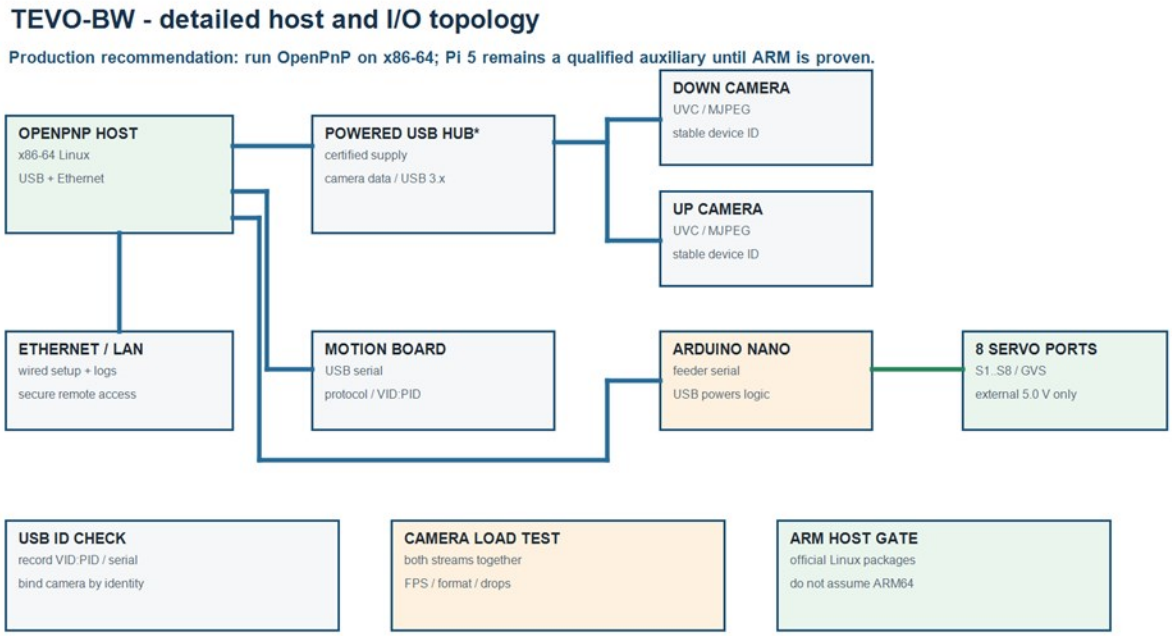
F1-F4 are reference IDs, not ratings. Choose protection and conductors from measured loads, module limits and installation rules.

*Switching of L/N follows local supply rules. PE is continuous, unswitched and unfused.

CONCEPT ONLY - verify actual hardware, protection and local electrical code before wiring.

AC/PE, protected DC branches, E-stop and feeder supply.

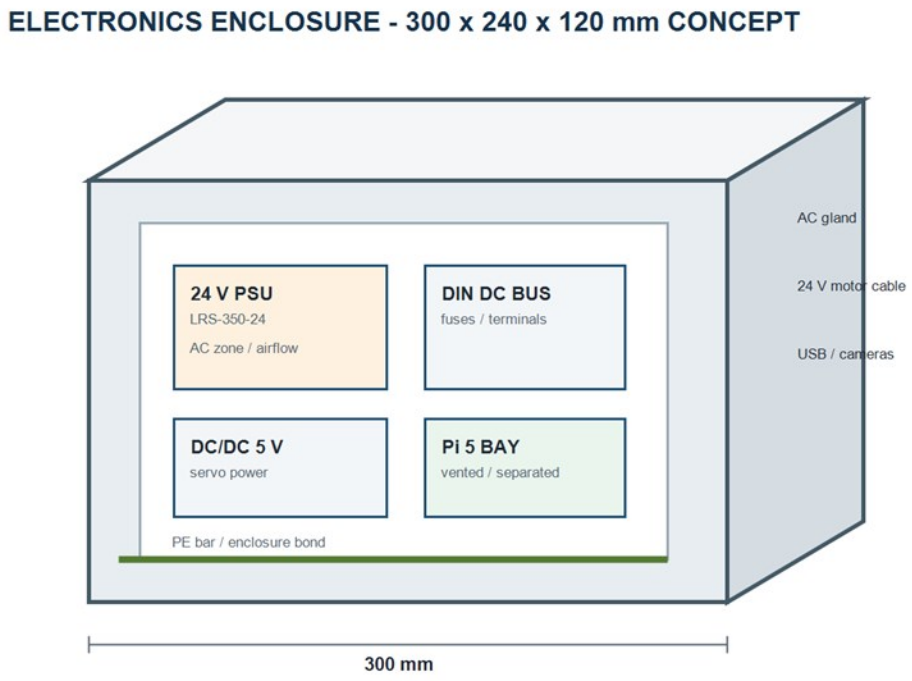
Источник: pnp/electronics/wiring_detail.png



OpenPnP host, UVC cameras, motion controller and feeder Nano.

Источник: pnp/electronics/host_io_detail.png

Electronics enclosure concept



Concept arrangement. Keep AC and low-voltage wiring apart; add finger-safe terminals, strain relief and PE bonding.

Concept arrangement of PSU, distribution, DC/DC and auxiliary SBC.

Источник: pnp/electronics/enclosure_render.png